

Introduction to Cloud Computing — Security & Identity Management (IAM)

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Environment: AWS

Project Link: [https://github.com/OluwaseunOa/DevOps-](https://github.com/OluwaseunOa/DevOps-Projects/blob/main/Cloud%20Computing/Security%20%26%20Identity%20Management%20(IAM)Security%20%26%20Identity%20Management%20(IAM)/README.mdes)

[Projects/blob/main/Cloud%20Computing/Security%20%26%20Identity%20Management%20\(IAM\)Security%20%26%20Identity%20Management%20\(IAM\)/README.mdes](https://github.com/OluwaseunOa/DevOps-Projects/blob/main/Cloud%20Computing/Security%20%26%20Identity%20Management%20(IAM)Security%20%26%20Identity%20Management%20(IAM)/README.mdes)

1. Introduction

This project focuses on **Amazon Web Services (AWS) Identity and Access Management (IAM)** and its application in securing cloud resources for Zappy e-Bank, a fintech startup.

The objective is to learn how IAM can help enforce **authentication and authorization** for users, ensuring that sensitive financial data is properly secured.

Through this project, I gained hands-on experience in creating **policies, groups, users, and multi-factor authentication (MFA)**, while applying the **principle of least privilege**.

2. Importance of IAM for Zappy e-Bank

For a fintech company like Zappy e-Bank, **security and compliance** are critical. IAM helps to:

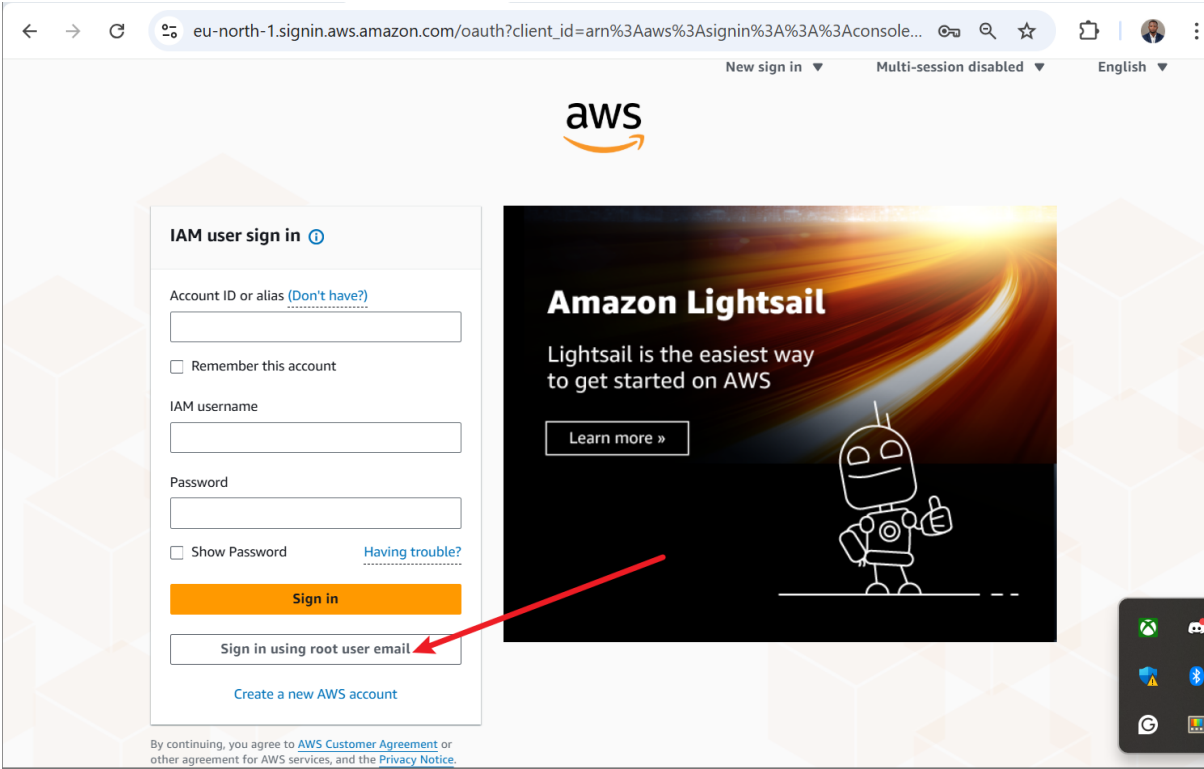
- Create and manage AWS **users and groups**.
- Assign fine-grained **permissions** through IAM **policies**.
- Implement **multi-factor authentication (MFA)** for stronger access control.
- Enforce the **principle of least privilege**, ensuring users only have access to what they need.

This reduces the risk of insider threats, unauthorized access, and compliance violations.

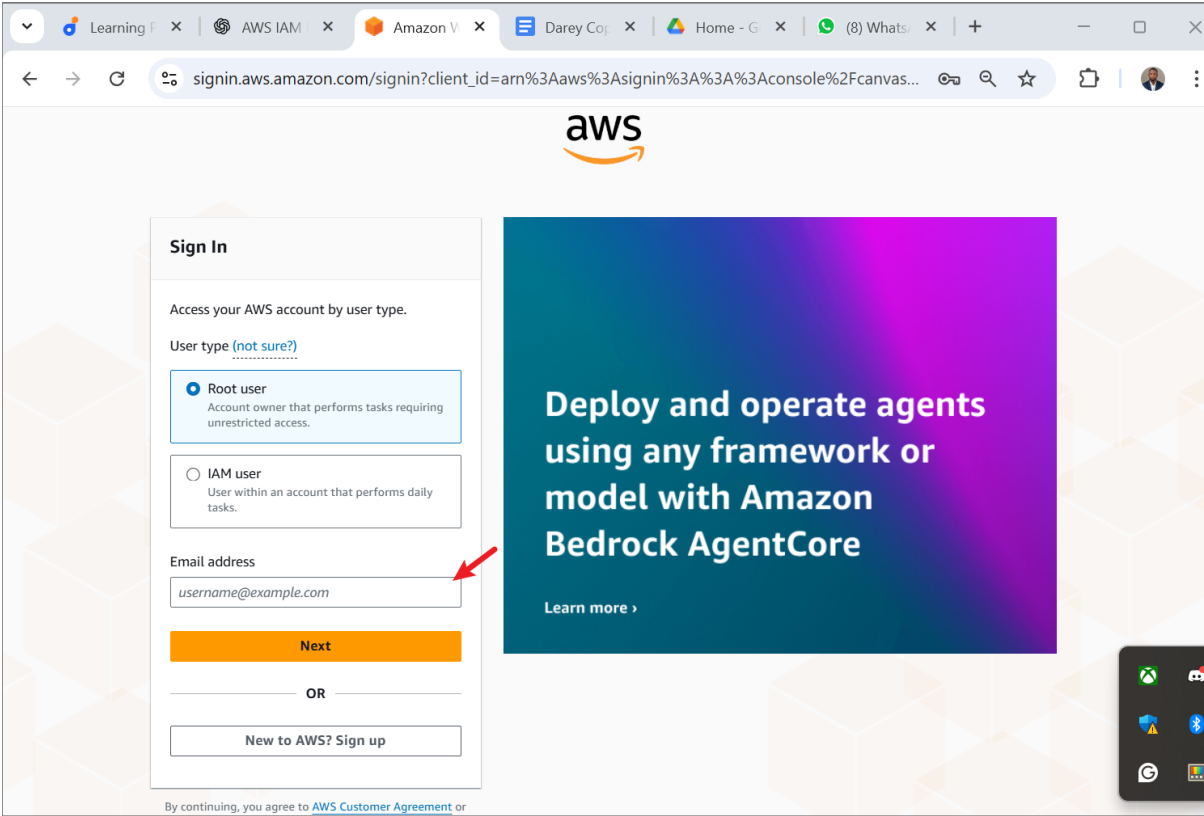
3. Project Setup

1. Logged into **AWS Management Console** using the administrator account.

- Log in using root email.



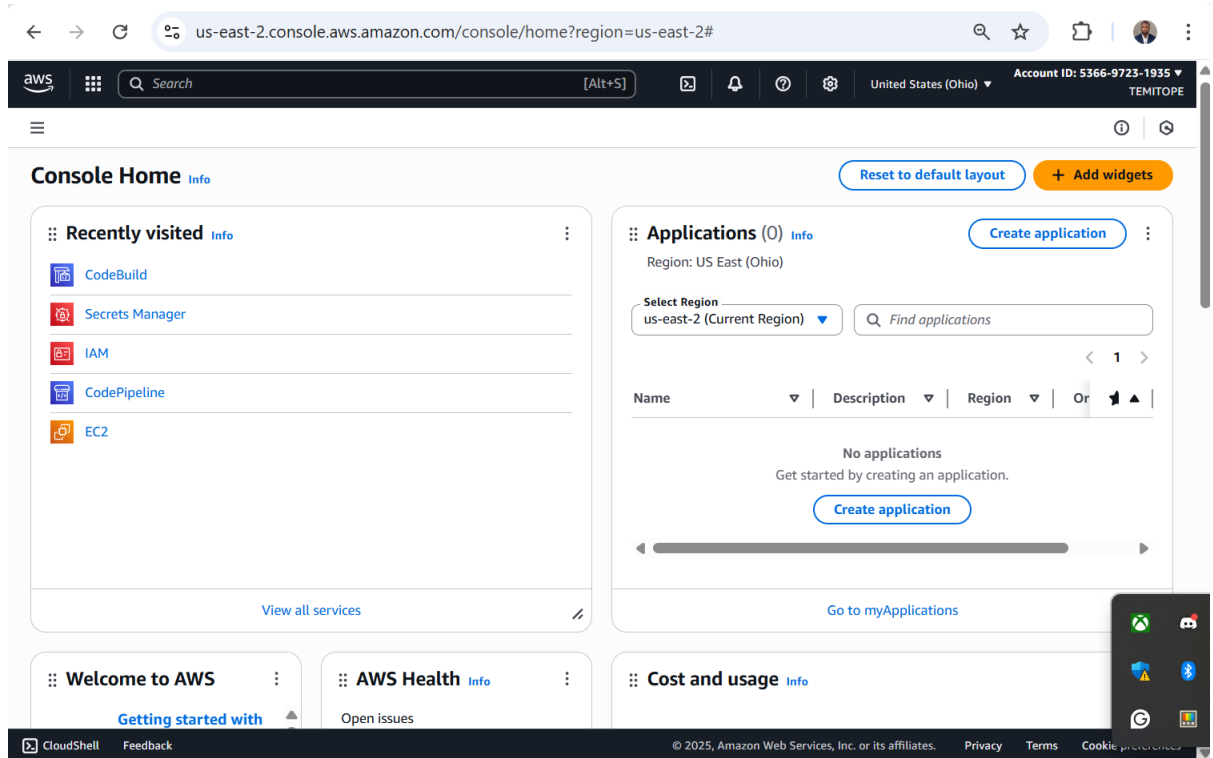
- Supply root user email.



- Supply root user password and click sign in.

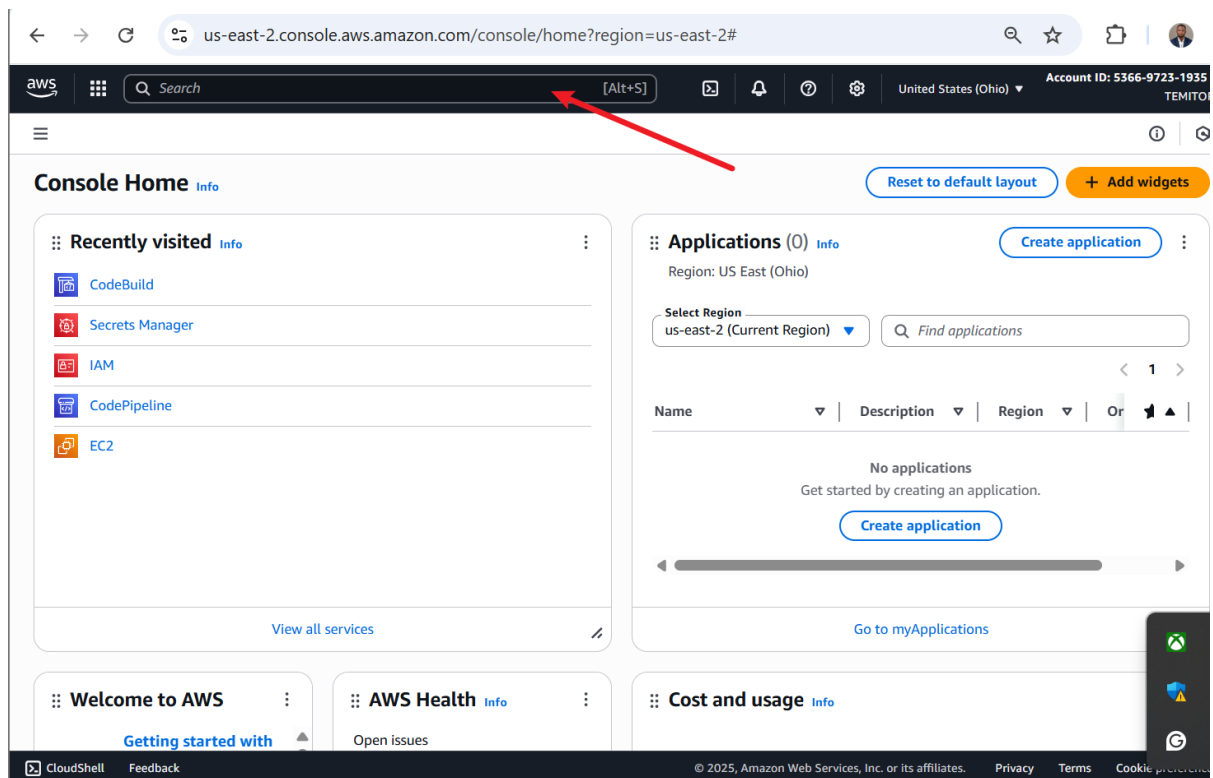
- Supply MFA code for additional verification.

- Landed on the console dashboard.

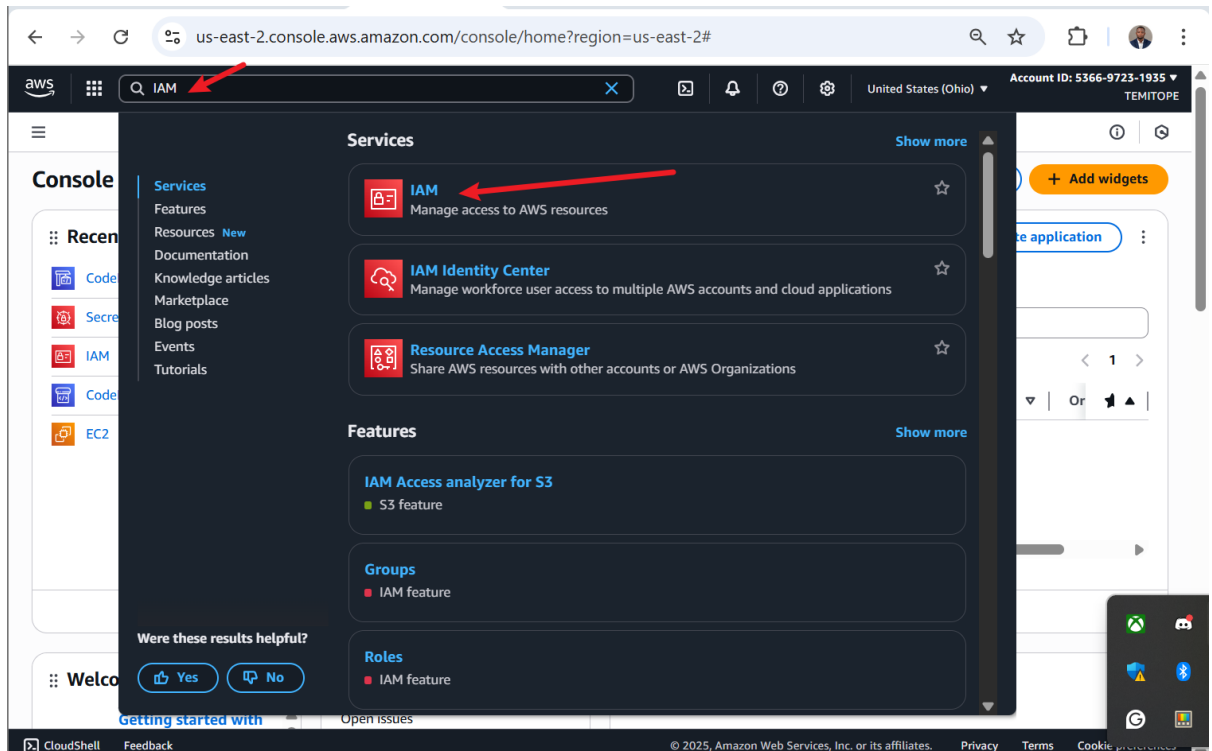


2. Navigated to the **IAM Dashboard**.

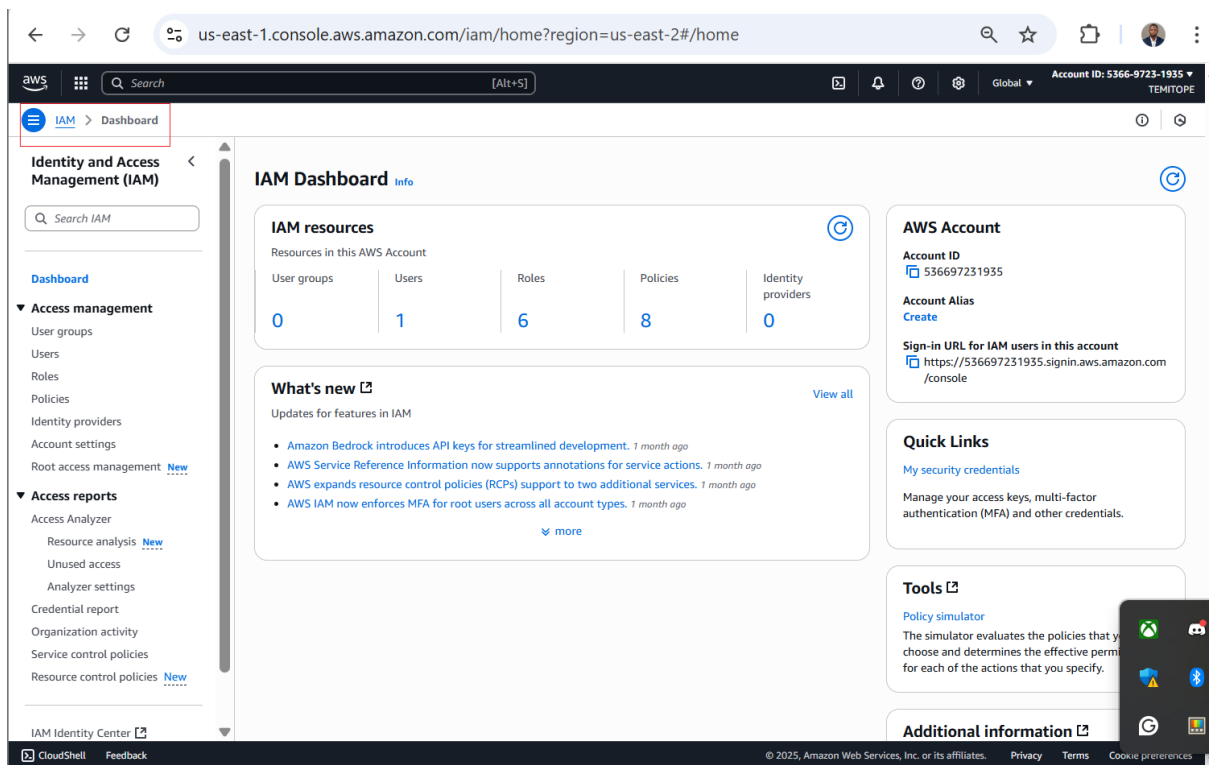
- Use the service search.



- Search for IAM and click on it.



- Now on IAM dashboard.



4. Creating IAM Policies

Developer Policy (EC2 Access)

- Created a **customer-managed policy** for backend developers.
 - Service: **EC2**.
 - Actions: **All EC2 actions**.
 - Resources: **All**.

- Policy Name: **developer-policy**.

Steps:

- Click on policies.

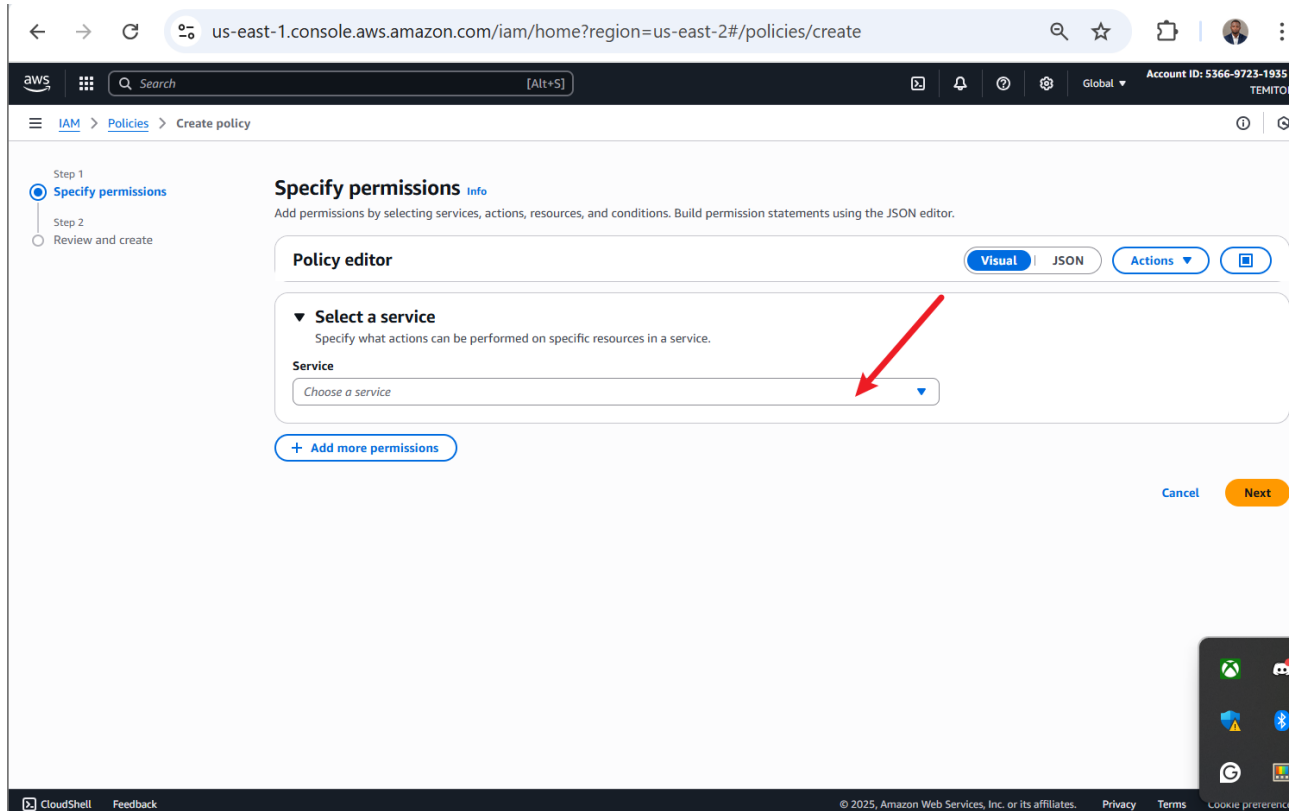
The screenshot shows the AWS IAM Dashboard in the console. The left sidebar under 'Identity and Access Management (IAM)' has a red arrow pointing to the 'Policies' link. The main content area displays the 'IAM Dashboard' with a summary of resources (0 user groups, 1 user, 6 roles, 8 policies, 0 identity providers) and a 'What's new' section. The right sidebar shows 'AWS Account' information and 'Quick Links'.

- Click on create policy.

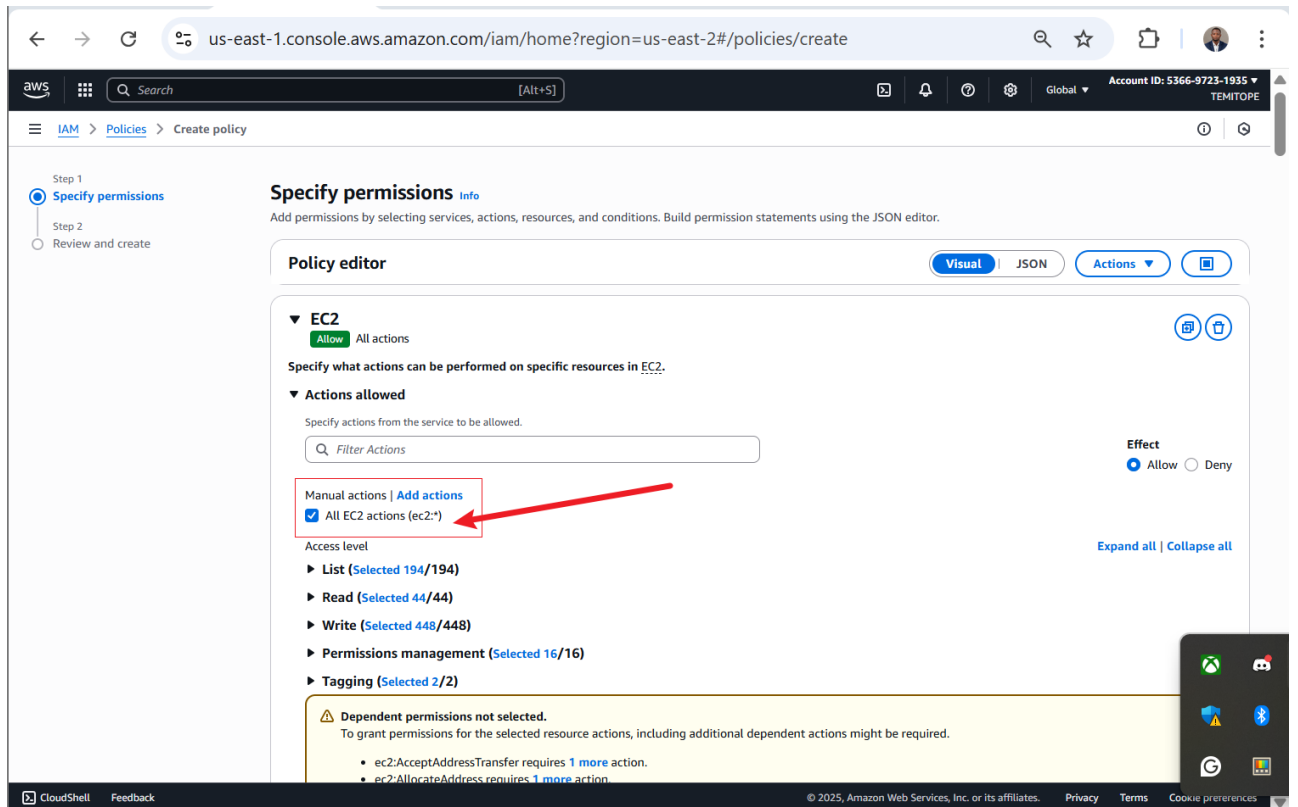
The screenshot shows the 'Policies (1387)' page in the AWS IAM console. A red arrow points to the 'Create policy' button in the top right corner. Below the button is a table listing various AWS managed policies. The table has columns for Policy name, Type, Used as, and Description.

Policy name	Type	Used as	Description
AccessAnalyzerServiceRole	AWS managed	None	Allow Access Analyzer to analyze resou...
AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services an...
AdministratorAccess-Amp...	AWS managed	None	Grants account administrative permis...
AdministratorAccess-AWS...	AWS managed	None	Grants account administrative permis...
AIOpsAssistantPolicy	AWS managed	None	Provides ReadOnly permissions requir...
AIOpsConsoleAdminPolicy	AWS managed	None	Grants full access to Amazon AI Opera...
AIOpsOperatorAccess	AWS managed	None	Grants access to the Amazon AI Opera...
AIOpsReadOnlyAccess	AWS managed	None	Grants ReadOnly permissions to the A...
AlexaForBusinessDeviceSe...	AWS managed	None	Provide device setup access to AlexaFo...
AlexaForBusinessFullAccess	AWS managed	None	Grants full access to AlexaForBusiness ...
AlexaForBusinessGateway...	AWS managed	None	Provide gateway execution
AlexaForBusinessLifeseD...	AWS managed	None	Provide access to Lifesize A
AlexaForBusinessNetwork...	AWS managed	None	This policy enables Alexa fo
AlexaForBusinessPolyDele...	AWS managed	None	Provide access to Poly AVS

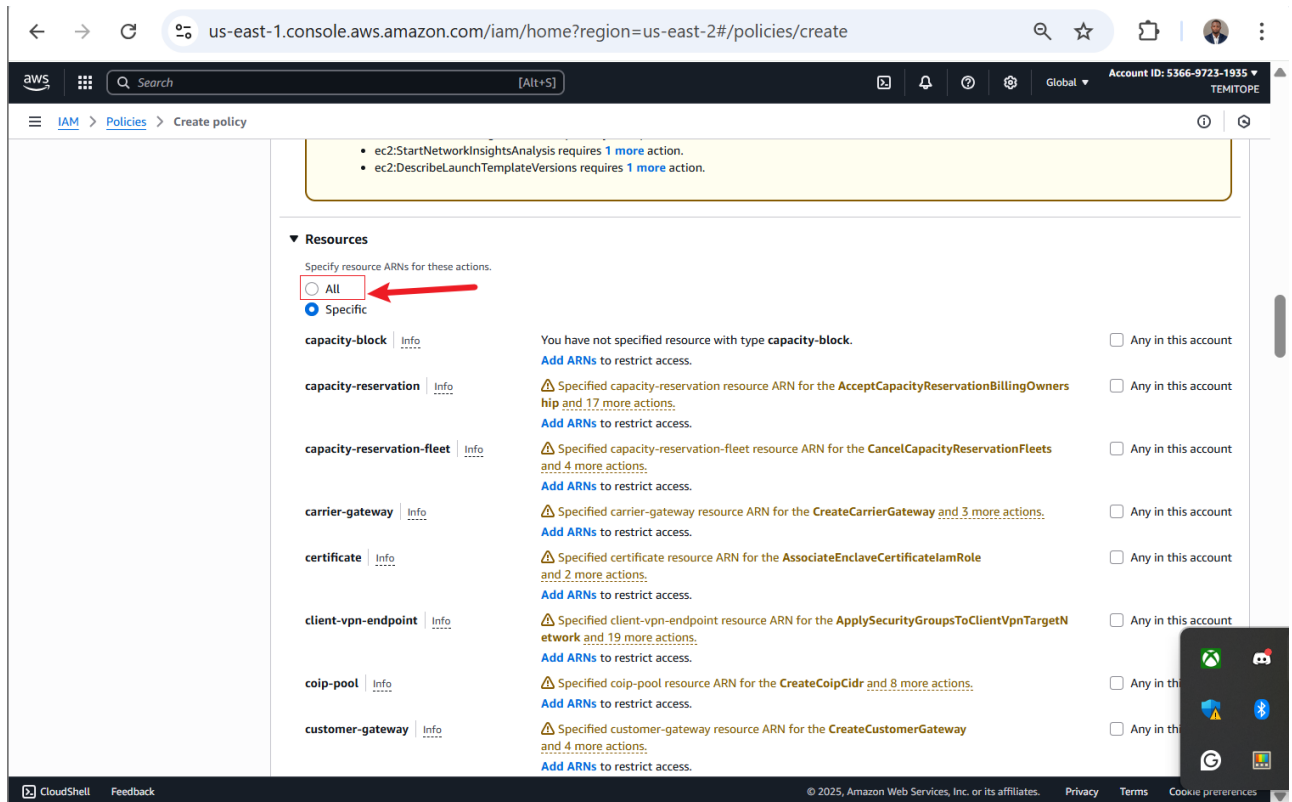
- Under select service section, choose EC2 service.



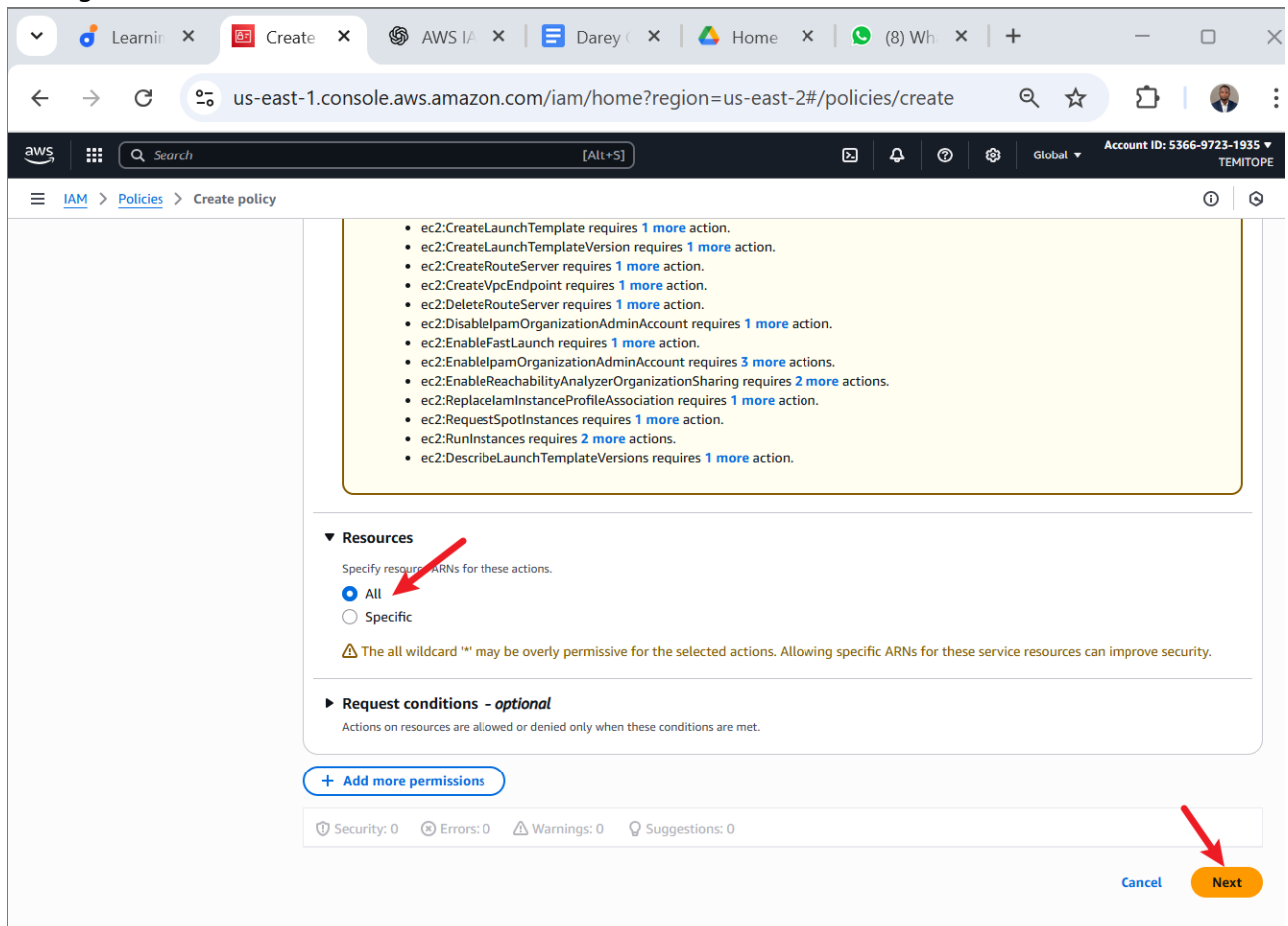
- Under action, check all EC2.



- Under resources, check all.



- Having checked all, click next.



- Name policy and click create policy.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/policies/create

Step 1: Specify permissions
Step 2: Review and create

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.
Maximum 128 characters. Use alphanumeric and '+', '=', '@', '-' characters.

Description - optional
Add a short explanation for this policy.
Maximum 1,000 characters. Use alphanumeric and '+', '=', '@', '-' characters.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Allow (1 of 449 services)

Service	Access level	Resource	Request condition
EC2	Full access	All resources	None

Add tags - optional
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources. No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create policy](#)

- Developer policy successfully created.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/policies

Policy developer-policy created. [View policy](#)

Policies (1388)

A policy is an object in AWS that defines permissions.

Policy name	Type	Used as	Description
AccessAnalyzerServiceRolePolicy	AWS managed	None	Allow Access Analyzer to analyze resou.
AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services an.
AdministratorAccess-Amplify	AWS managed	None	Grants account administrative permissi.
AdministratorAccess-AWSElasticBeanstalk	AWS managed	None	Grants account administrative permissi.
AIOpsAssistantPolicy	AWS managed	None	Provides ReadOnly permissions require.
AIOpsConsoleAdminPolicy	AWS managed	None	Grants full access to Amazon AI Opera..
AIOpsOperatorAccess	AWS managed	None	Grants access to the Amazon AI Opera..
AIOpsReadOnlyAccess	AWS managed	None	Grants ReadOnly permissions to the A..
AlexaForBusinessDeviceSetup	AWS managed	None	Provide device setup access to AlexaFo.
AlexaForBusinessFullAccess	AWS managed	None	Grants full access to AlexaForBusiness .
AlexaForBusinessGatewayExecution	AWS managed	None	Provide gateway execution access to AL.
AlexaForBusinessLifesizeDelegatedAccess...	AWS managed	None	Provide access to Lifesize AVS devices
AlexaForBusinessNetworkProfileServicePo...	AWS managed	None	This policy enables Alexa for Business ..
AlexaForBusinessPolyDelegatedAccessPolicy	AWS managed	None	Provide access to Poly AVS devices
AlexaForBusinessReadOnlyAccess	AWS managed	None	Provide read only access to AlexaForBu.
AmazonAPIGatewayAdministrator	AWS managed	None	Provides full access to create/edit/dele.
AmazonAPIGatewayInvokeFullAccess	AWS managed	None	Provides full access to invoke APIs in A.
AmazonAPIGatewayPushToCloudWatchLogs	AWS managed	None	Allows API Gateway to push logs to us..
AmazonAppFlowFullAccess	AWS managed	None	Provides full access to Amazon AppFlo.
AmazonAppFlowReadOnlyAccess	AWS managed	None	Provides read only access to Amazon A

Analyst Policy (S3 Access)

- Created a **customer-managed policy** for data analysts.
 - Service: **S3**.
 - Actions: **All S3 actions**.
 - Resources: **All**.
 - Policy Name: **analyst-policy**.

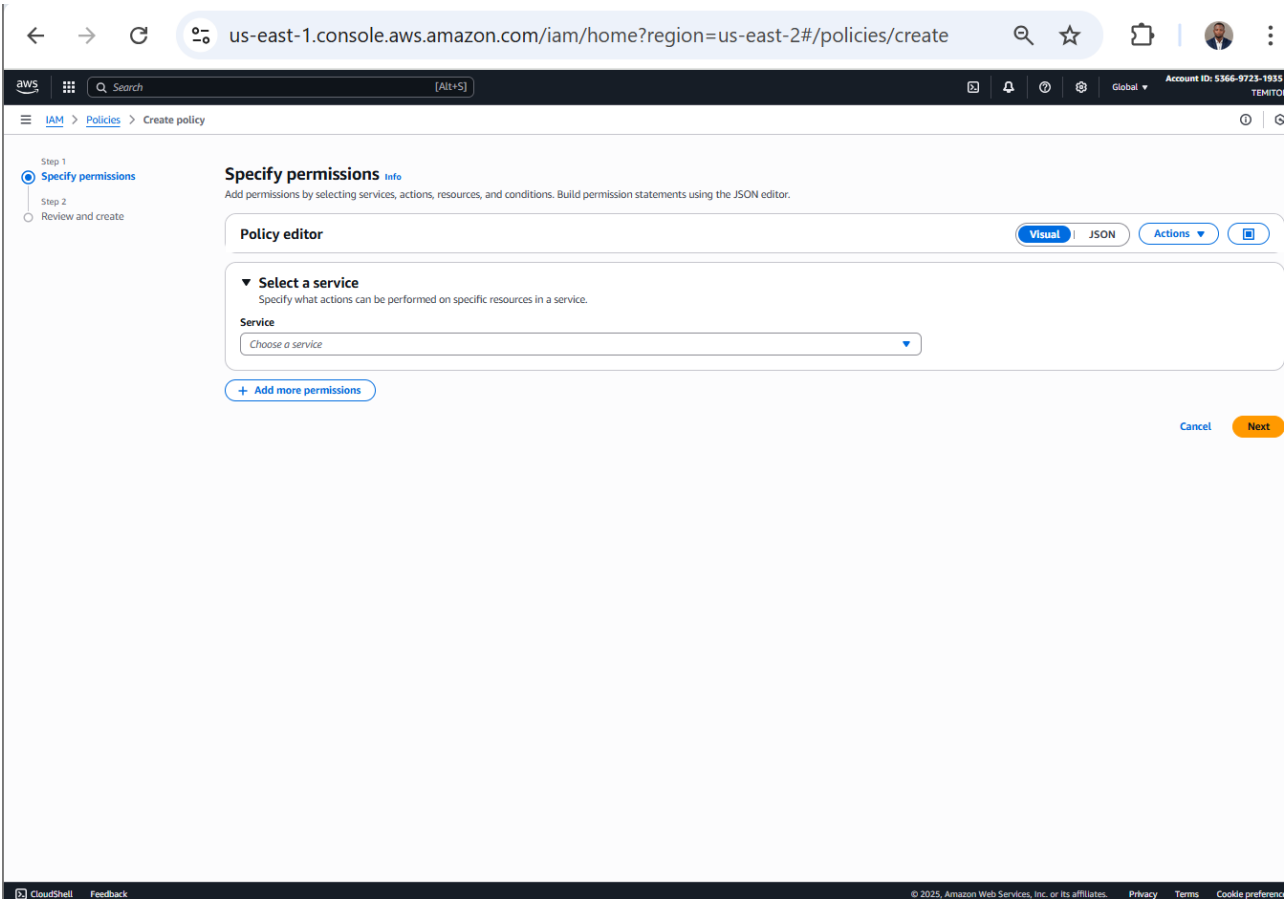
Steps:

- Click create policy to create another one.

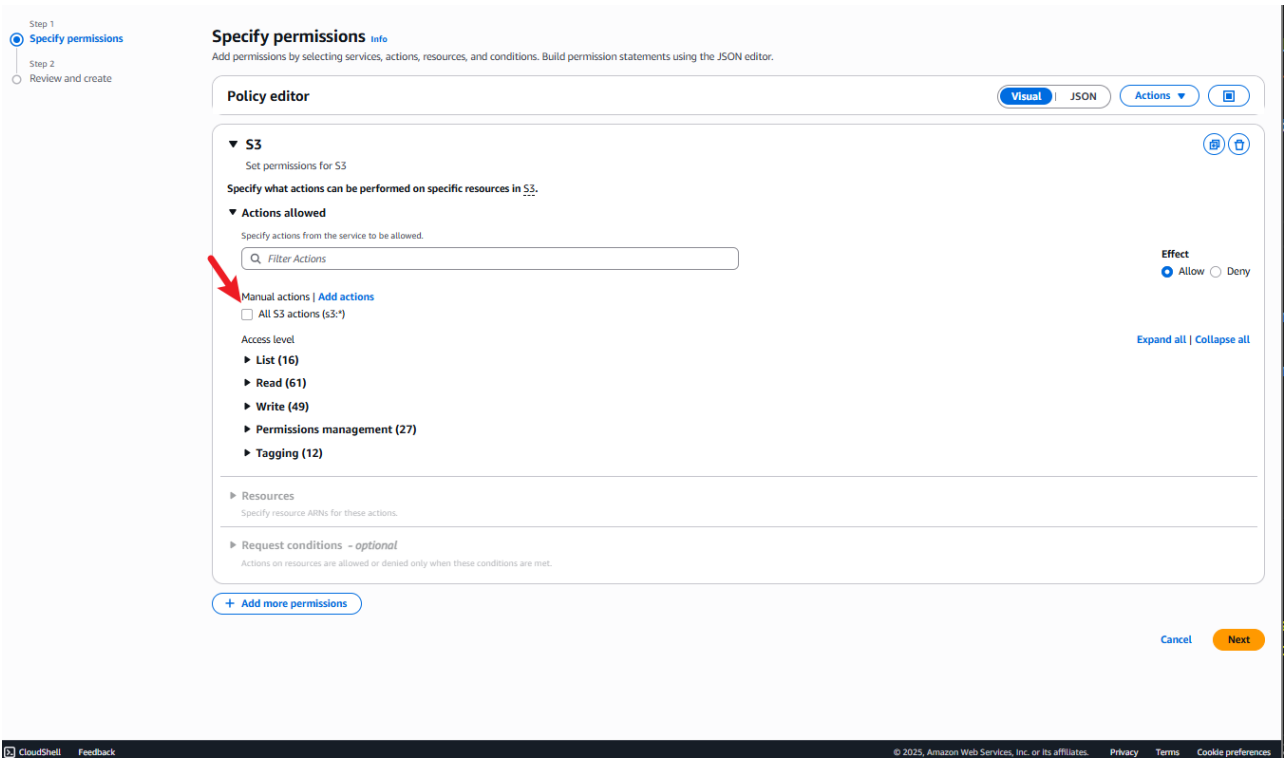
The screenshot shows the AWS IAM console interface. At the top, a green banner indicates a policy was created. Below this, a 'Create policy' button is highlighted with a red arrow. The main content area displays a table of existing policies.

Policy name	Type	Used as	Description
AccessAnalyzerServiceRolePolicy	AWS managed	None	Allow Access Analyzer to analyze resou.
AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services an.
AdministratorAccess-Amplify	AWS managed	None	Grants account administrative permissi.
AdministratorAccess-AWSElasticBeanstalk	AWS managed	None	Grants account administrative permissi.
AIAssistantPolicy	AWS managed	None	Provides ReadOnly permissions require.
AIConsoleAdminPolicy	AWS managed	None	Grants full access to Amazon AI Opera.
AIOperatorAccess	AWS managed	None	Grants access to the Amazon AI Opera.
AIReadOnlyAccess	AWS managed	None	Grants ReadOnly permissions to the A..
AlexaForBusinessDeviceSetup	AWS managed	None	Provide device setup access to AlexaFo.
AlexaForBusinessFullAccess	AWS managed	None	Grants full access to AlexaForBusiness .
AlexaForBusinessGatewayExecution	AWS managed	None	Provide gateway execution access to AL
AlexaForBusinessLifsizeDelegatedAccess...	AWS managed	None	Provide access to Lifesize AVS devices
AlexaForBusinessNetworkProfileServicePo...	AWS managed	None	This policy enables Alexa for Business ..
AlexaForBusinessPolyDelegatedAccessPolicy	AWS managed	None	Provide access to Poly AVS devices
AlexaForBusinessReadOnlyAccess	AWS managed	None	Provide read only access to AlexaForBu.
AmazonAPIGatewayAdministrator	AWS managed	None	Provides full access to create/edit/dele.
AmazonAPIGatewayInvokeFullAccess	AWS managed	None	Provides full access to invoke APIs in A.
AmazonAPIGatewayPushToCloudWatchLogs	AWS managed	None	Allows API Gateway to push logs to us..
AmazonAppFlowFullAccess	AWS managed	None	Provides full access to Amazon AppFlo.
AWSDataSyncFullAccess	AWS managed	None	Provides read-only access to Amazon A

- Under select a service, choose S3.



- Under action, check all S3.



- Under resources, check all.

The screenshot shows the 'Create policy' page in the AWS IAM console. The 'Resources' section is active, and the 'Specific' radio button is selected. A red arrow points to the 'Specific' radio button. The 'Resources' list includes: accessgrant, accessgrantsinstance, accessgrantslocation, accesspoint, bucket, job, multiregionaccesspoint, multiregionaccesspointrequestarn, object, objectlambdaccesspoint, storagelensconfiguration, and storagelensgroup. Each resource has a description of the actions it allows and a checkbox to select it. The 'Request conditions - optional' section is also visible.

- Having checked all, click next.

The screenshot shows the 'Create policy' page in the AWS IAM console, now in the 'Actions allowed' section. The 'All S3 actions (s3*)' checkbox is checked. The 'Access level' section shows a list of actions: List (Selected 16/16), Read (Selected 61/61), Write (Selected 49/49), Permissions management (Selected 27/27), and Tagging (Selected 12/12). A warning message states: 'Dependent permissions not selected. To grant permissions for the selected resource actions, including additional dependent actions might be required.' The 'Resources' section is visible at the bottom, with the 'All' radio button selected. A red arrow points to the 'Next' button at the bottom right.

- Name policy and click create policy.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/policies/create

Step 1: Specify permissions
Step 2: Review and create

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.
analyst-policy
Maximum 128 characters. Use alphanumeric and '+', '-', '@', '_', '.' characters.

Description - optional
Add a short explanation for this policy.
Maximum 1,000 characters. Use alphanumeric and '+', '-', '@', '_', '.' characters.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Allow (1 of 449 services)

Service	Access level	Resource	Request condition
S3	Full access	All resources	None

Show remaining 448 services

Add tags - optional

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Previous **Create policy**

- Analyst policy successfully created.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/policies

Policy analyst-policy created. View policy

Policies (1389)

A policy is an object in AWS that defines permissions.

Filter by Type: All types

Policy name	Type	Used as	Description
AccessAnalyzerServiceRolePolicy	AWS managed	None	Allow Access Analyzer to analyze resou.
AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services an.
AdministratorAccess-Amplify	AWS managed	None	Grants account administrative permissi.
AdministratorAccess-AWSElasticBeanstalk	AWS managed	None	Grants account administrative permissi.
AIOPsAssistantPolicy	AWS managed	None	Provides ReadOnly permissions require.
AIOPsConsoleAdminPolicy	AWS managed	None	Grants full access to the Amazon AI Opera..
AIOPsOperatorAccess	AWS managed	None	Grants access to the Amazon AI Opera..
AIOPsReadOnlyAccess	AWS managed	None	Grants ReadOnly permissions to the A..
AlexaForBusinessDeviceSetup	AWS managed	None	Provide device setup access to AlexaFo.
AlexaForBusinessFullAccess	AWS managed	None	Grants full access to AlexaForBusiness .
AlexaForBusinessGatewayExecution	AWS managed	None	Provide gateway execution access to AI.
AlexaForBusinessLifesizeDelegatedAccess...	AWS managed	None	Provide access to Lifesize AWS devices
AlexaForBusinessNetworkProfileServicePo...	AWS managed	None	This policy enables Alexa for Business ..
AlexaForBusinessPolyDelegatedAccessPolicy	AWS managed	None	Provide access to Poly AVS devices
AlexaForBusinessReadOnlyAccess	AWS managed	None	Provide read only access to AlexaForBu.
AmazonAPIGatewayAdministrator	AWS managed	None	Provides full access to create/edit/dele.
AmazonAPIGatewayInvokeFullAccess	AWS managed	None	Provides full access to invoke APIs in A.
AmazonAPIGatewayPushToCloudWatchLogs	AWS managed	None	Allows API Gateway to push logs to us..
AmazonAppFlowFullAccess	AWS managed	None	Provides full access to Amazon AppFlo.
AmazonAppFlowOnlyAccess	AWS managed	None	Provides read only access to Amazon A

5. Creating IAM Groups

- **Development-Team** → Attached **developer-policy**.
- **Analyst-Team** → Attached **analyst-policy**.

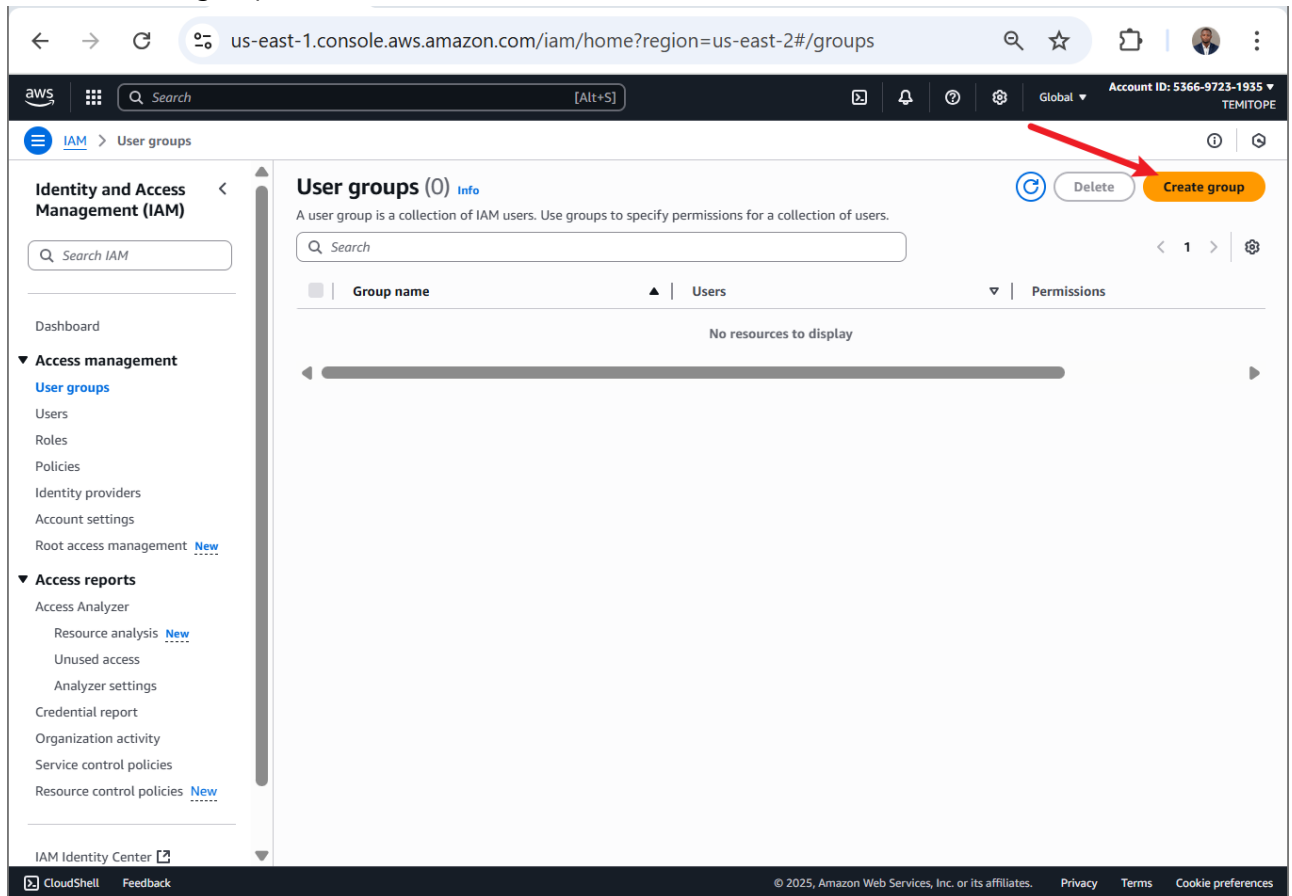
Steps for Development-Team:

- Click on user group.

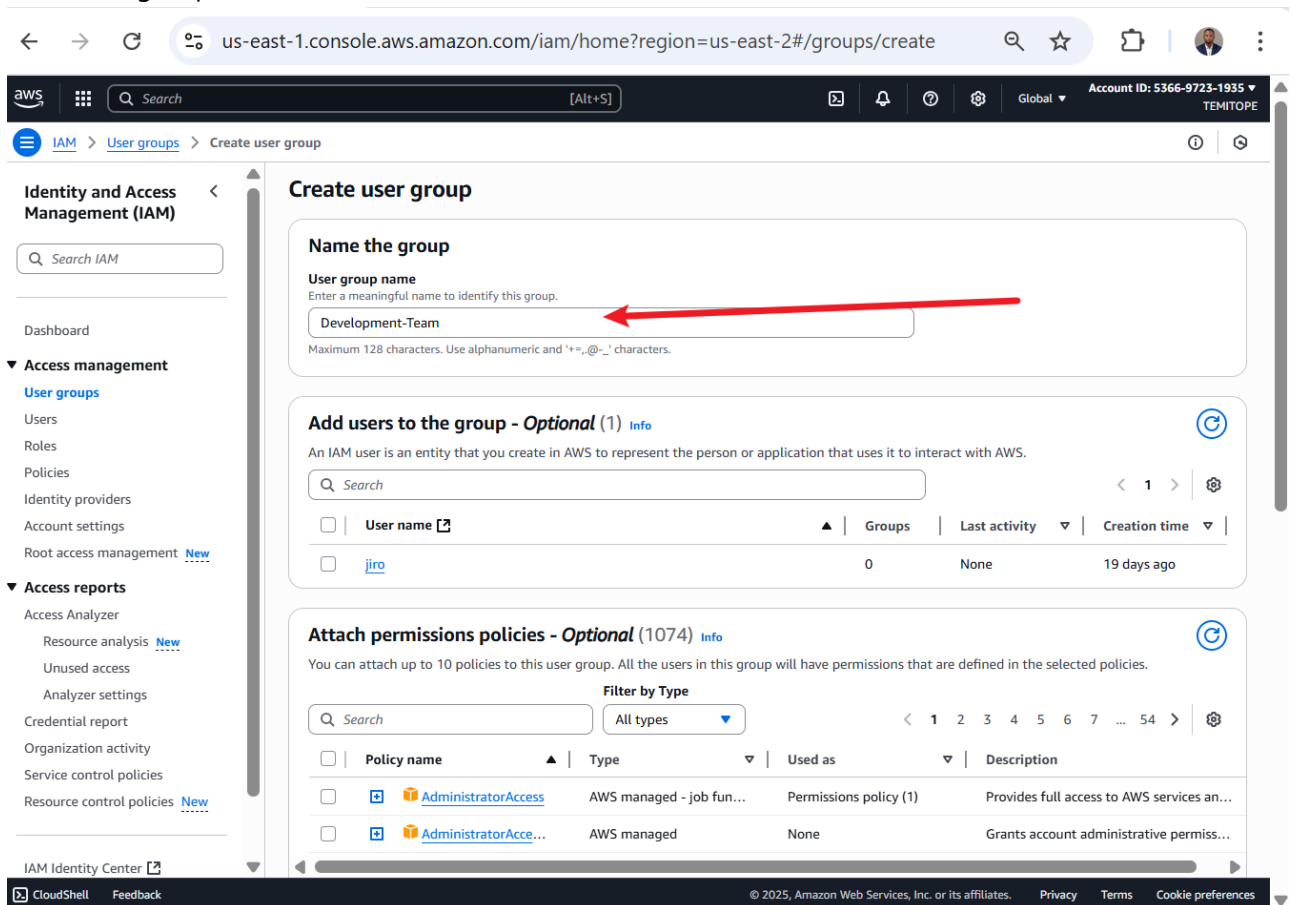
The screenshot shows the AWS IAM console interface. The left sidebar is titled 'Identity and Access Management (IAM)' and contains a search bar and a list of navigation items. A red arrow points to 'User groups' under the 'Access management' section. The main content area displays a green notification banner at the top stating 'Policy analyst-policy created.' Below this, the 'Policies (1389)' section is shown, including a search bar, a 'Filter by Type' dropdown set to 'All types', and a table of policies. The table has columns for Policy name, Type, Used as, and Description. The first few policies listed are 'AccessAnalyzerServiceRolePolicy', 'AdministratorAccess', 'AdministratorAccess-Amplify', and 'AdministratorAccess-AWSElasticBeanstalk'.

Policy name	Type	Used as	Description
AccessAnalyzerServiceRolePolicy	AWS managed	None	Allow Access Analyzer to analyze resou.
AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services an.
AdministratorAccess-Amplify	AWS managed	None	Grants account administrative permisi.
AdministratorAccess-AWSElasticBeanstalk	AWS managed	None	Grants account administrative permisi.
AIOpsAssistantPolicy	AWS managed	None	Provides ReadOnly permissions require.
AIOpsConsoleAdminPolicy	AWS managed	None	Grants full access to Amazon AI Opera..
AIOpsOperatorAccess	AWS managed	None	Grants access to the Amazon AI Opera..
AIOpsReadOnlyAccess	AWS managed	None	Grants ReadOnly permissions to the A..
AlexaForBusinessDeviceSetup	AWS managed	None	Provide device setup access to AlexaFo.
AlexaForBusinessFullAccess	AWS managed	None	Grants full access to AlexaForBusiness .
AlexaForBusinessGatewayExecution	AWS managed	None	Provide gateway execution access to AL.
AlexaForBusinessLifesizeDelegatedAccess...	AWS managed	None	Provide access to Lifesize AVS devices
AlexaForBusinessNetworkProfileServicePo...	AWS managed	None	This policy enables Alexa for Business ..
AlexaForBusinessPolyDelegatedAccessPolicy	AWS managed	None	Provide access to Poly AVS devices
AlexaForBusinessReadOnlyAccess	AWS managed	None	Provide read only access to AlexaForBu.
AmazonAPIGatewayAdministrator	AWS managed	None	Provides full access to create/edit/dele.
AmazonAPIGatewayInvokeFullAccess	AWS managed	None	Provides full access to invoke APIs in A.
AmazonAPIGatewayPushToCloudWatchLogs	AWS managed	None	Allows API Gateway to push logs to us..
AmazonAppFlowFullAccess	AWS managed	None	Provides full access to Amazon AppFlo.
AmazonAppFlowReadOnlyAccess	AWS managed	None	Provides read-only access to Amazon A

- Click on create group.



- Name the group based on their role.



- Attach developer policy and click create group.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/groups/create

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

▼ Access reports

- Access Analyzer
- Resource analysis [New](#)
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies [New](#)

IAM Identity Center [?](#)

Maximum 128 characters. Use alphanumeric and "+", "@", "-" characters.

Add users to the group - Optional (1) [Info](#)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name	Groups	Last activity	Creation time
☐	jiro	0	None	19 days ago

Attach permissions policies - Optional (1/1074) [Info](#)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Search develop X Filter by Type All types 5 matches

☐	Policy name	Type	Used as	Description
☐	AmazonCognitoDe...	AWS managed	None	Provides access to Amazon Cognito AP...
☐	AmazonQDevelope...	AWS managed	None	Provides developer access to enable in...
☐	AWSCodeBuildDev...	AWS managed	None	Provides access to AWS CodeBuild via ...
☐	AWSProtonDevelo...	AWS managed	None	Provides access to the AWS Proton API...
☒	developer-policy	Customer managed	None	-

[Cancel](#) [Create user group](#)

- Development user group successfully created.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/groups

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

▼ Access reports

- Access Analyzer
- Resource analysis [New](#)
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies [New](#)

IAM Identity Center [?](#)

Development-Team user group created. [View group](#) X

User groups (1) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

☐	Group name	Users	Permissions	Creation time
☐	Development-Team	0	Defined	Now

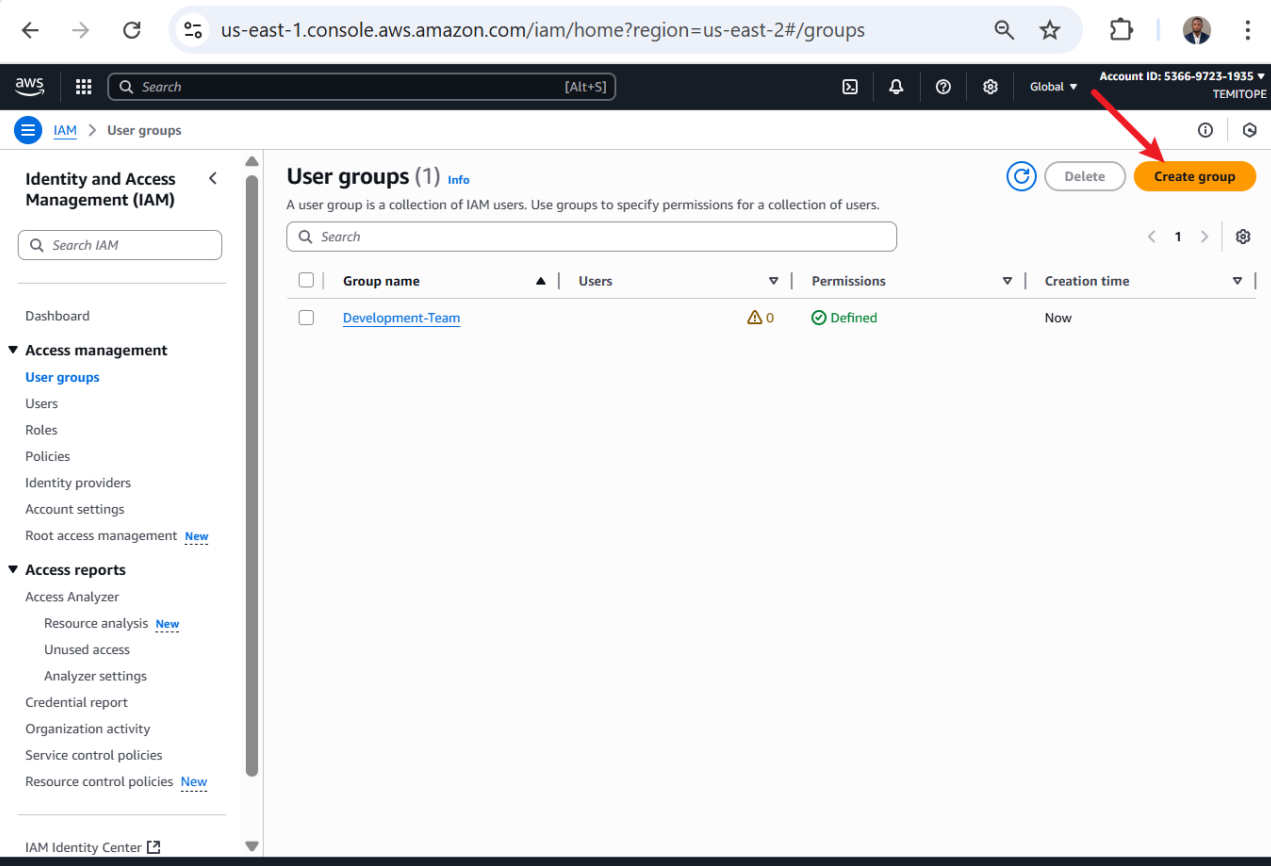
[Delete](#) [Create group](#)

CloudShell Feedback

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Steps for Analyst-Team:

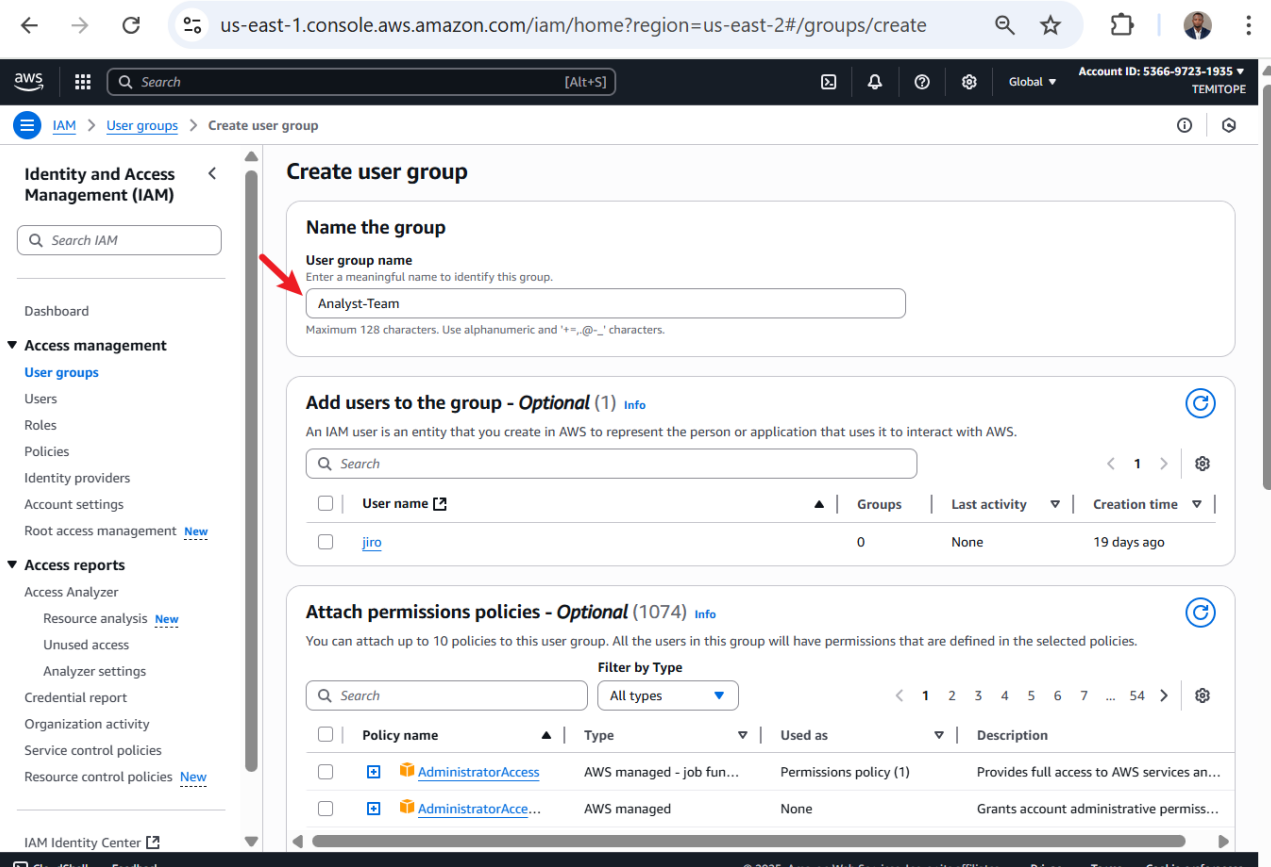
- Click on create group to create one more.



The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, and Access reports. The main content area is titled 'User groups (1)' and shows a table with one user group, 'Development-Team'. A red arrow points to the 'Create group' button in the top right corner of the main content area.

Group name	Users	Permissions	Creation time
Development-Team	0	Defined	Now

- Name the group based on their role.



The screenshot shows the 'Create user group' page in the AWS IAM console. The 'Name the group' section is highlighted with a red arrow. The 'User group name' field contains 'Analyst-Team'. Below this, there is a section for 'Add users to the group - Optional' and a section for 'Attach permissions policies - Optional'.

Name the group

User group name
Enter a meaningful name to identify this group.

Maximum 128 characters. Use alphanumeric and '+-@_.' characters.

Add users to the group - Optional (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

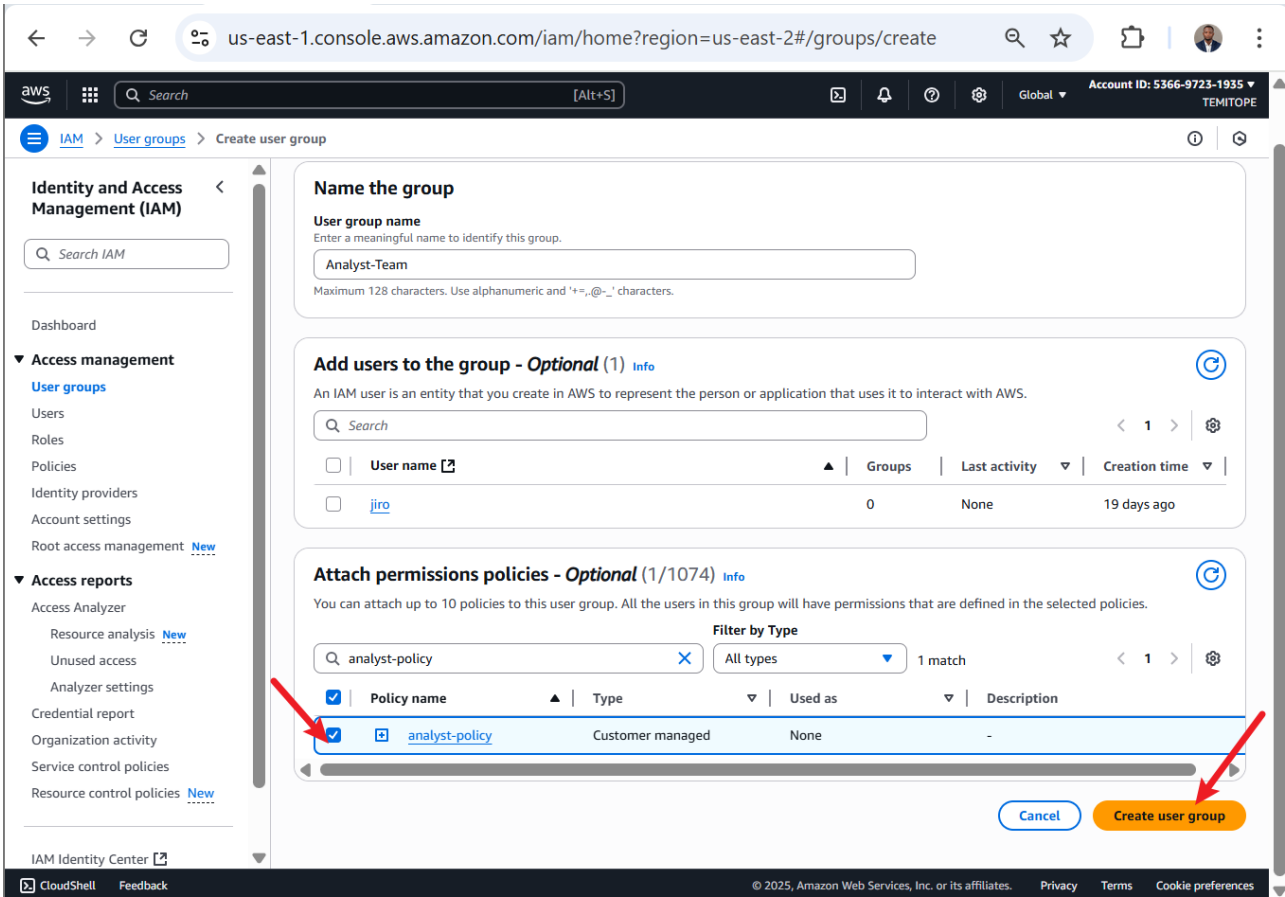
User name	Groups	Last activity	Creation time
jiro	0	None	19 days ago

Attach permissions policies - Optional (1074)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

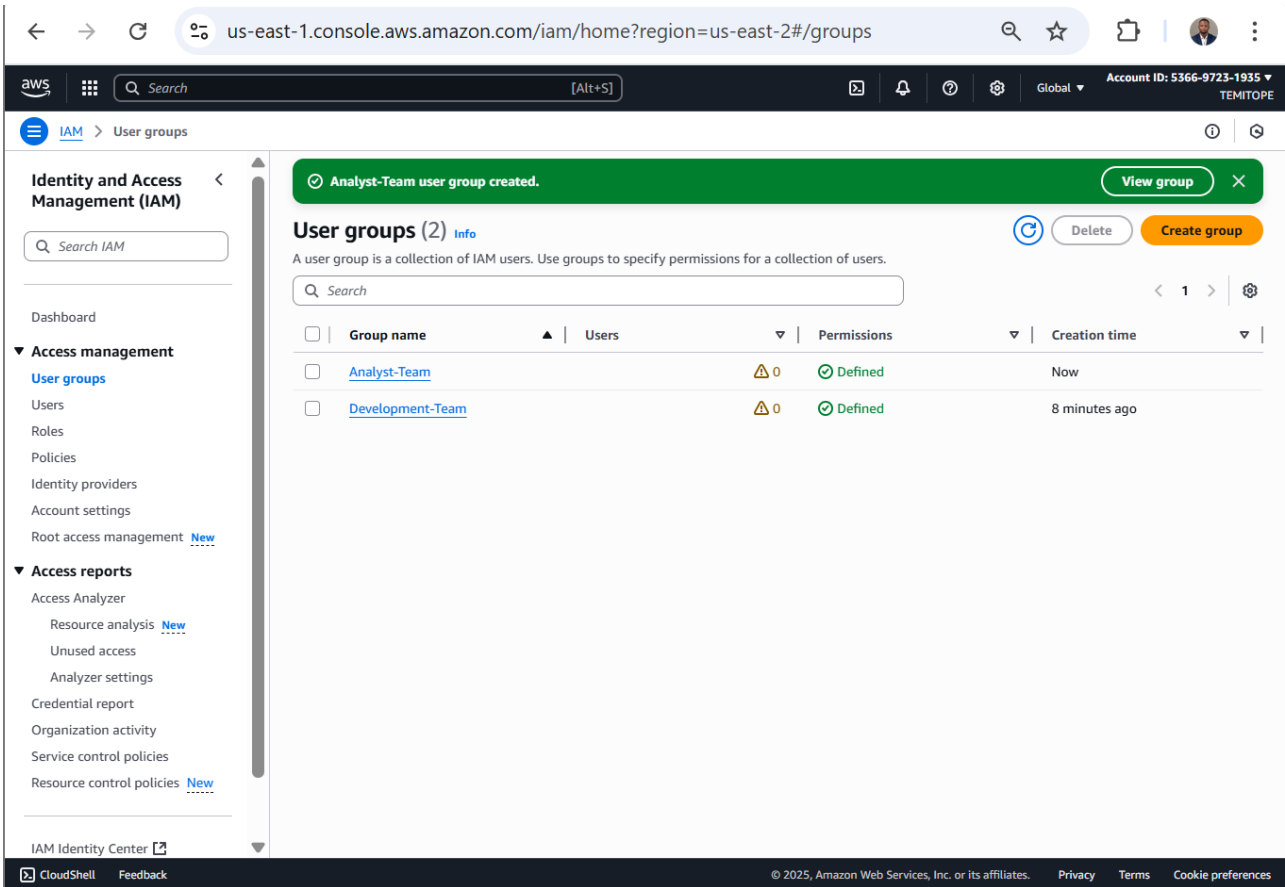
Policy name	Type	Used as	Description
AdministratorAccess	AWS managed - job fun...	Permissions policy (1)	Provides full access to AWS services an...
AdministratorAcce...	AWS managed	None	Grants account administrative permis...

- Attach analyst policy and click create group.



(Note: Filename references developer policy, but context indicates analyst policy attachment.)

- Analyst user group successfully created.



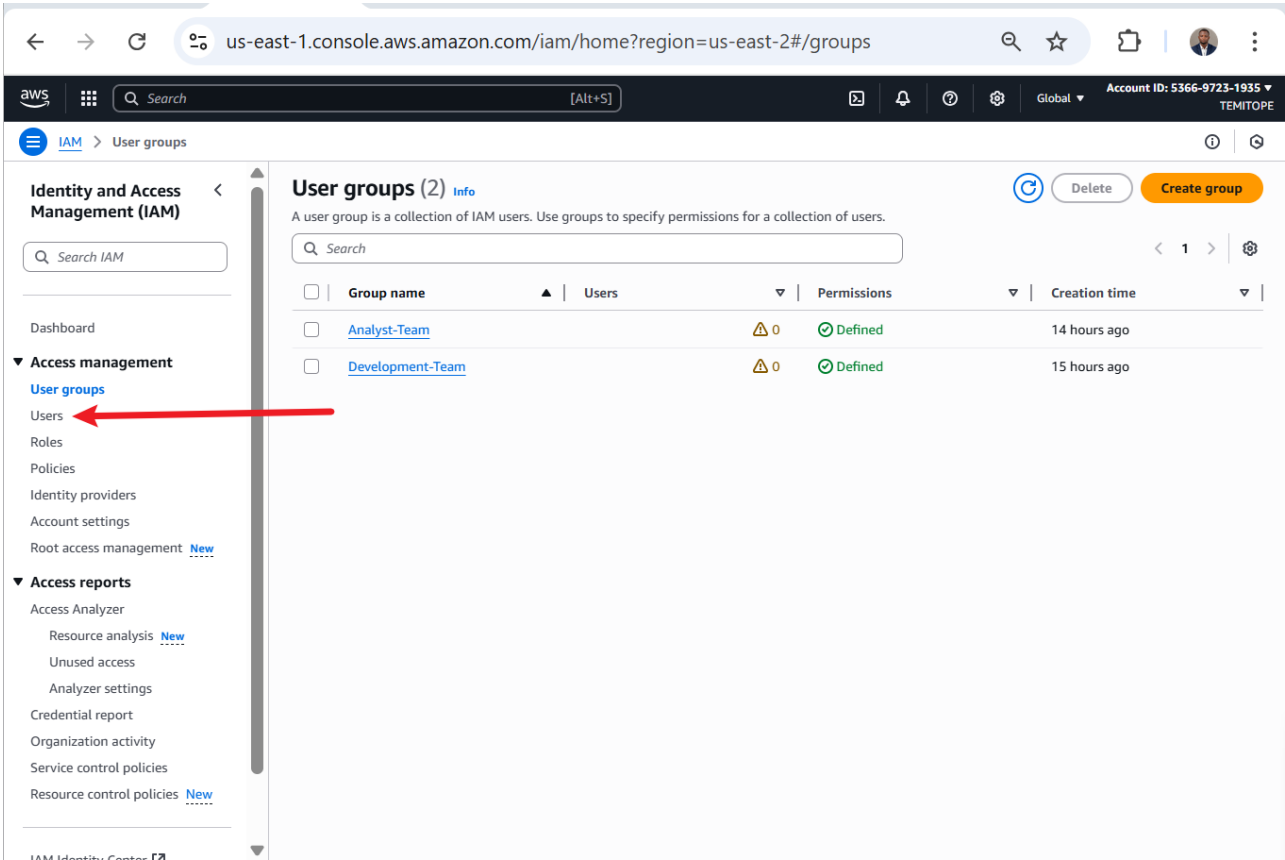
6. Creating IAM Users

John (Backend Developer)

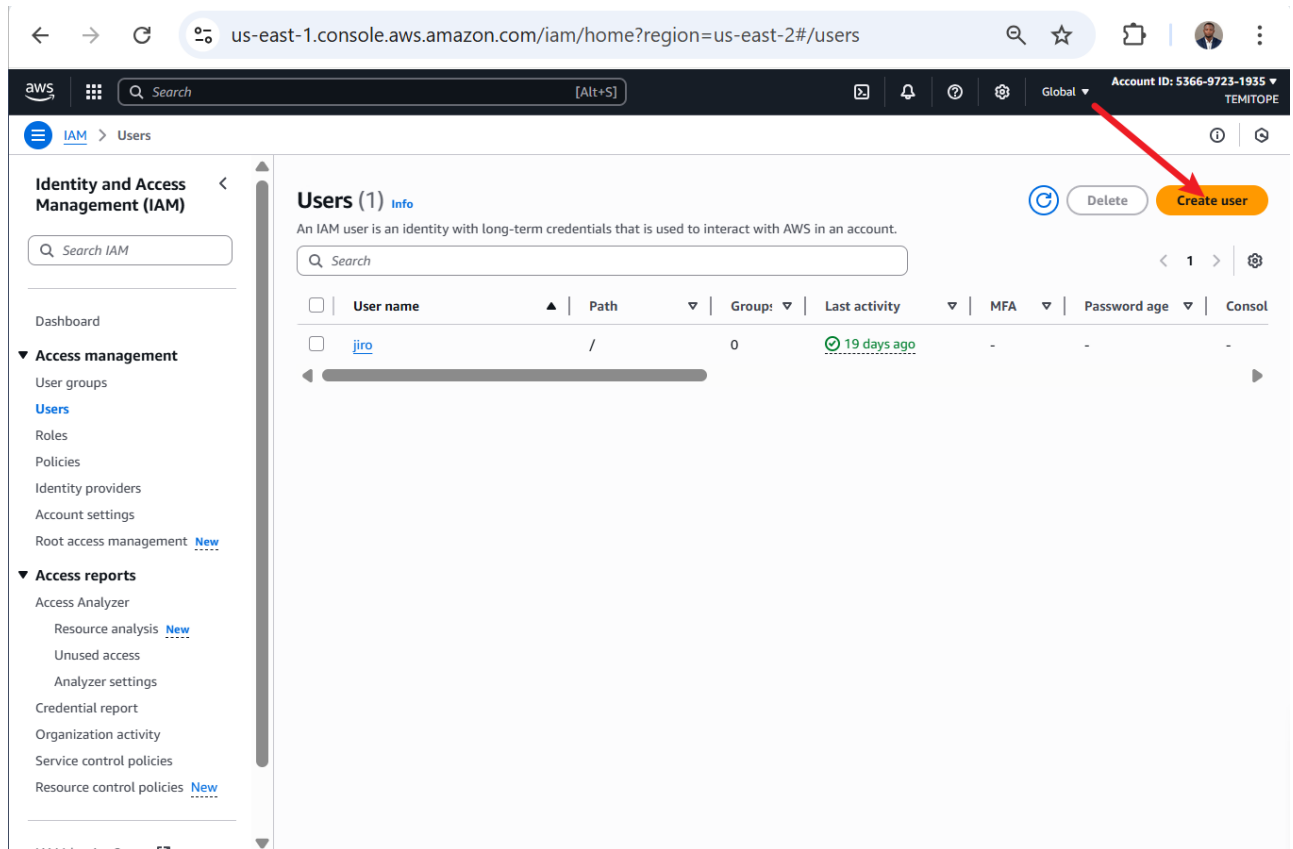
- User created: **John**.
- Added to **Development-Team** group.
- Console login enabled.
- Credentials downloaded.

Steps:

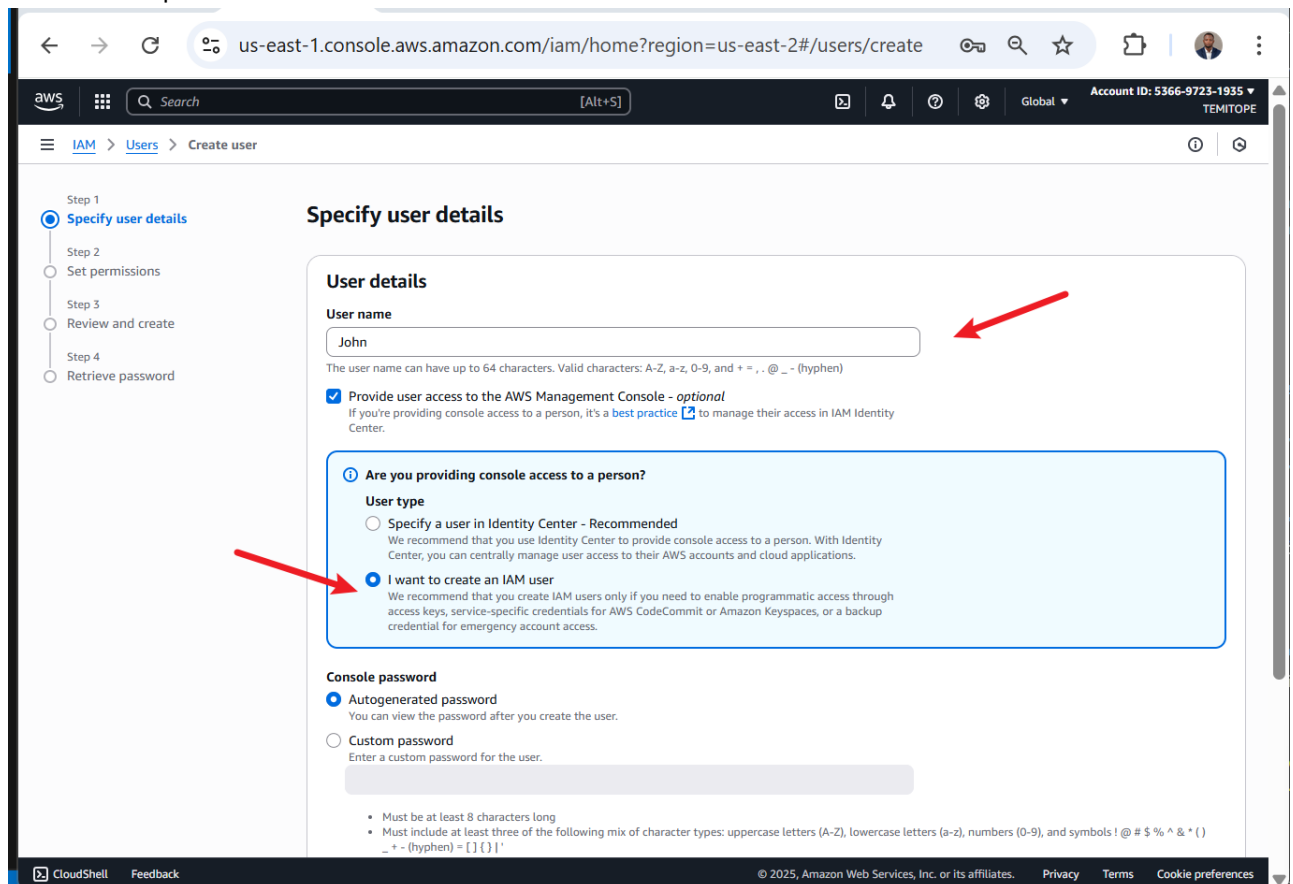
- Click users to create users.



- Click create user.



- Enter name, provide access to console, and check IAM user.



- Choose password, uncheck password change on next login.

I want to create an IAM user
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ Custom password
Enter a custom password for the user.

user@12345

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' "

☒ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

- Save password for next login.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/users/create

aws IAM > Users > Create user

Step 1: Specify user details
Step 2: **Set permissions**
Step 3: Review and create
Step 4: Retrieve password

Set permissions
Add user to an existing group or create a new one. Using

Permissions options

☒ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

User groups (2)

Group name	Users
☐ Analyst-Team	0
☐ Development-Team	0

Set permissions boundary - optional

Save password?

Username: John

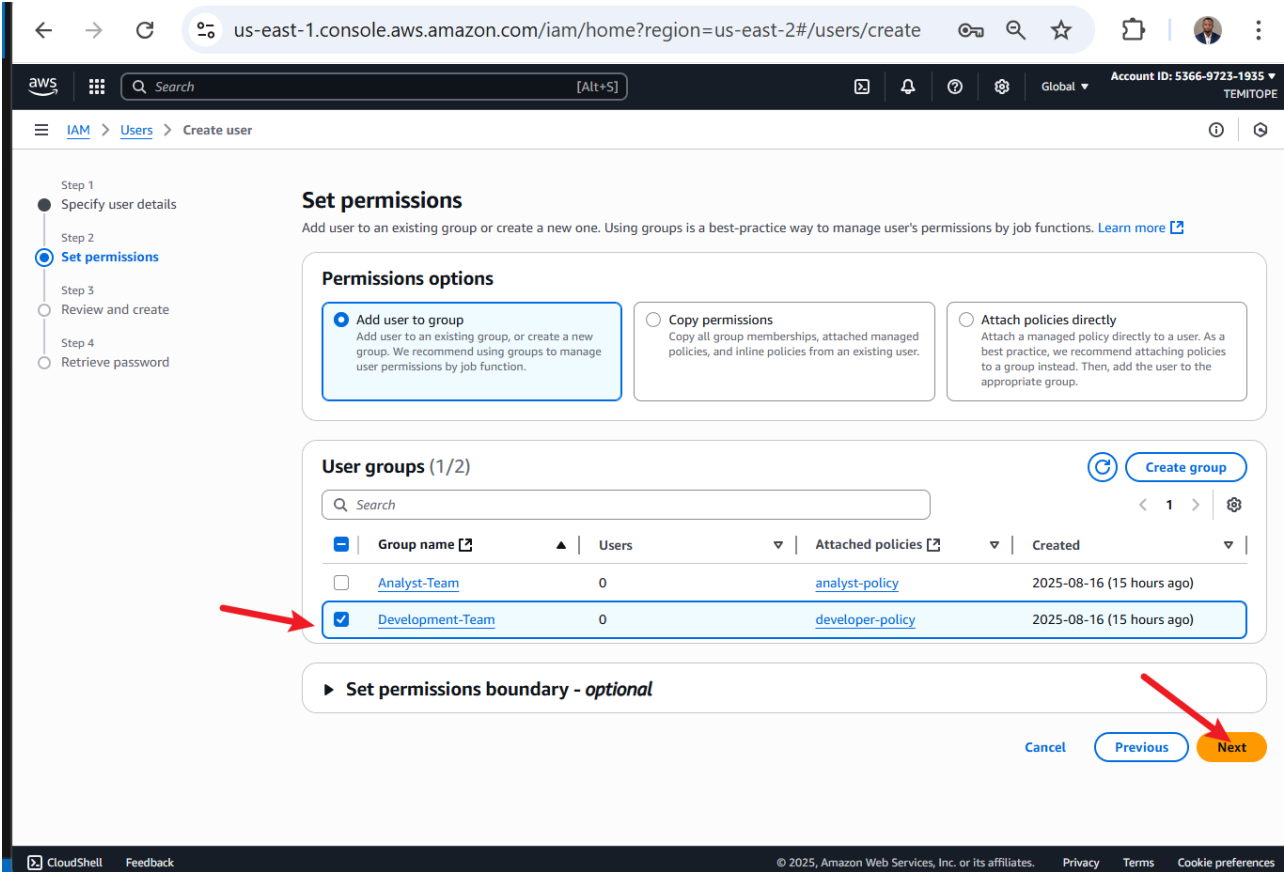
Password:

Save Never

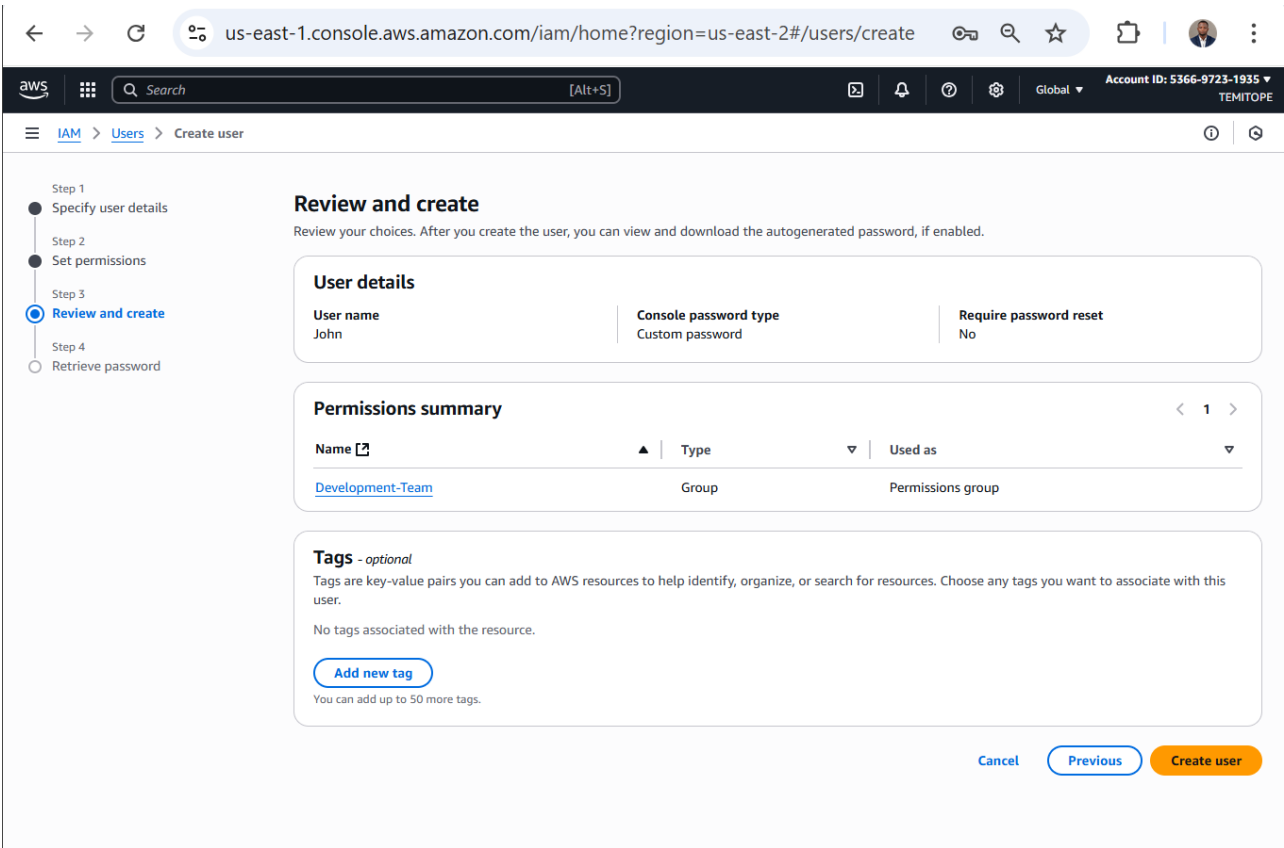
You can use saved passwords on any device. They're saved to [Google Password Manager](#) for oluwaseun.beicks@gmail.com.

Cancel Previous Next

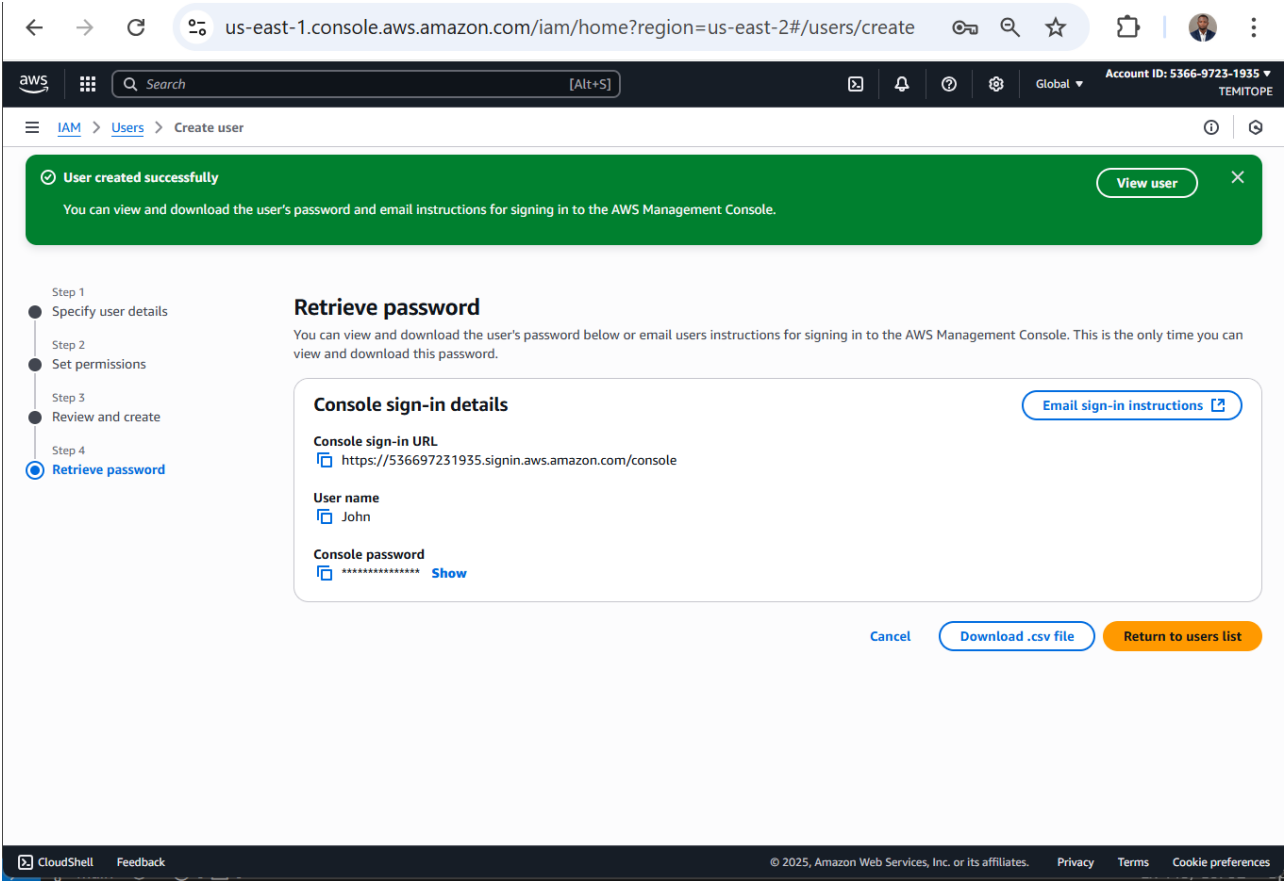
- Check to add to development group and click next.



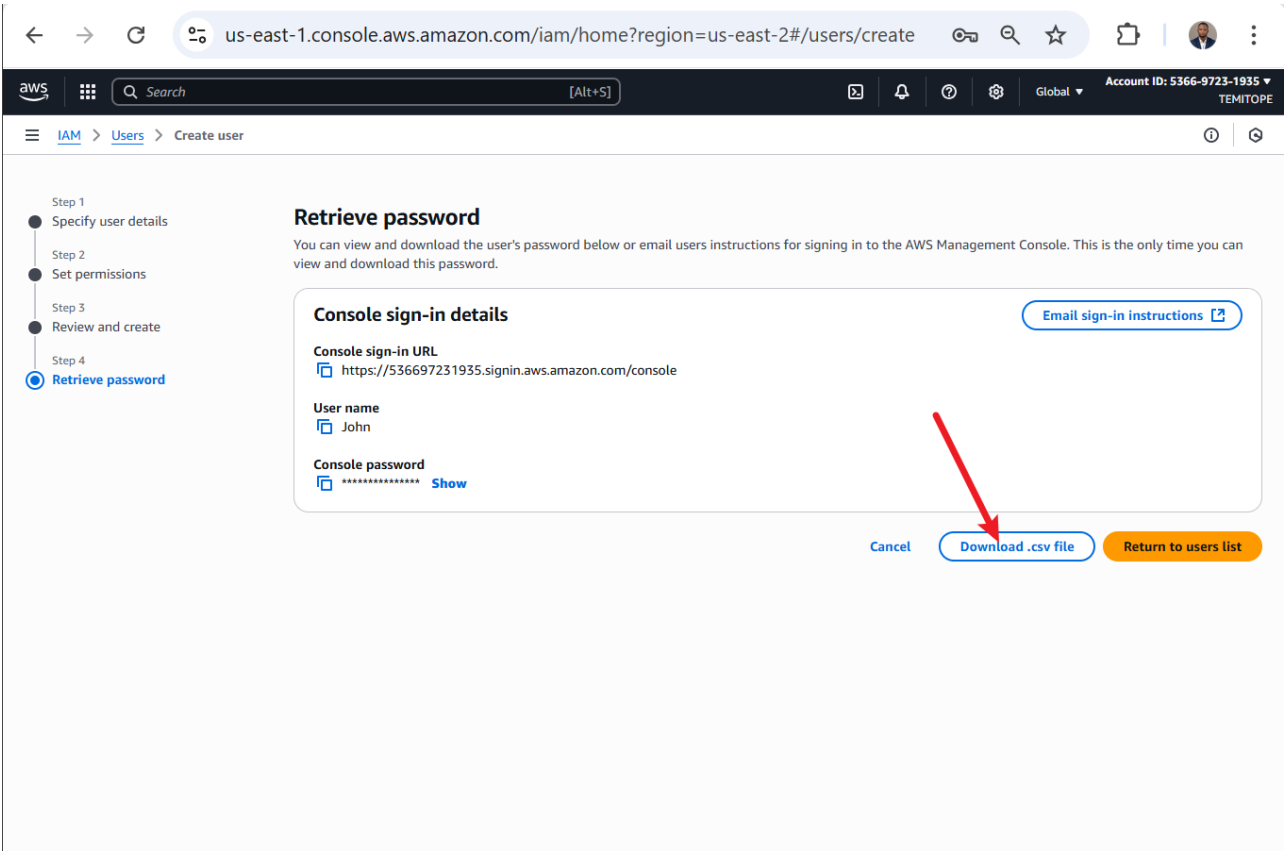
- Review and create user.



- Successfully create user.



- Download user login CSV.



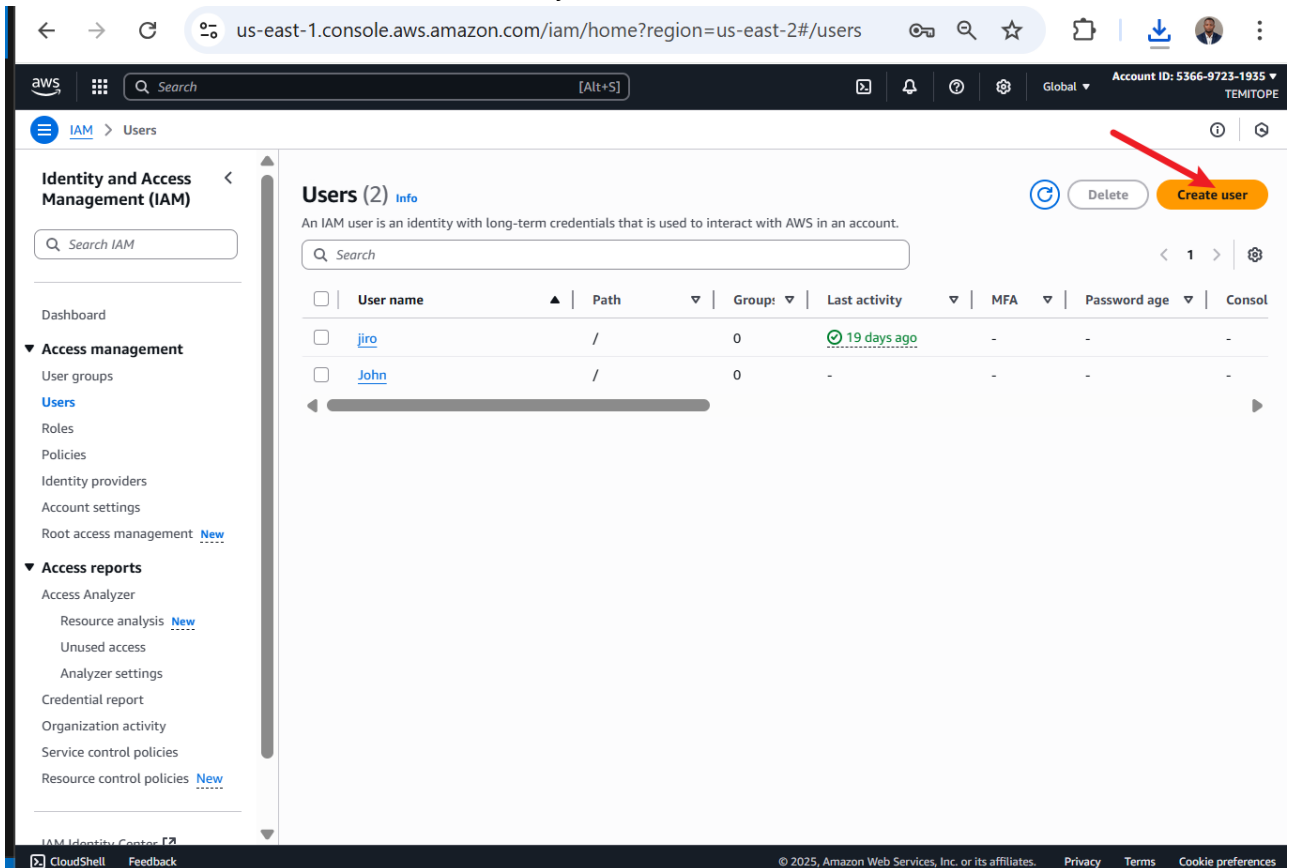
Mary (Data Analyst)

- User created: **Mary**.
- Added to **Analyst-Team** group.

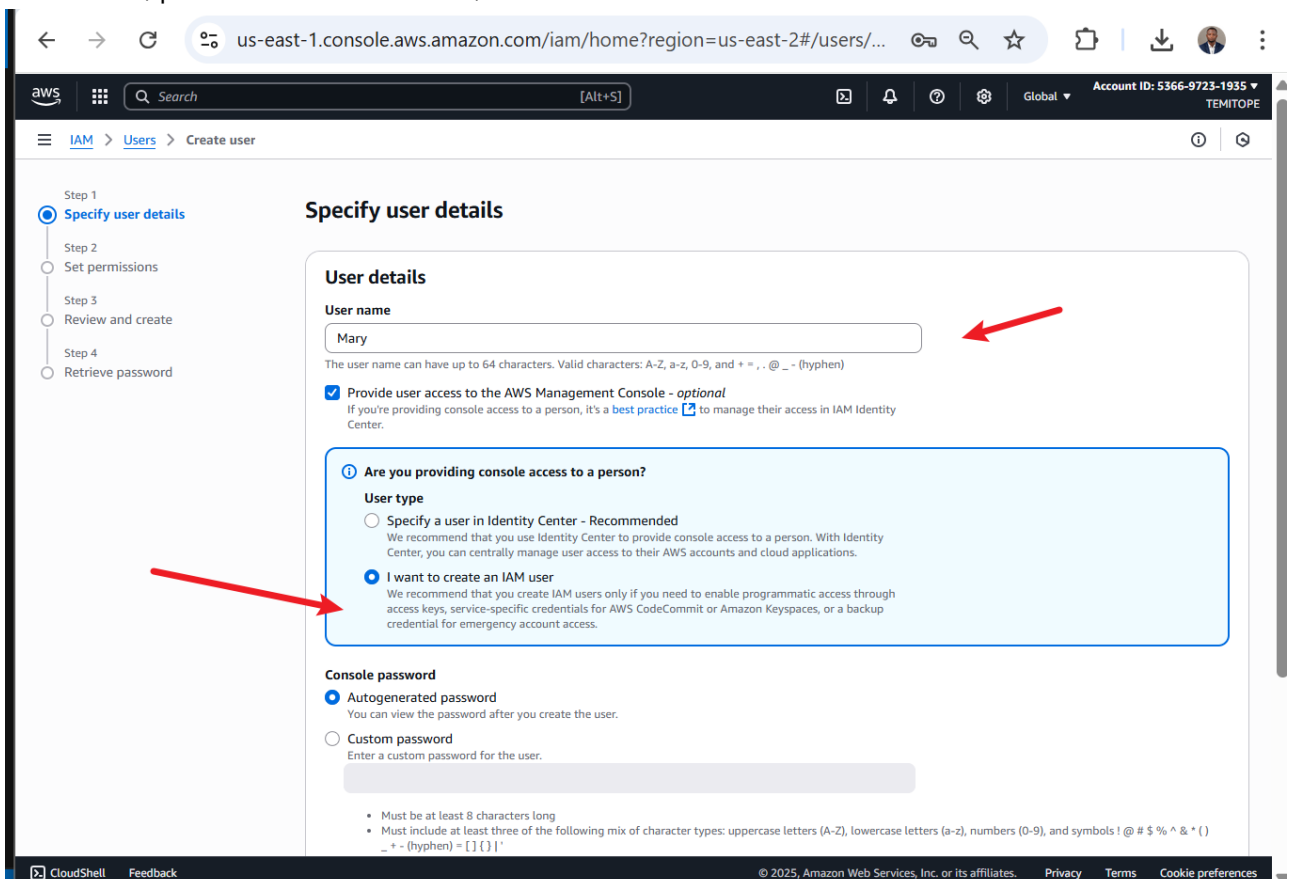
- Console login enabled.
- Credentials downloaded.

Steps:

- Click create new user to create new user Mary.



- Enter name, provide access to console, and check IAM user.



- Choose password, uncheck password change on next login.

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ I want to create an IAM user
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ Custom password
Enter a custom password for the user.

☒ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

[Cancel](#) [Next](#)

- Save password for next login.

Set permissions
Add user to an existing group or create a new one

Permissions options

☒ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

User groups (2)

Group name	Policy	Created
☐ Analyst-Team	analyst-policy	2025-08-16 (16 hours ago)
☐ Development-Team	developer-policy	2025-08-16 (16 hours ago)

Set permissions boundary - optional

[Cancel](#) [Previous](#) [Next](#)

Save password?

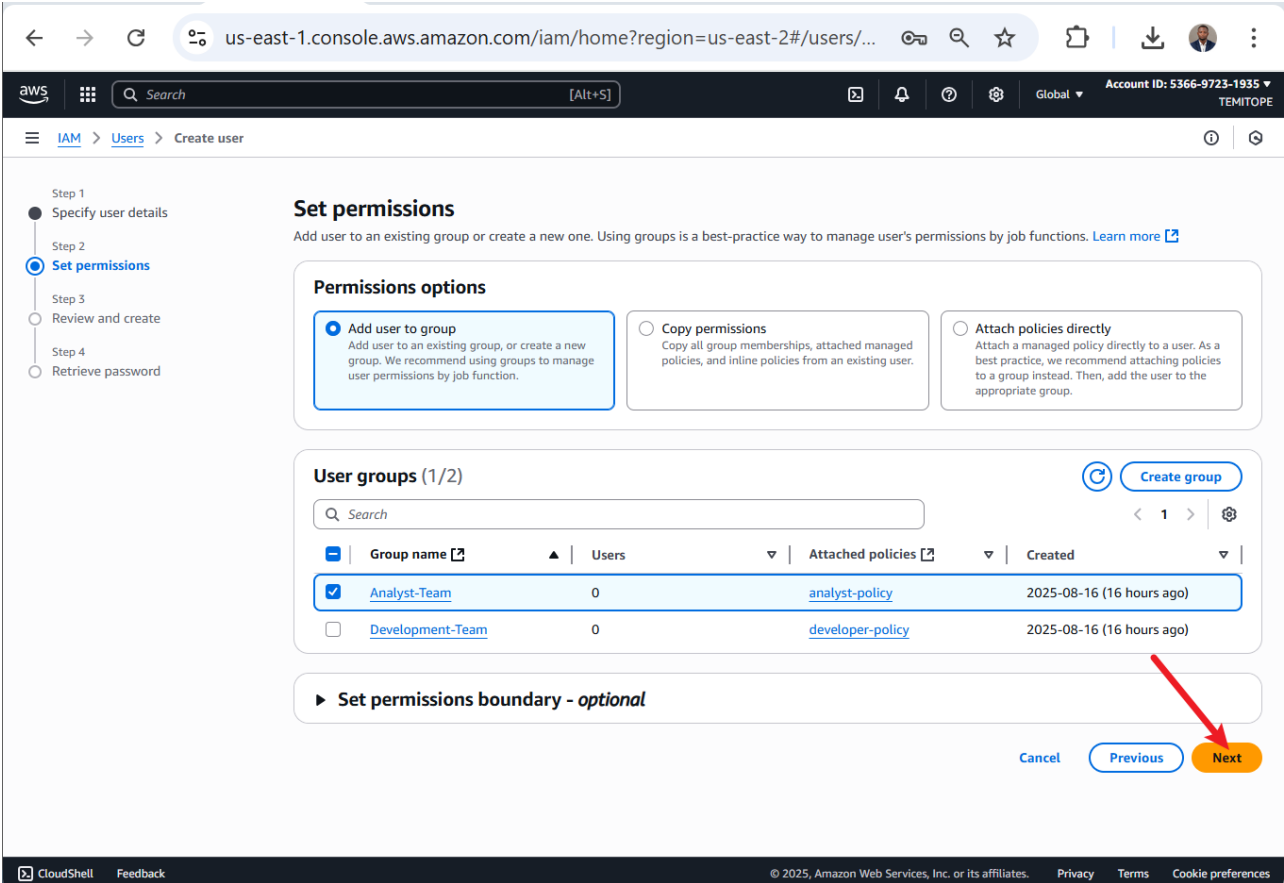
Username:

Password:

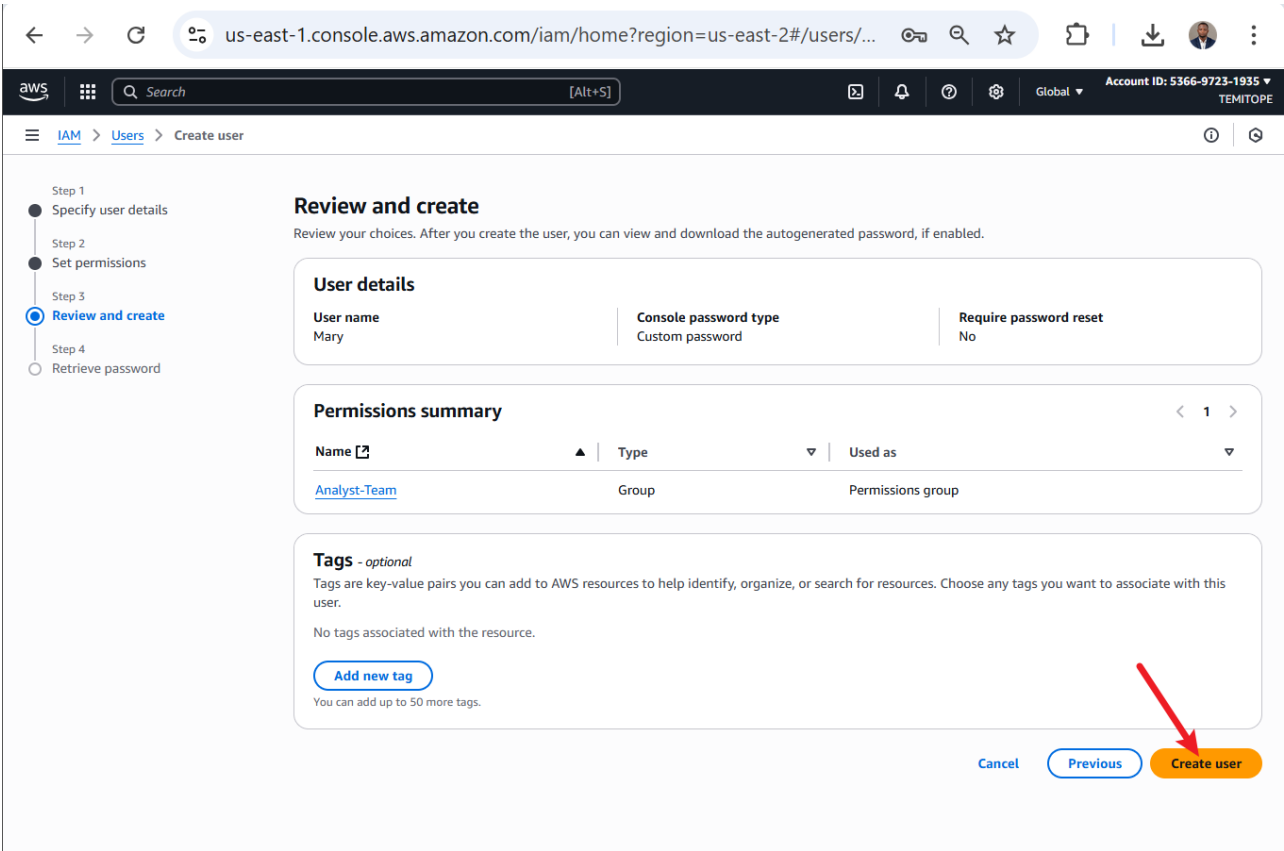
[Save](#) [Never](#)

You can use saved passwords on any device. They're saved to [Google Password Manager](#) for oluwaseun.beicks@gmail.com.

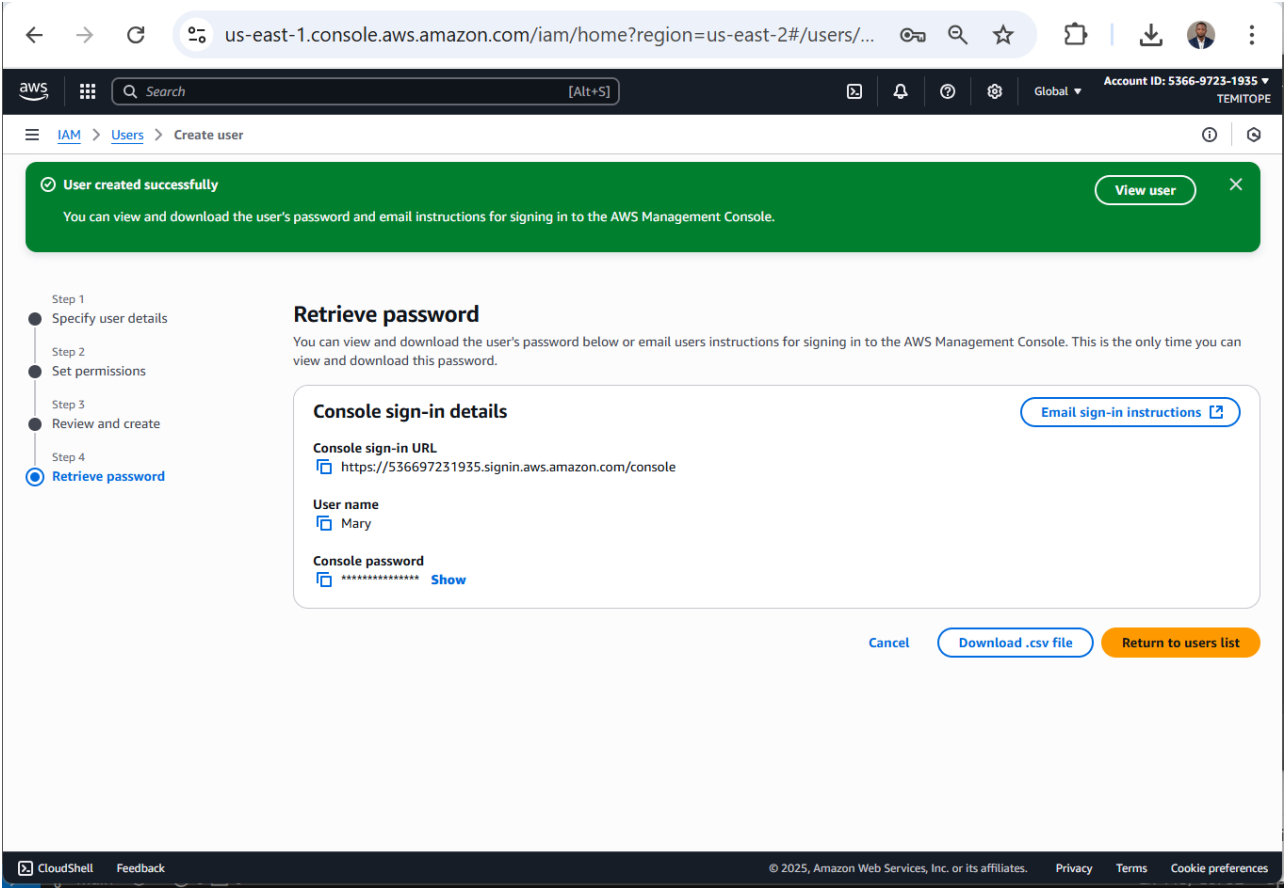
- Check to add to analyst group and click next.



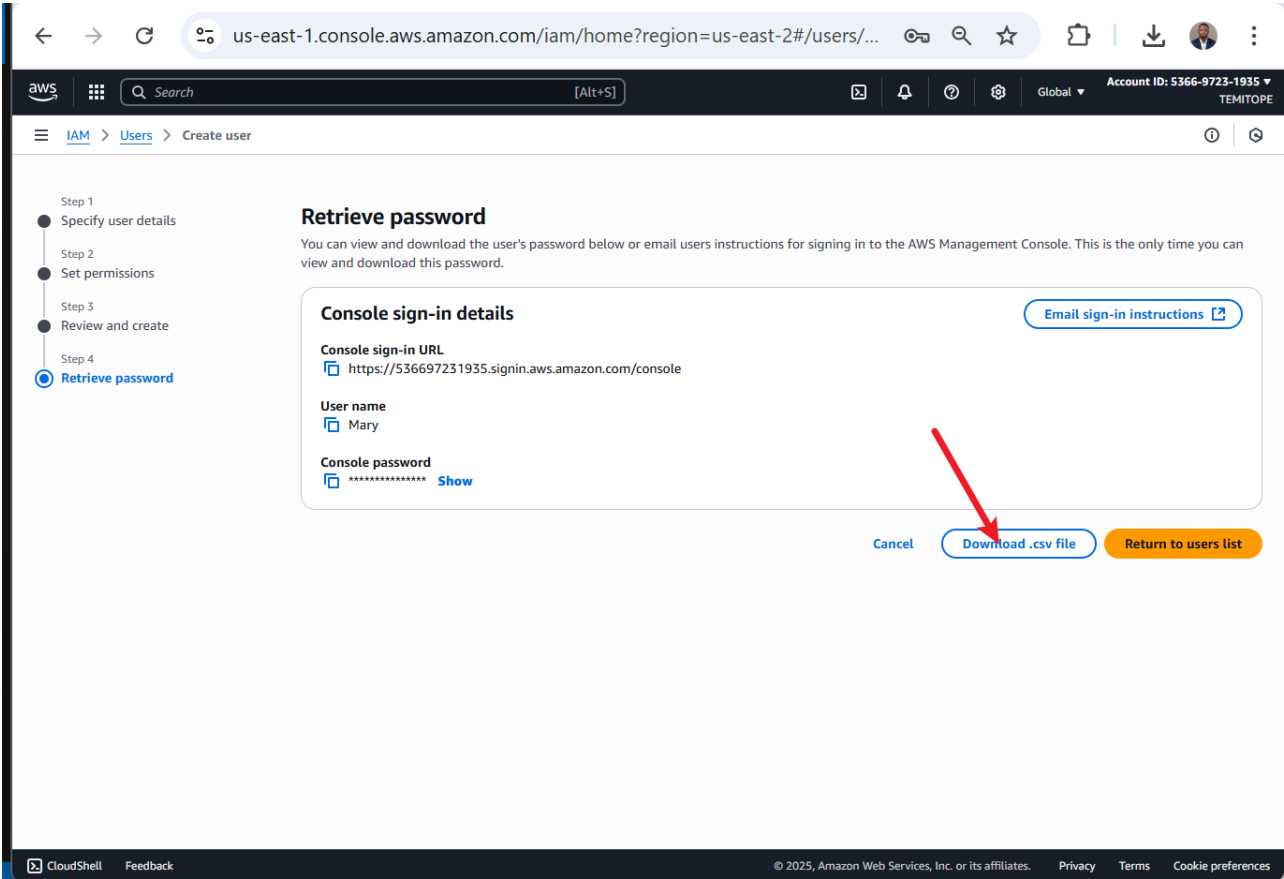
- Review and create user.



- Successfully create user.

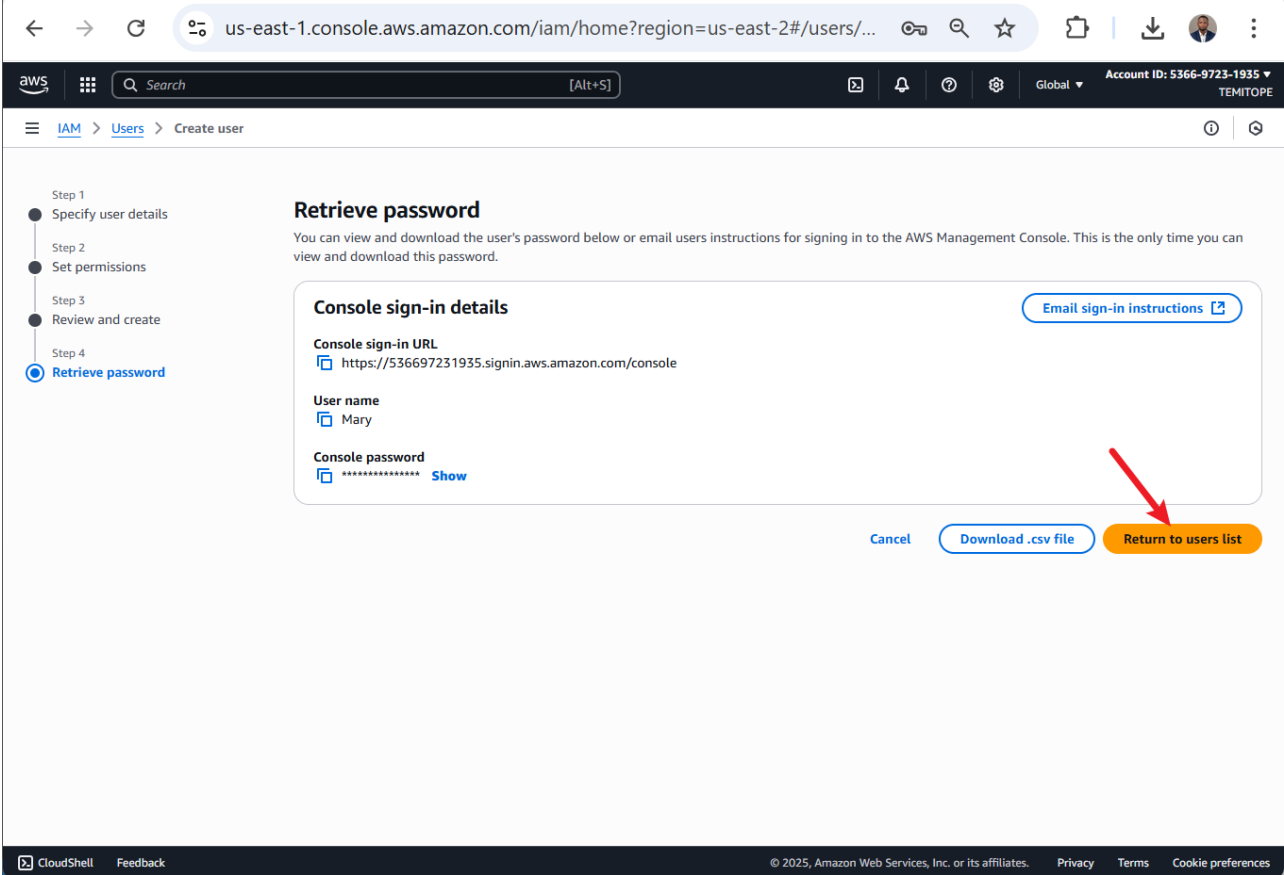


- Download user login CSV.

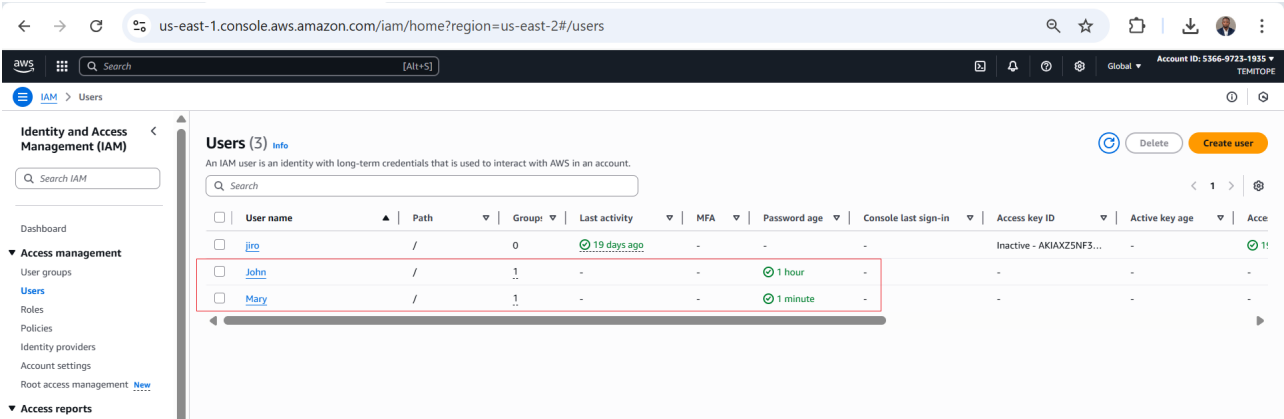


7. Testing and Validation

- Return user list.



- Landed on user list page.

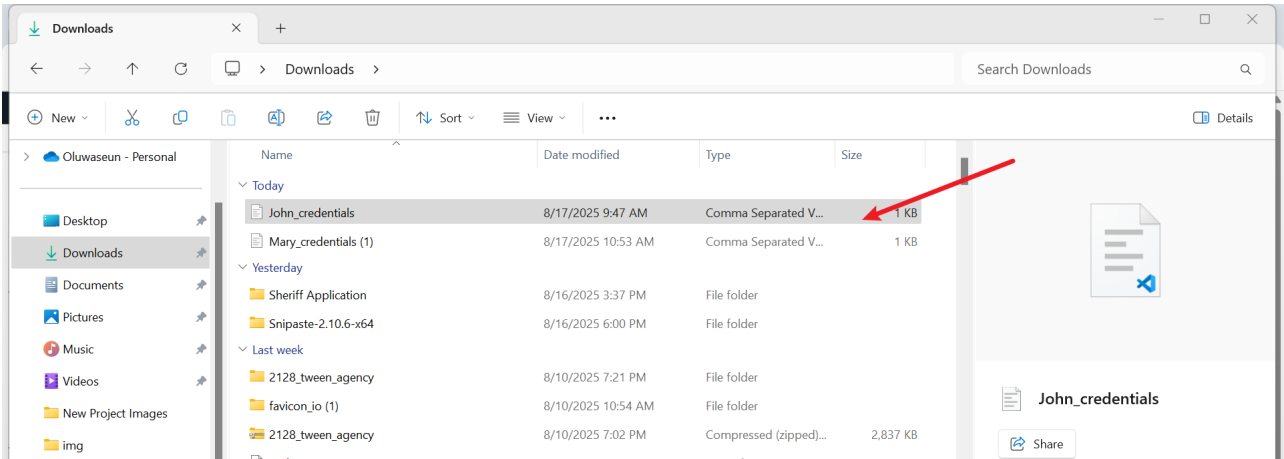


John's Access

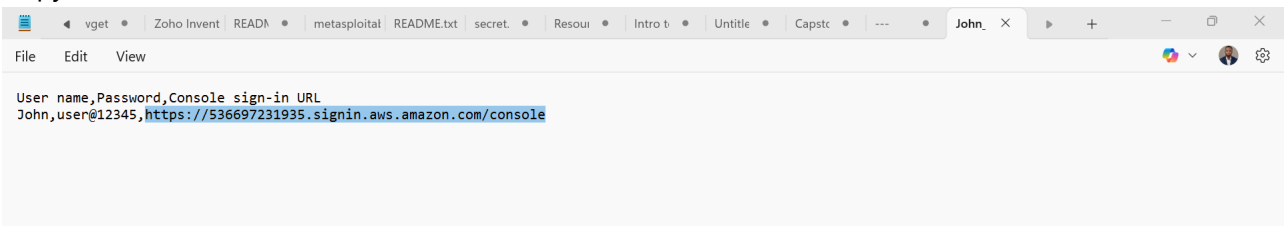
- Logged in as John.
- Accessed **EC2 Dashboard** ☒.
- Attempted to access **S3 Dashboard** ☒ (denied).

Steps:

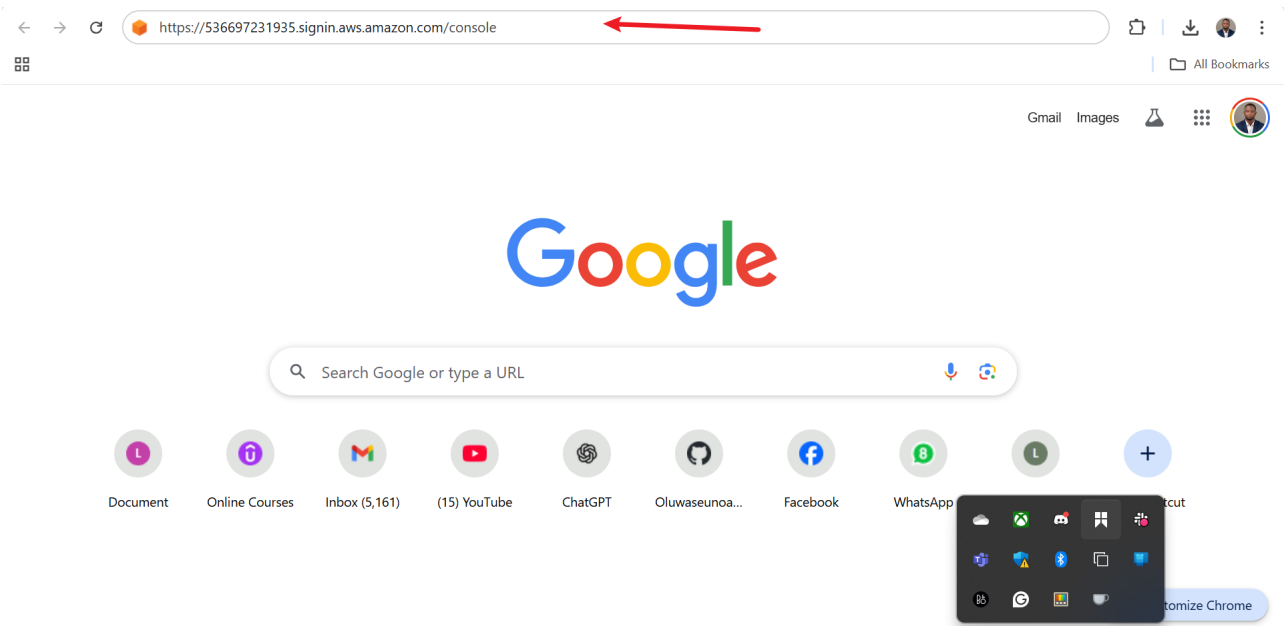
- Locate and open John downloaded CSV file.



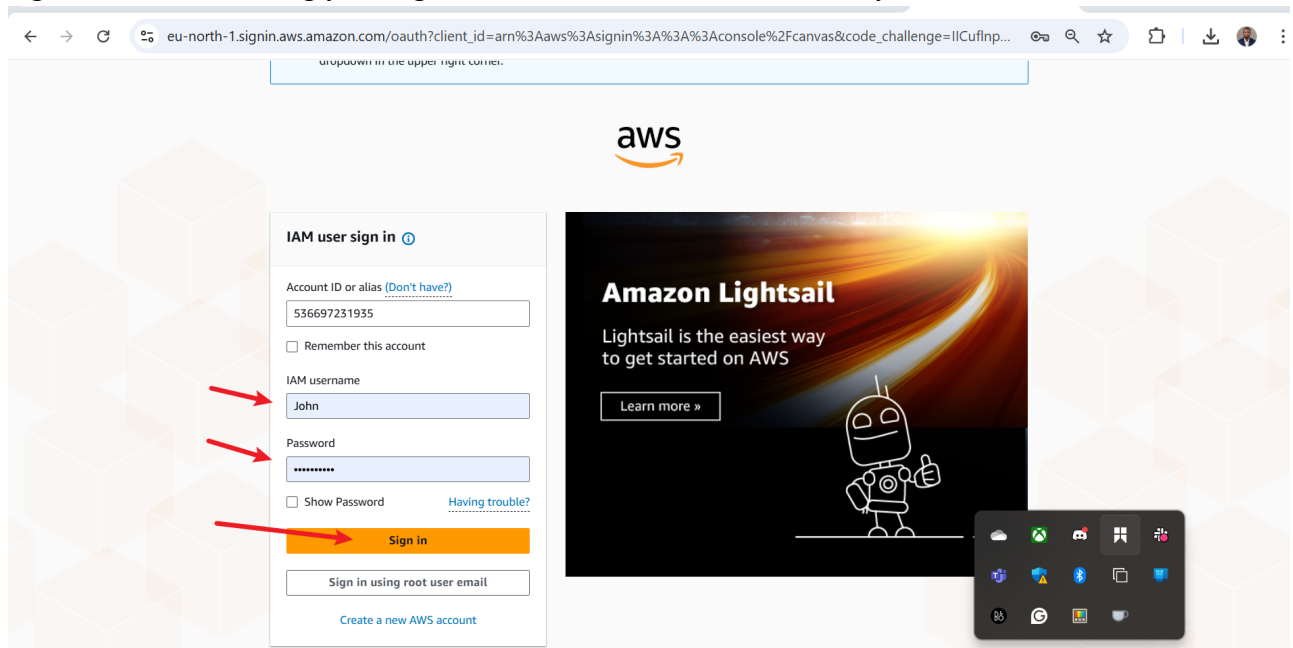
- Copy the HTTPS link.



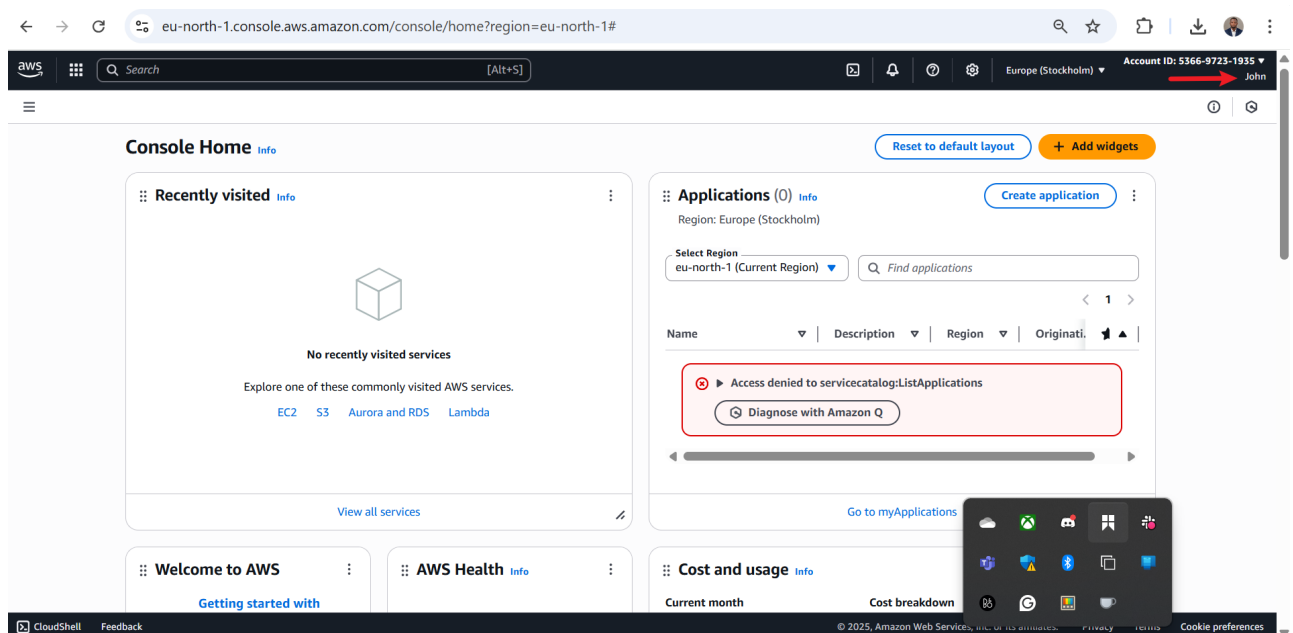
- Paste it on the browser.



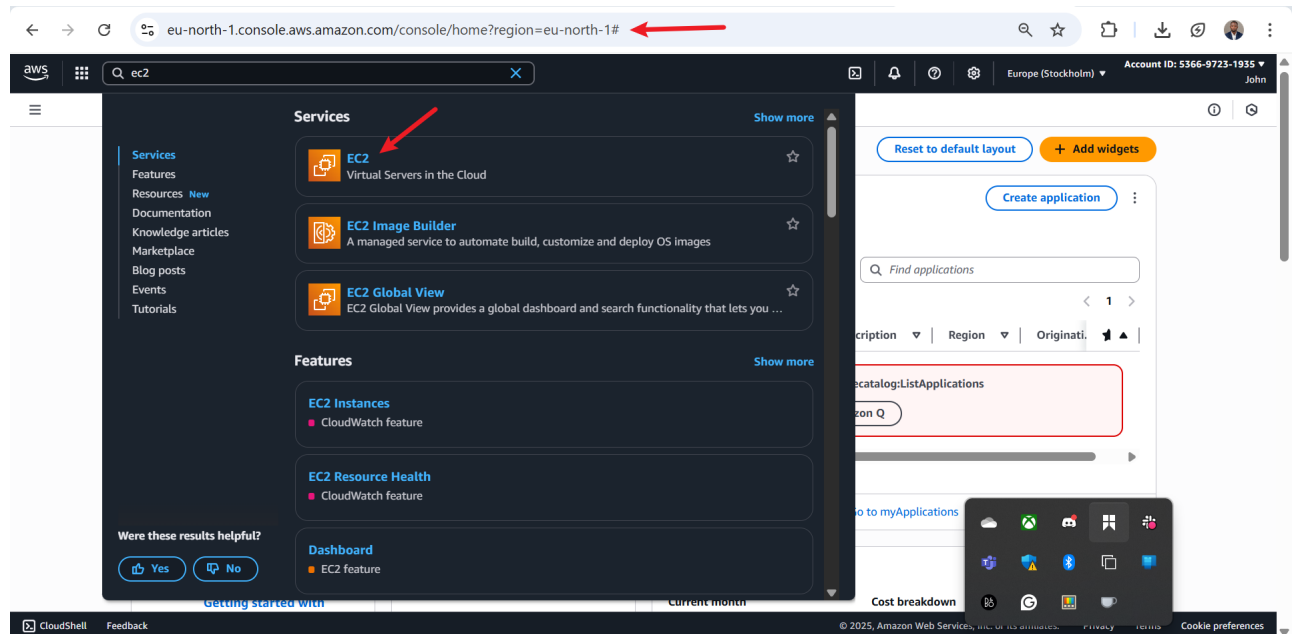
- Sign in as IAM user using your login and account ID which is fetched by the link.



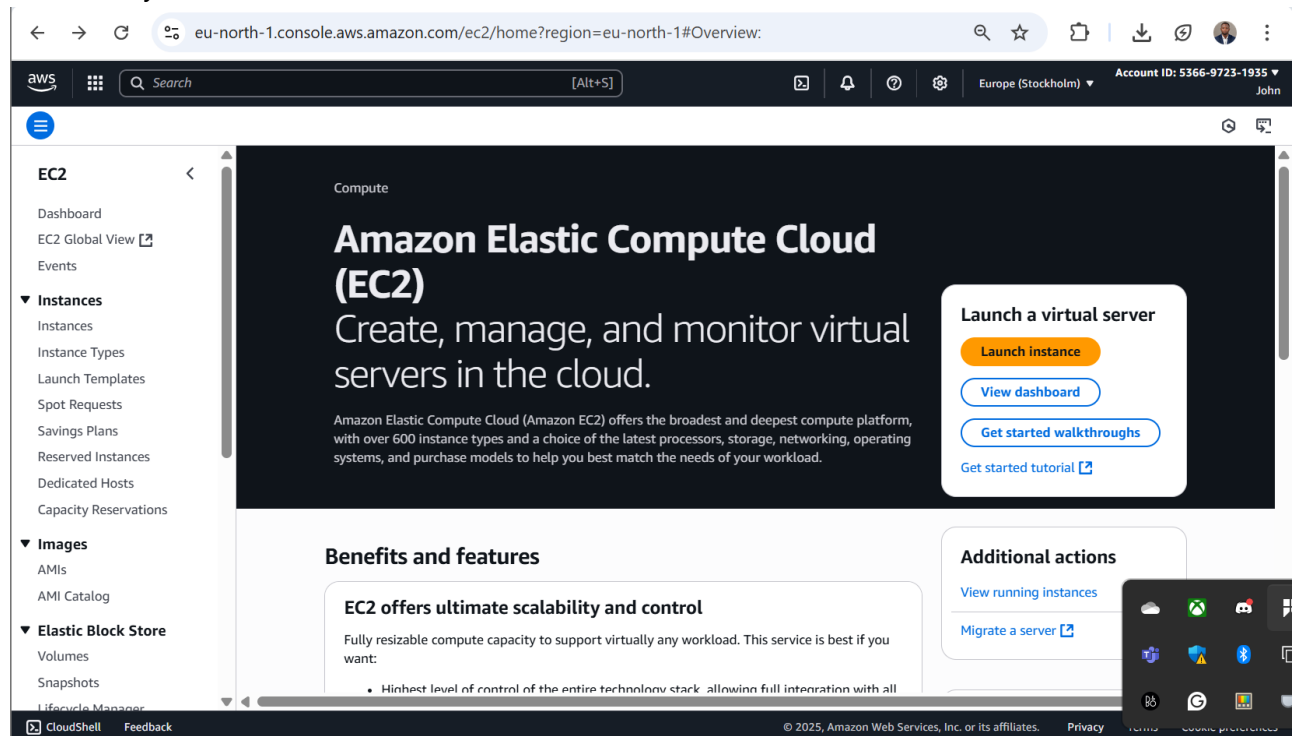
- Logged in as John.



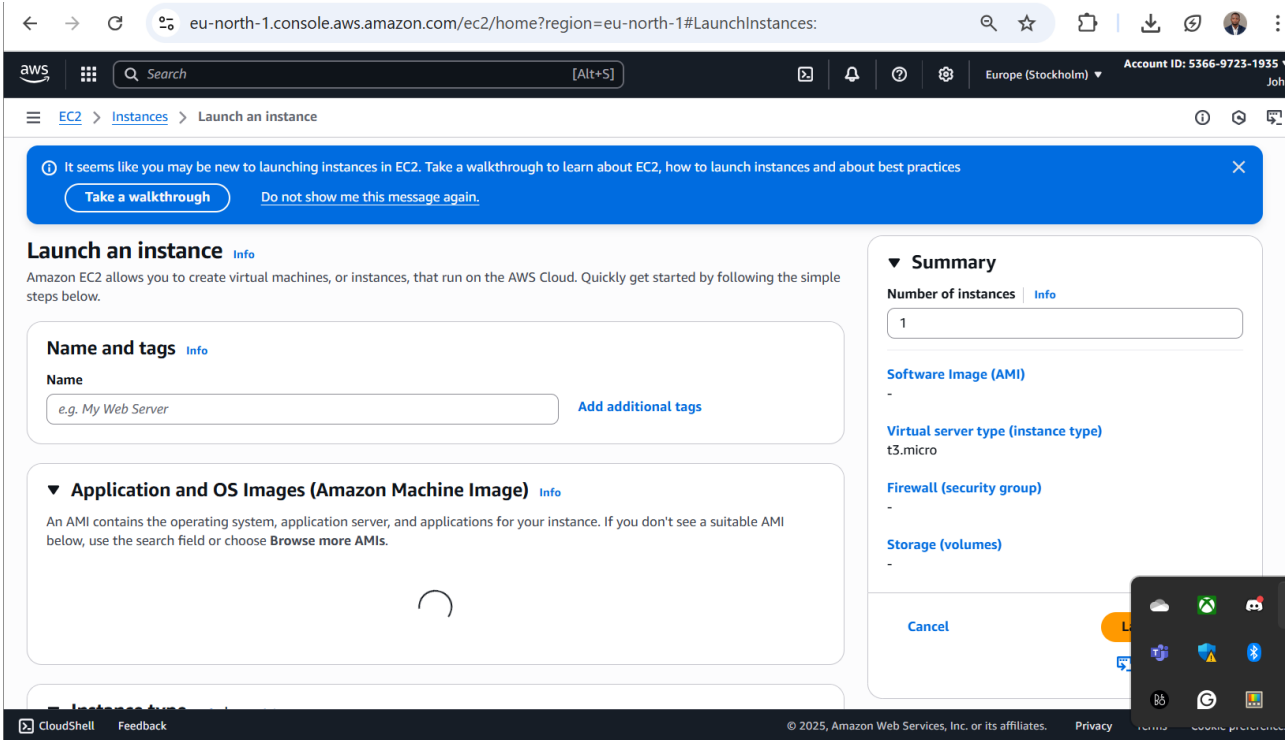
- Search for EC2 and click on it.



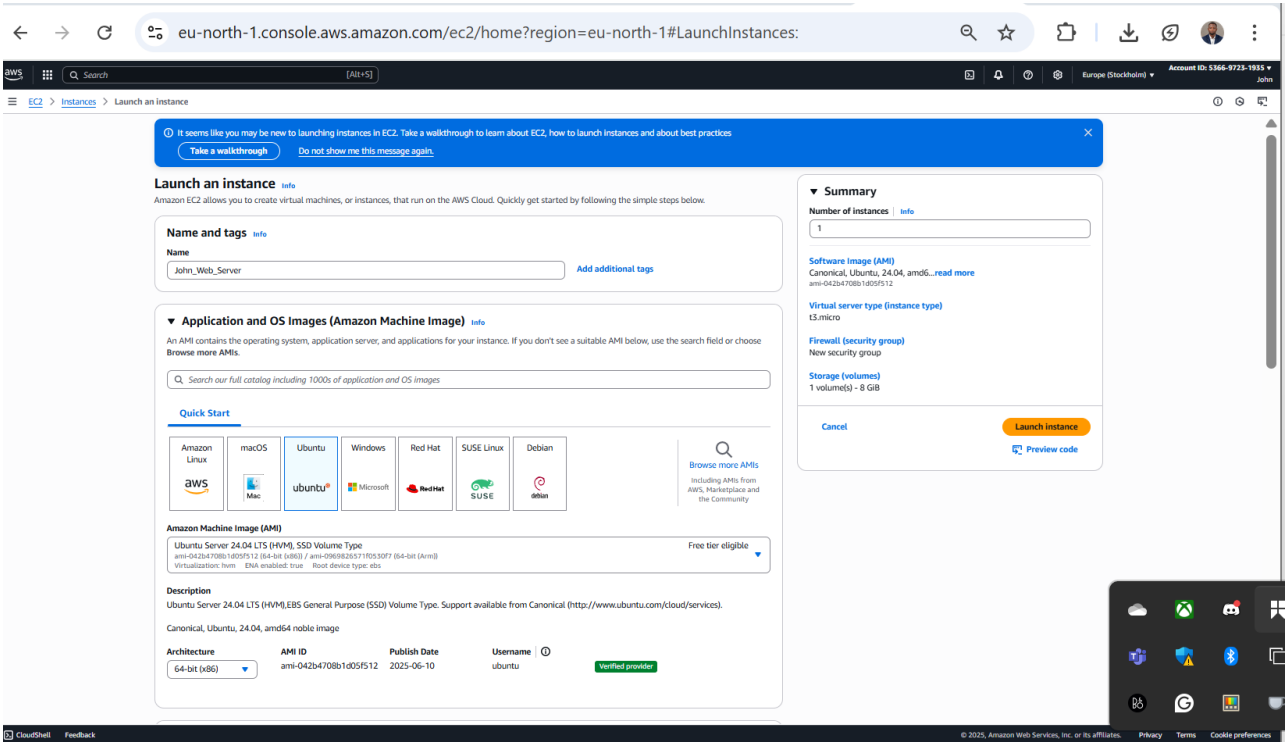
- Successfully access EC2 dashboard, click launch instance.



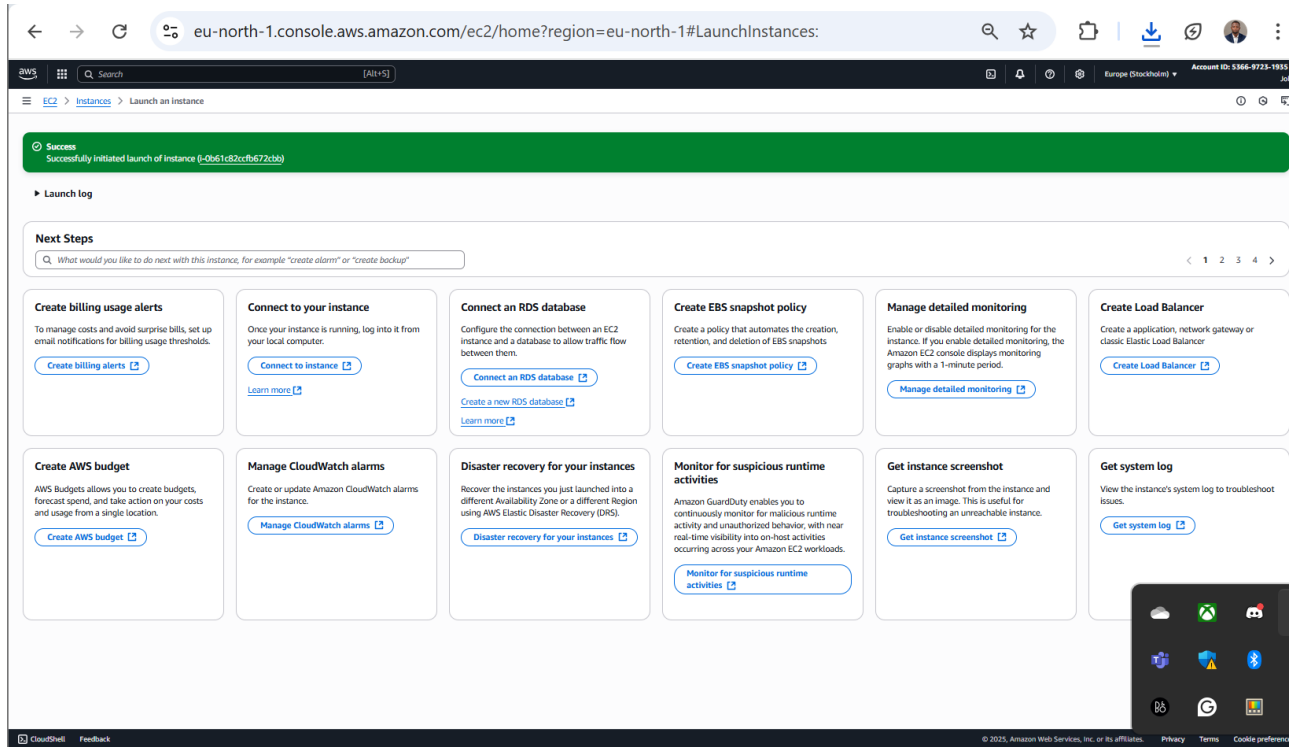
- Launch an instance page appear.



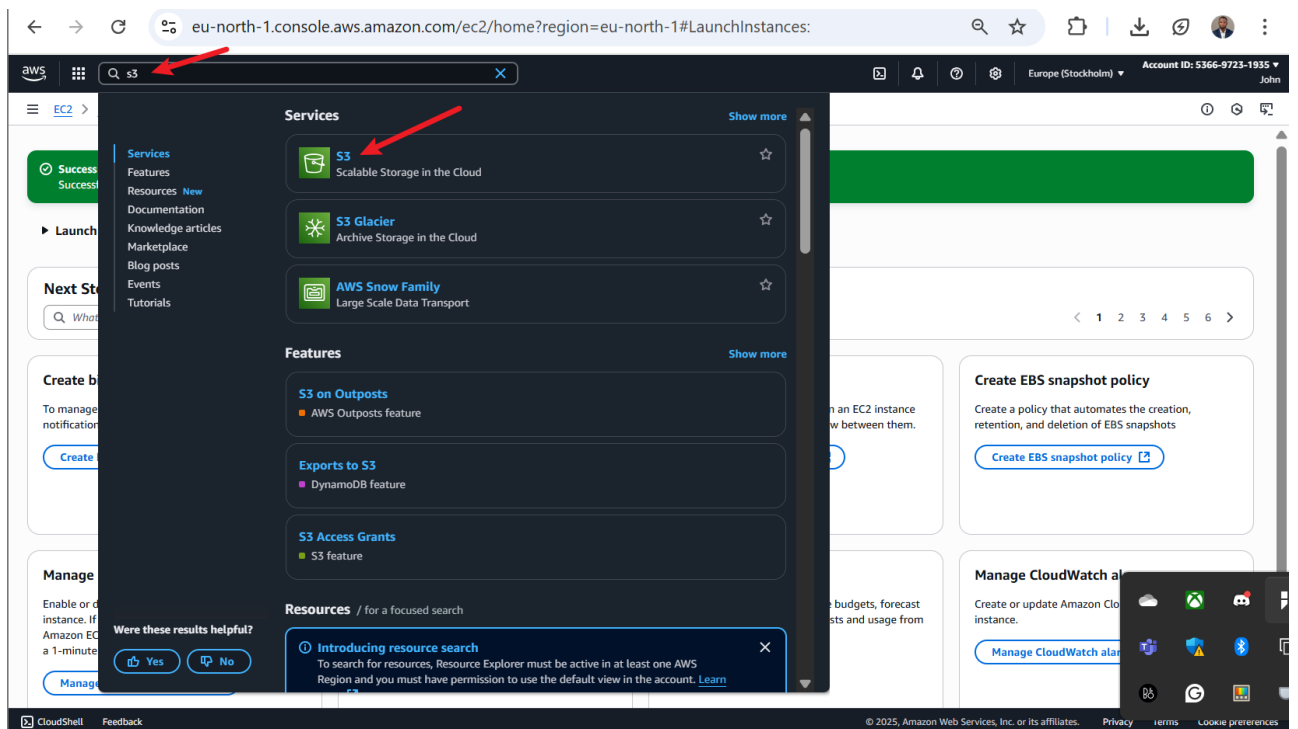
- Name server and set it up, also set key pairs and launch instance.



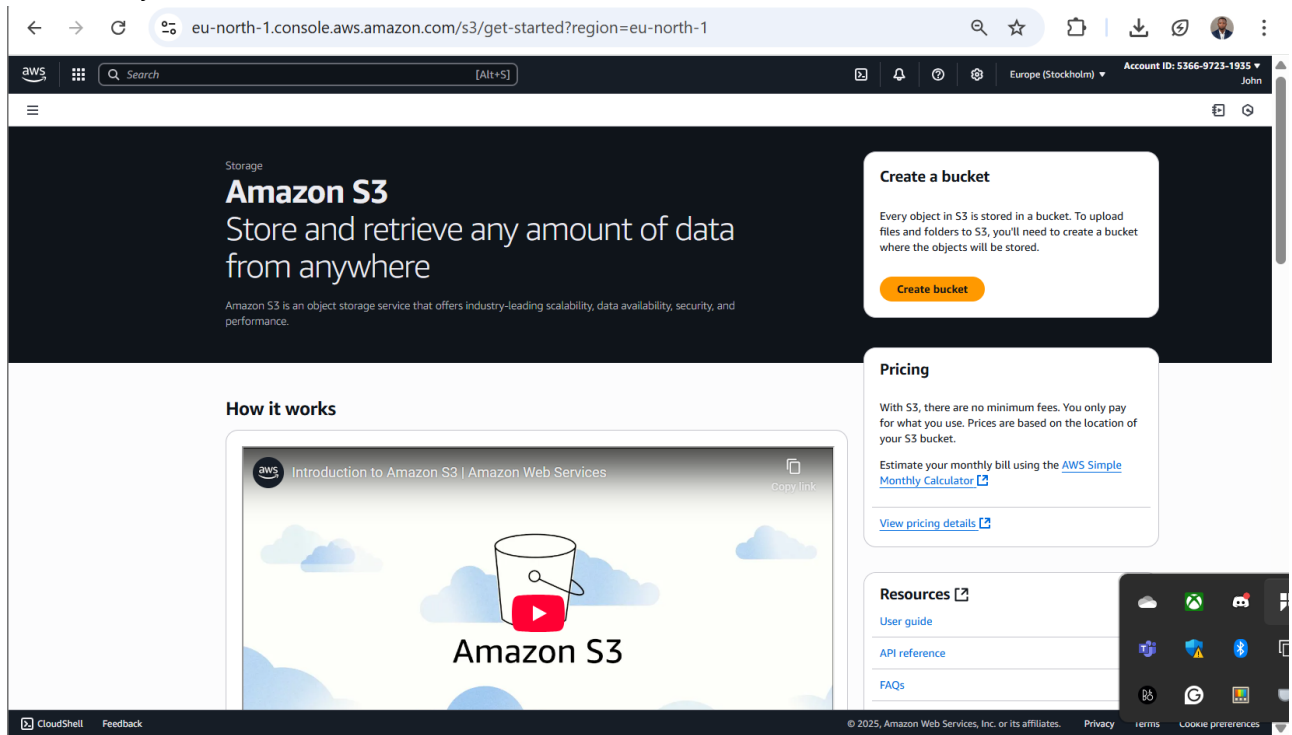
- Instance successfully launched.



- Search for S3 and click on it.

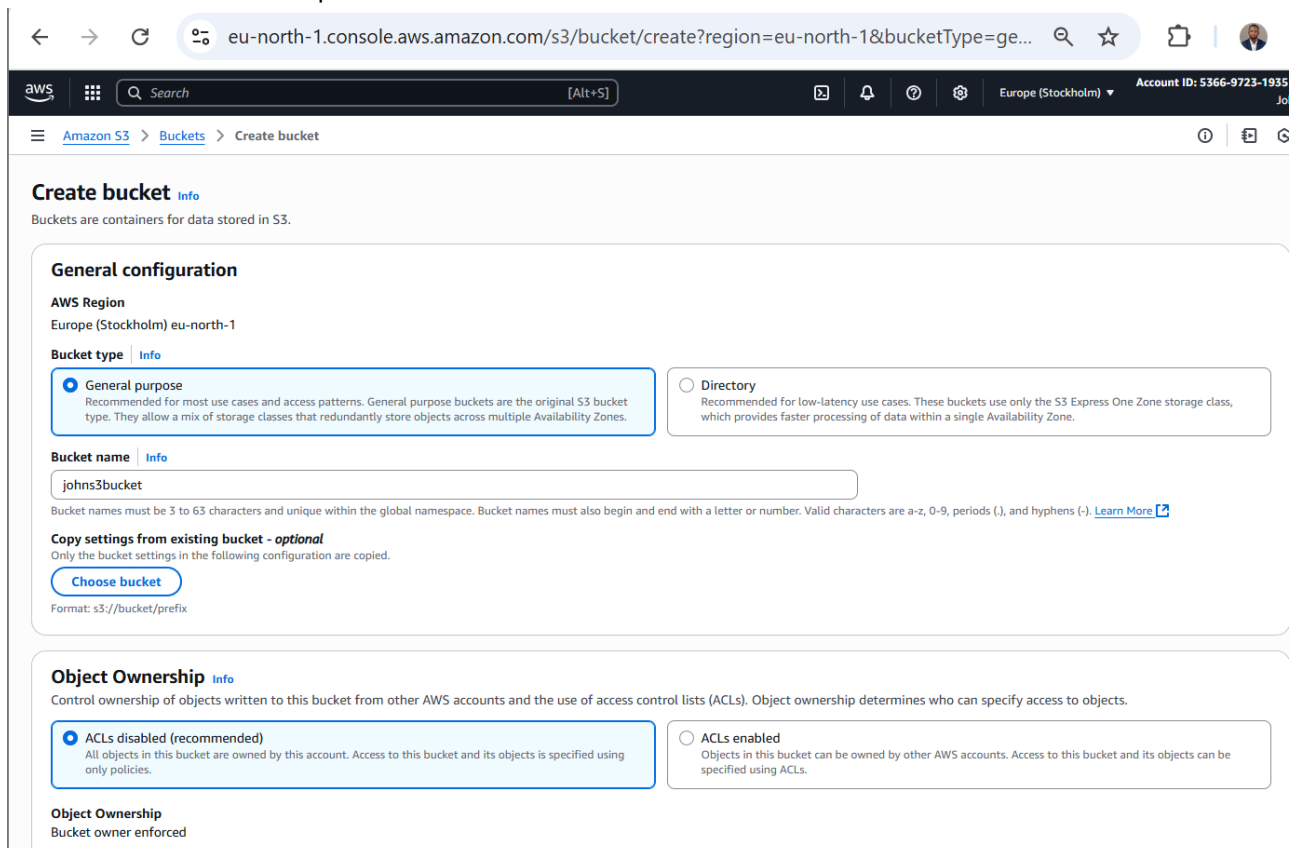


- Successfully access S3 dashboard, click create bucket.



(Note: Context indicates access attempt leading to denial.)

- Name bucket and set it up.



- Scroll down and click create bucket.

The screenshot shows the AWS console 'Create bucket' page for the eu-north-1 region. The page includes sections for 'Tags', 'Default encryption', and 'Bucket Key'. The 'Encryption type' is set to 'Server-side encryption with Amazon S3 managed keys (SSE-S3)'. The 'Bucket Key' is set to 'Enable'. The 'Advanced settings' section is expanded, showing a message: 'After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.' At the bottom right, there are 'Cancel' and 'Create bucket' buttons. A red arrow points to the 'Create bucket' button.

- Failed to create bucket.

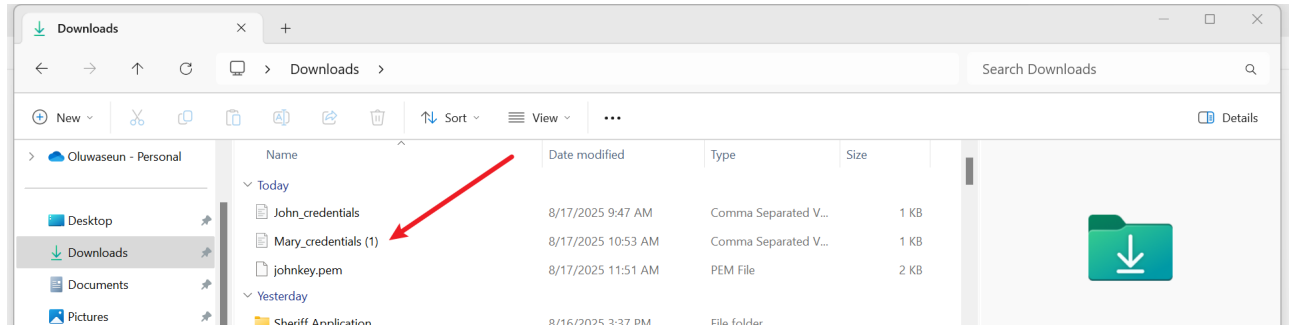
The screenshot shows the AWS console 'Create bucket' page for the eu-north-1 region. The configuration is the same as the previous screenshot. However, at the bottom, there is a red-bordered error box with the message: 'Failed to create bucket. To create a bucket, the s3:CreateBucket permission is required.' Below this message, there are links to 'View your permissions in the IAM console' and 'Identity and Access Management in Amazon S3'. At the bottom right, there are 'Cancel' and 'Create bucket' buttons. A red arrow points to the error message box.

Mary's Access

- Logged in as Mary.
- Accessed **S3 Dashboard** ☒.
- Attempted to access **EC2 Dashboard** **X** (denied).

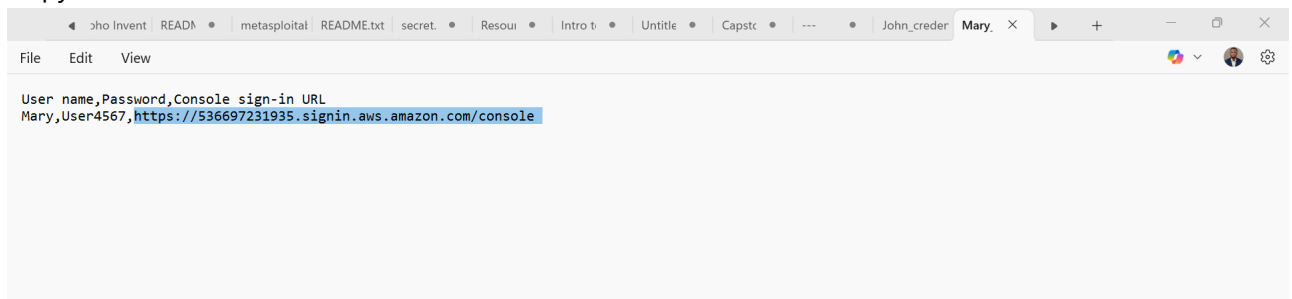
Steps:

- Locate and open Mary downloaded CSV file.

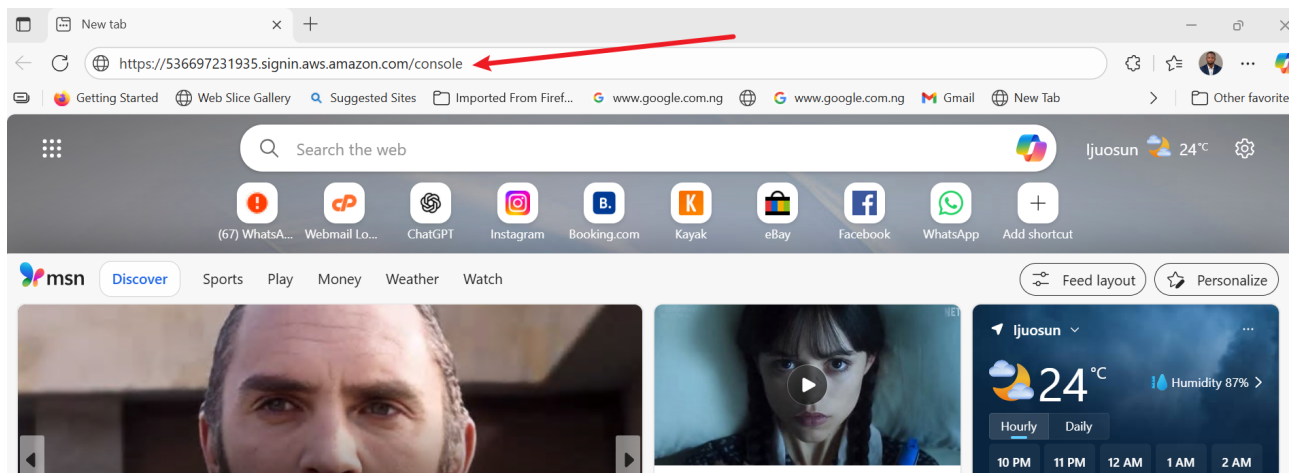


(Note: Filename references John, but context indicates Mary's file.)

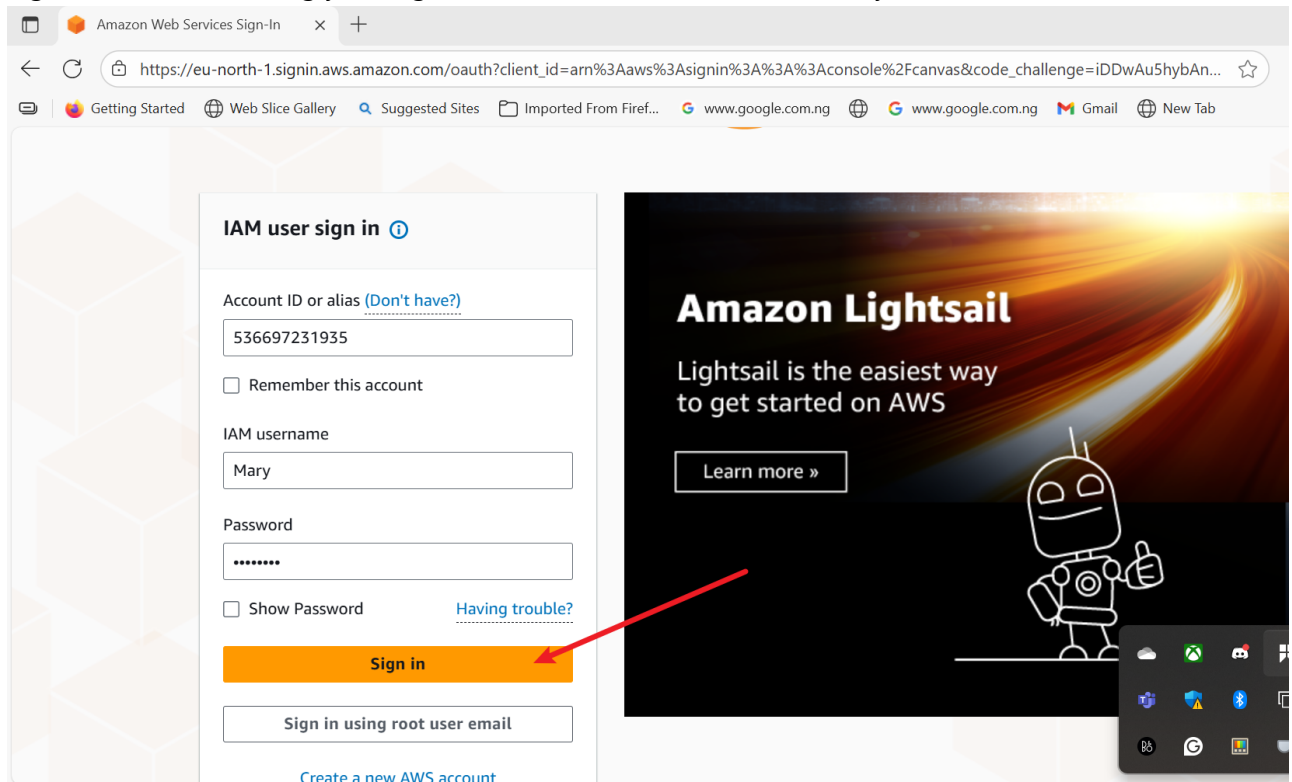
- Copy the HTTPS link.



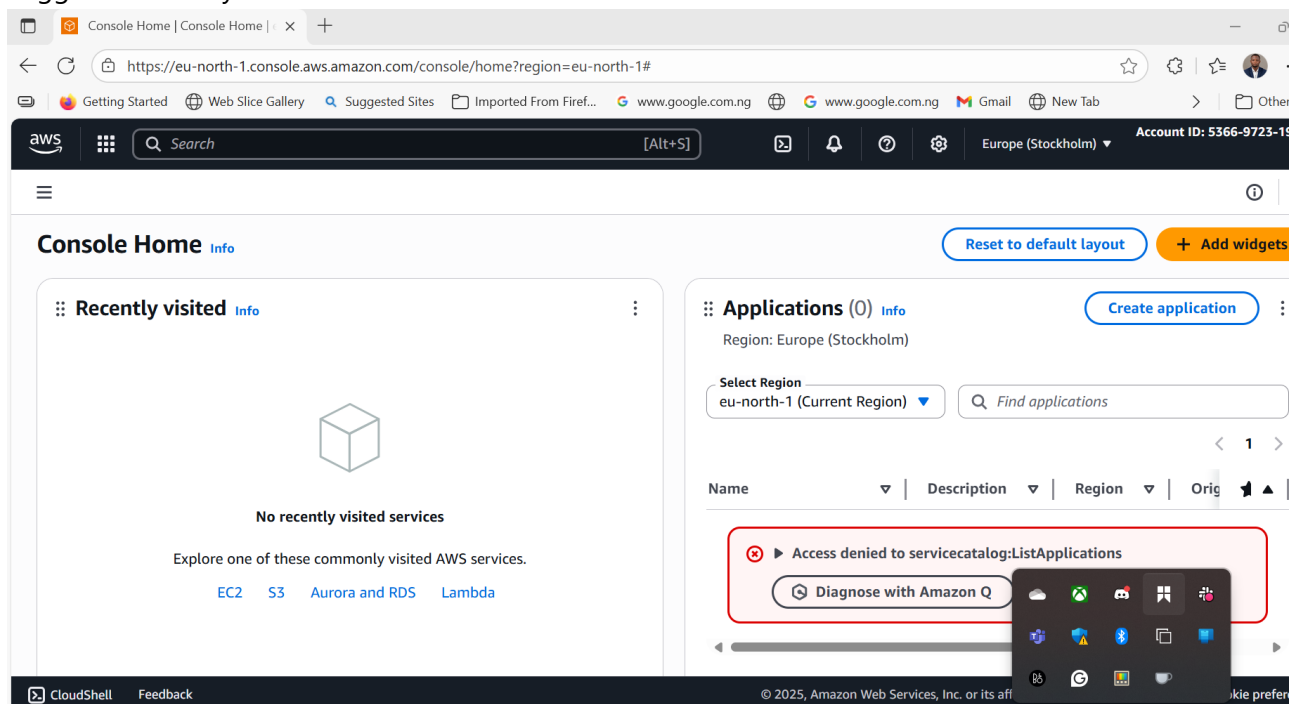
- Paste it on the browser.



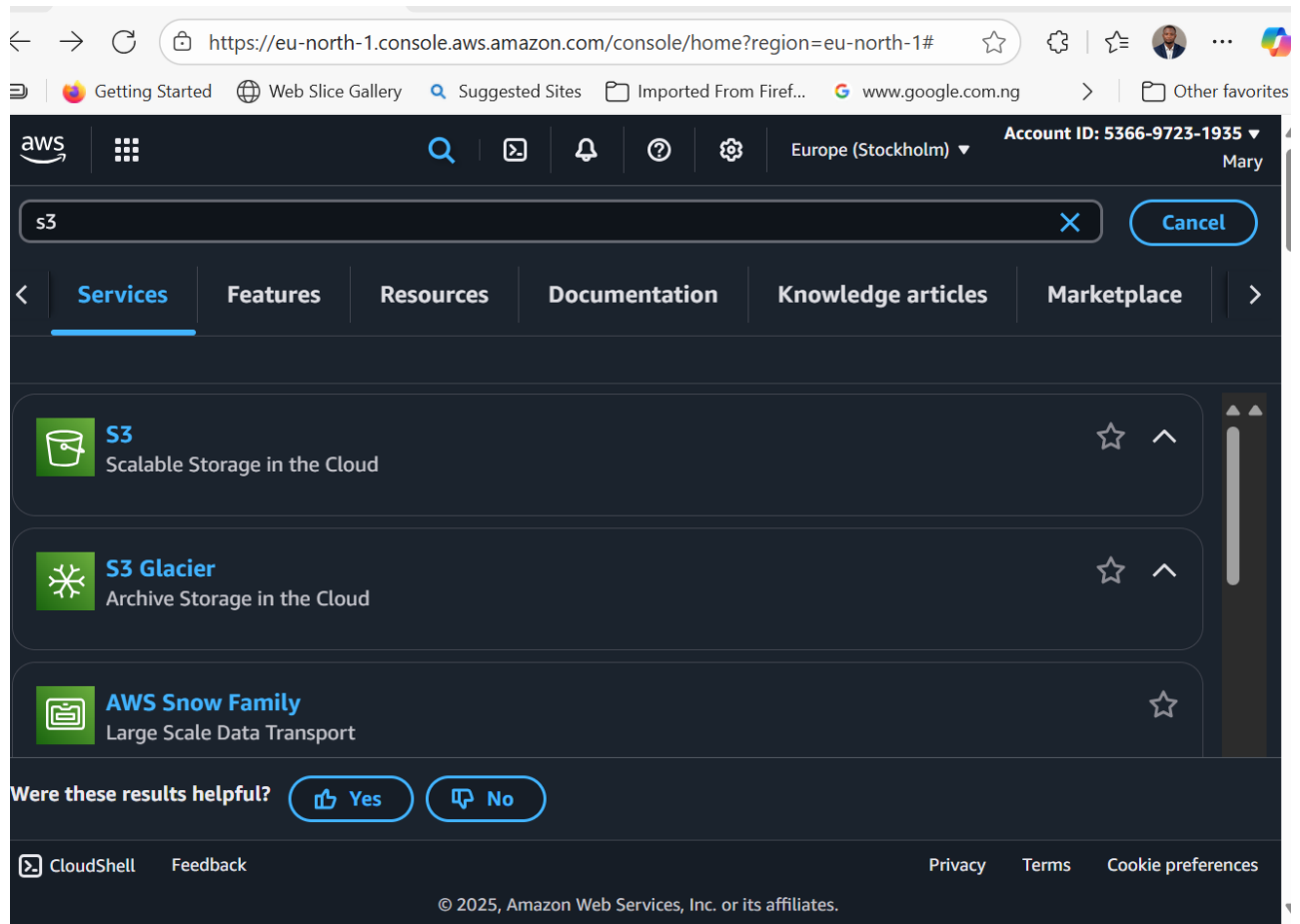
- Sign in as IAM user using your login and account ID which is fetched by the link.



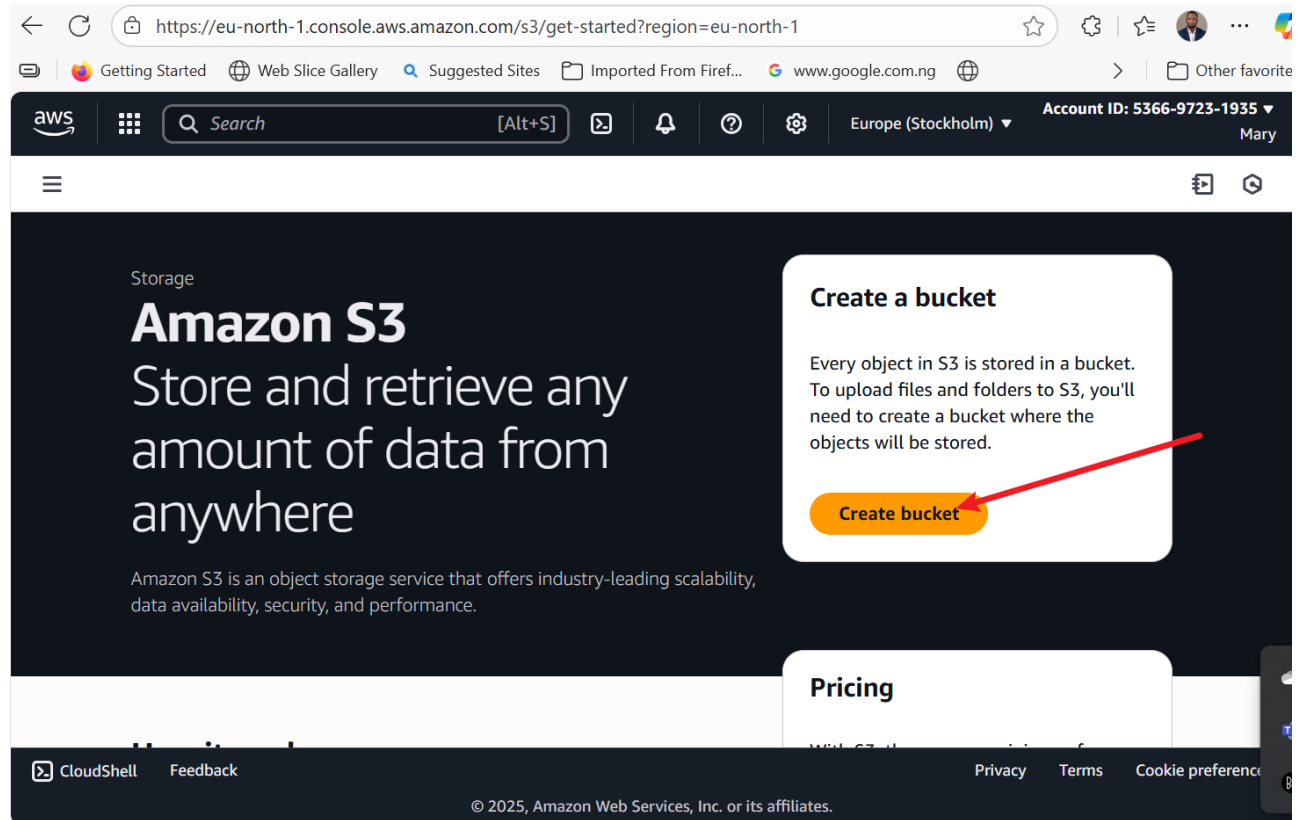
- Logged in as Mary.



- Search for S3 and click on it.



- Successfully access S3 dashboard, click create bucket.



- Name bucket and set it up.

https://eu-north-1.console.aws.amazon.com/s3/bucket/create?region=eu-north-1&bucketType=g...

Getting Started Web Slice Gallery Suggested Sites Imported From Firef... www.google.com.ng

Search [Alt+S] Europe (Stockholm) Account ID: 5366-9723-1935

Amazon S3 Buckets Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Europe (Stockholm) eu-north-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)
marys3bucket1234

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

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- Scroll down and click create bucket.

Create S3 bucket | S3 | eu-north-1

https://eu-north-1.console.aws.amazon.com/s3/bucket/create?region=eu-north-1&bucketType=g...

Getting Started Web Slice Gallery Suggested Sites Imported From Firef... www.google.com.ng

Search [Alt+S] Europe (Stockholm) Account ID: 5366-9723-1935 Mary

Amazon S3 Buckets Create bucket

You can add up to 50 tags.

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

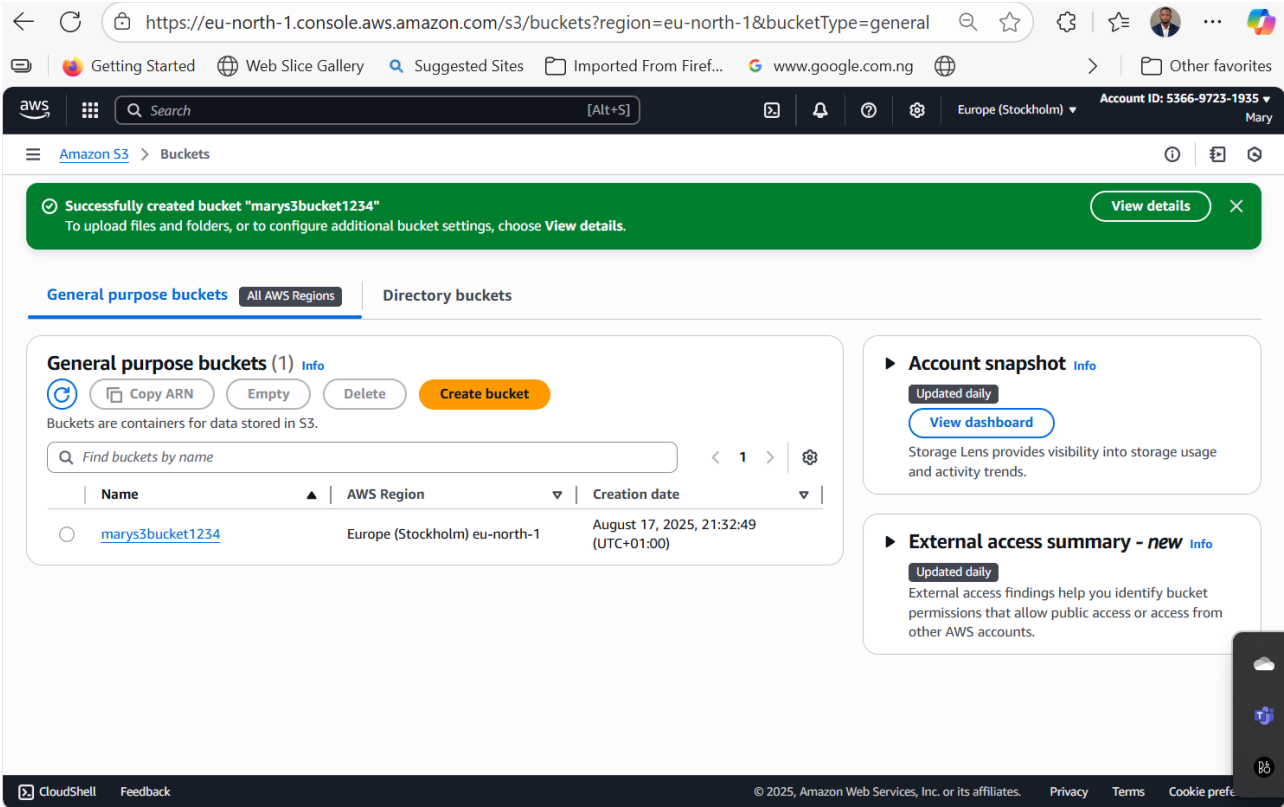
☒ Enable

Advanced settings

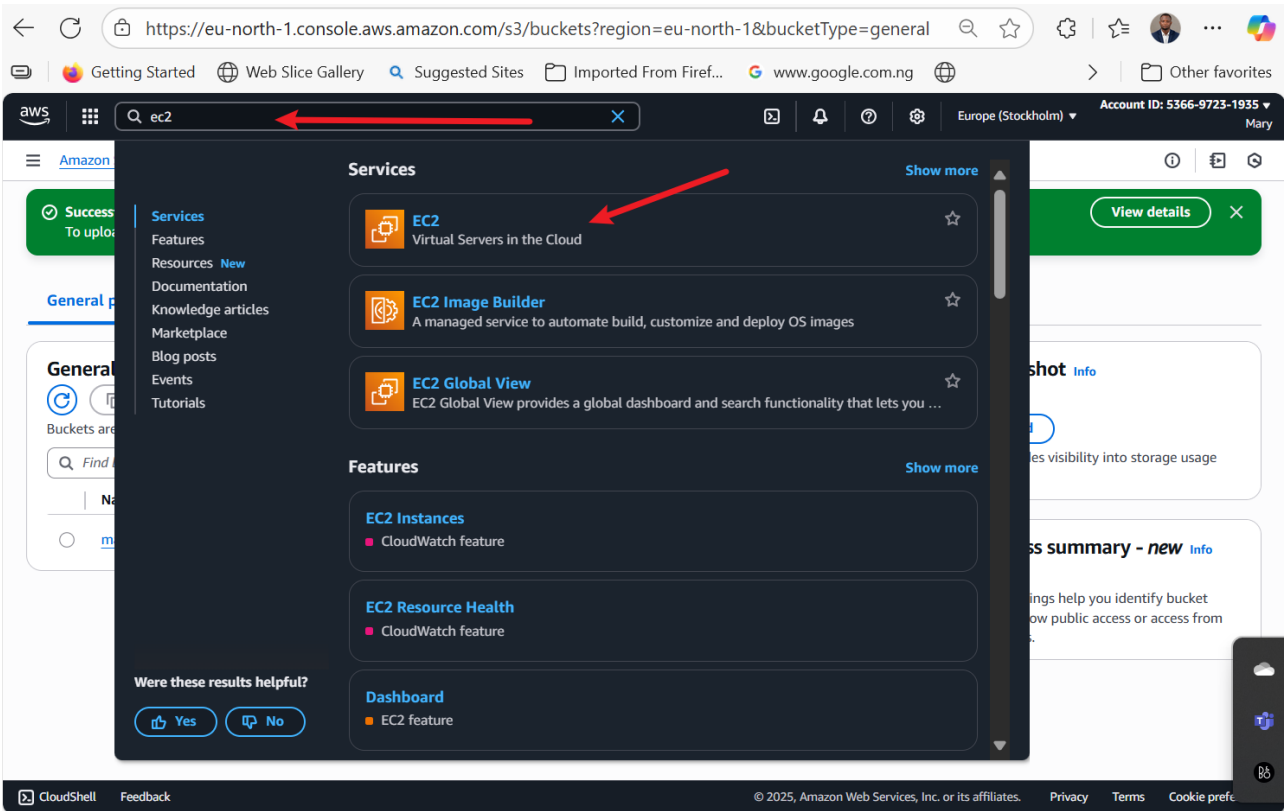
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel [Create bucket](#)

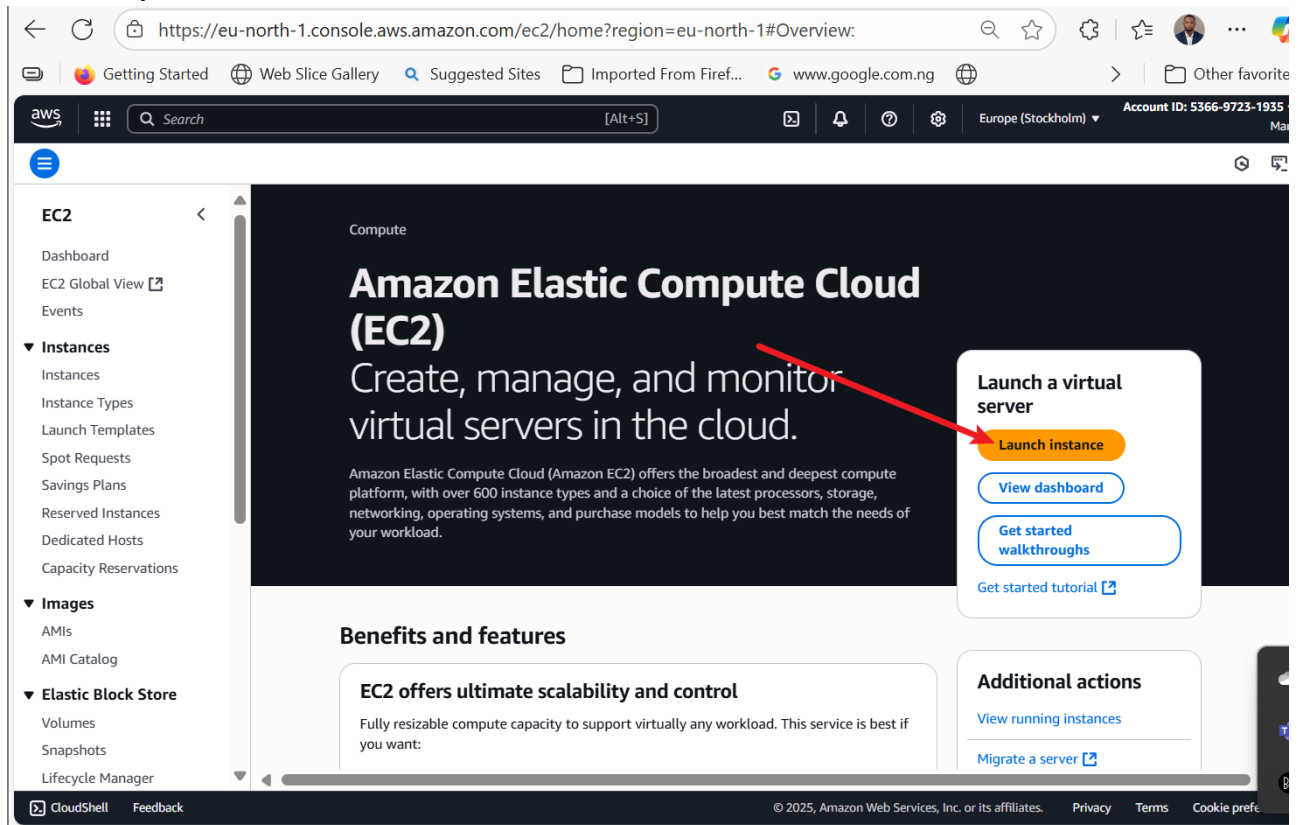
- Bucket successfully created.



- Search for EC2 and click on it.

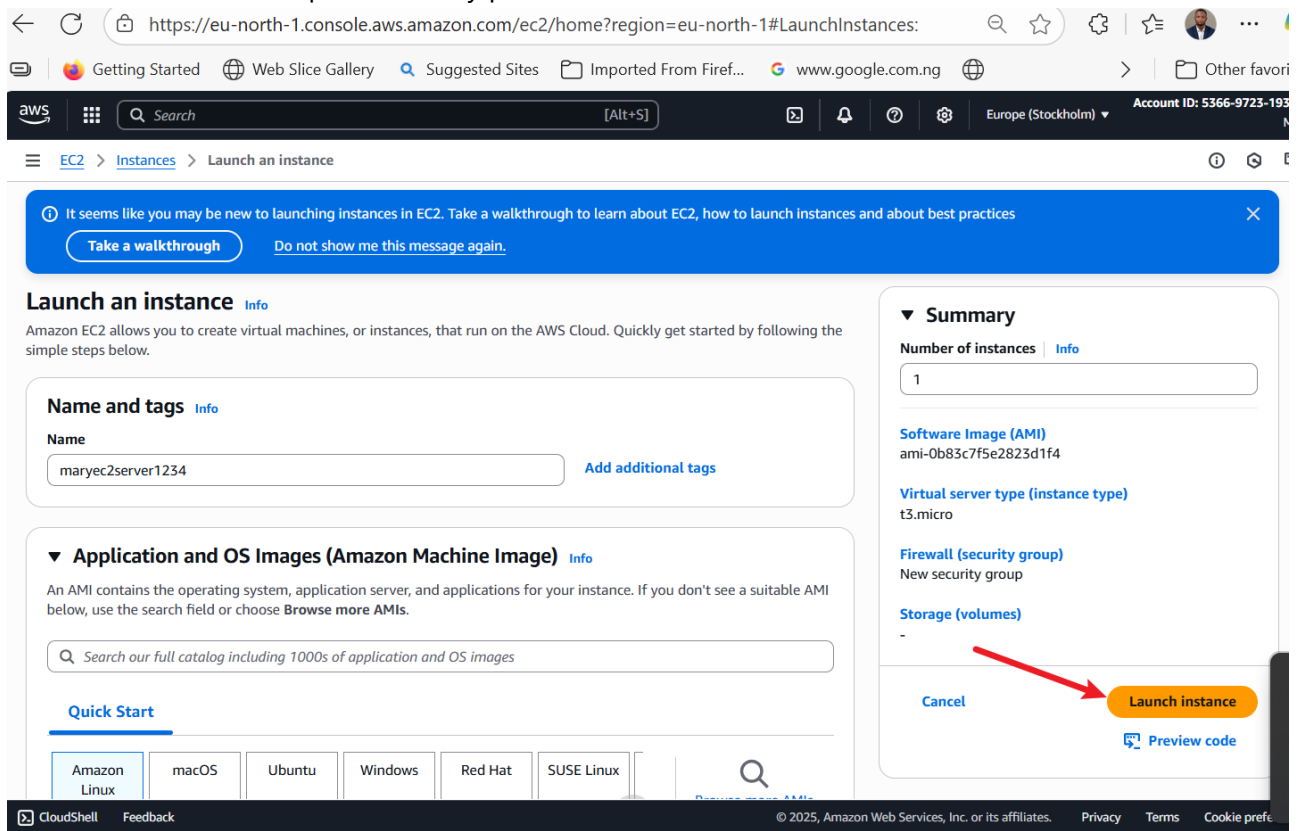


- Successfully access EC2 dashboard, click launch instance.

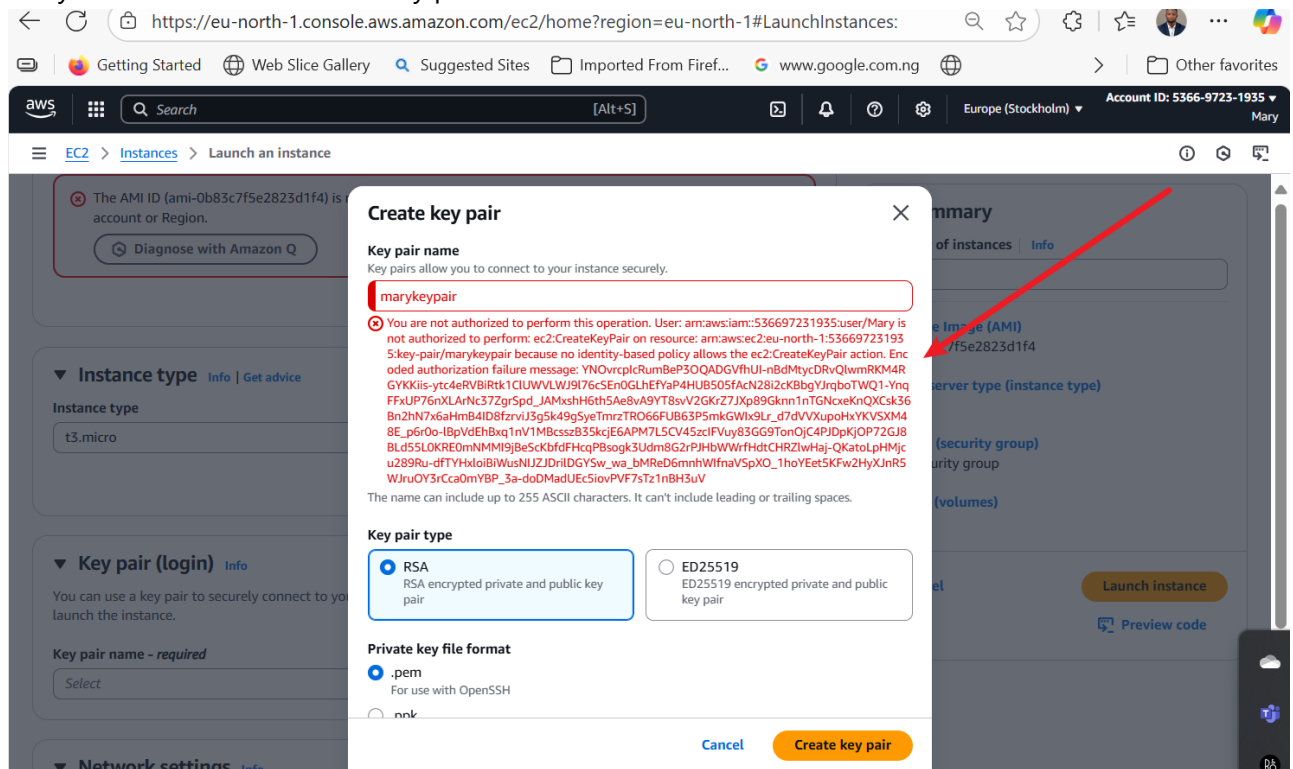


(Note: Context indicates access attempt leading to denial.)

- Name server and set it up, also set key pairs and launch instance.



- Mary not authorized to create key pair.



This confirmed that users had **role-specific permissions** aligned with the principle of least privilege.

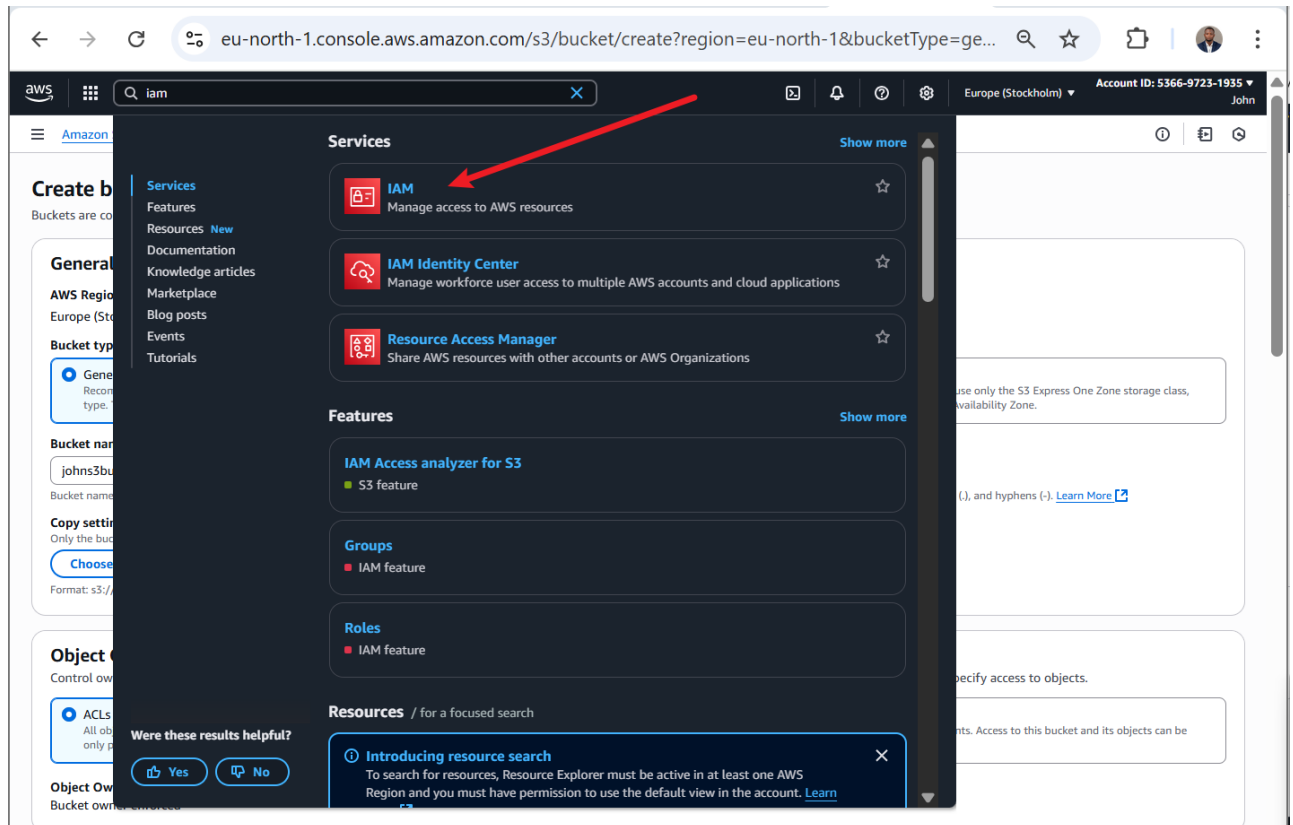
8. Enabling Multi-Factor Authentication (MFA)

John (Backend Developer)

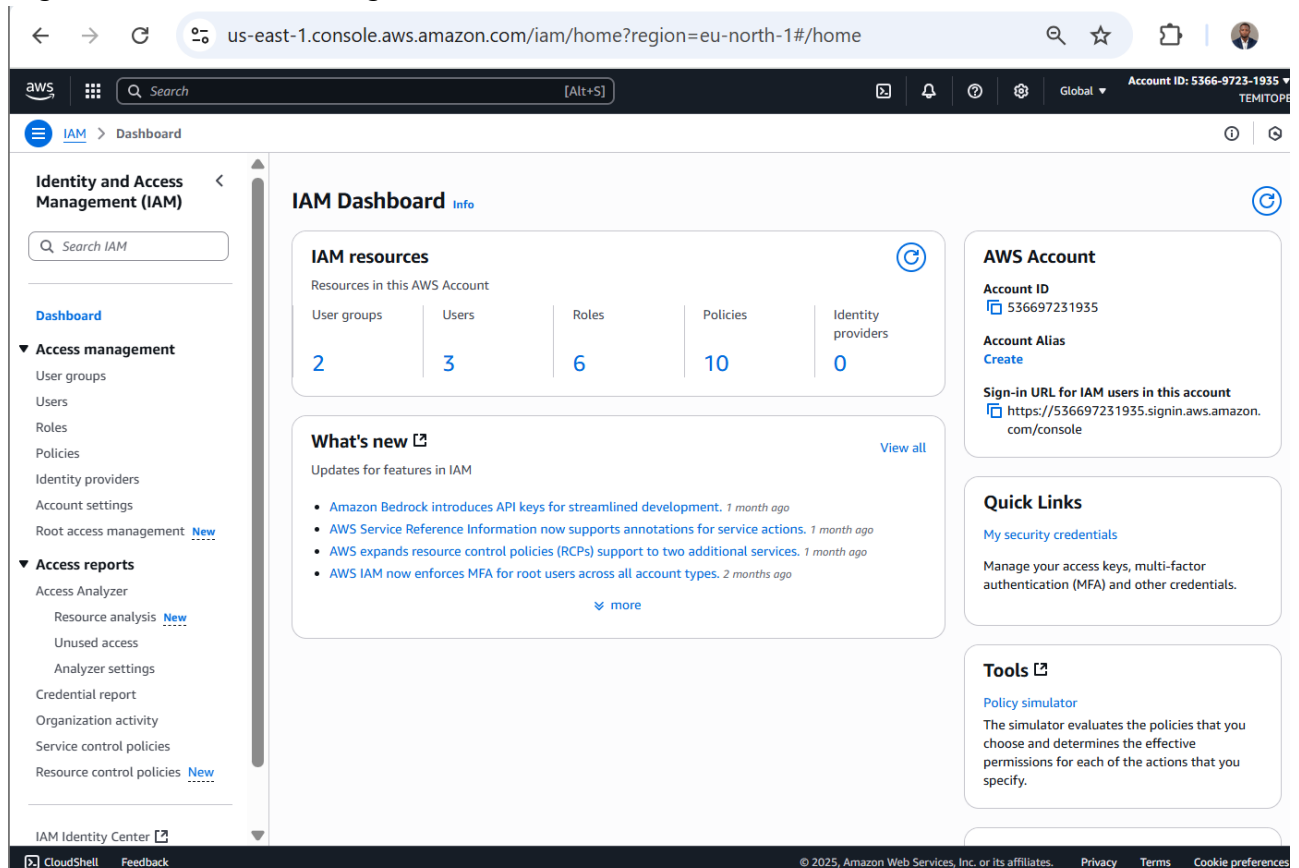
- Enabled MFA.
- Linked with **Google Authenticator**.
- Tested successful login with MFA.

Steps:

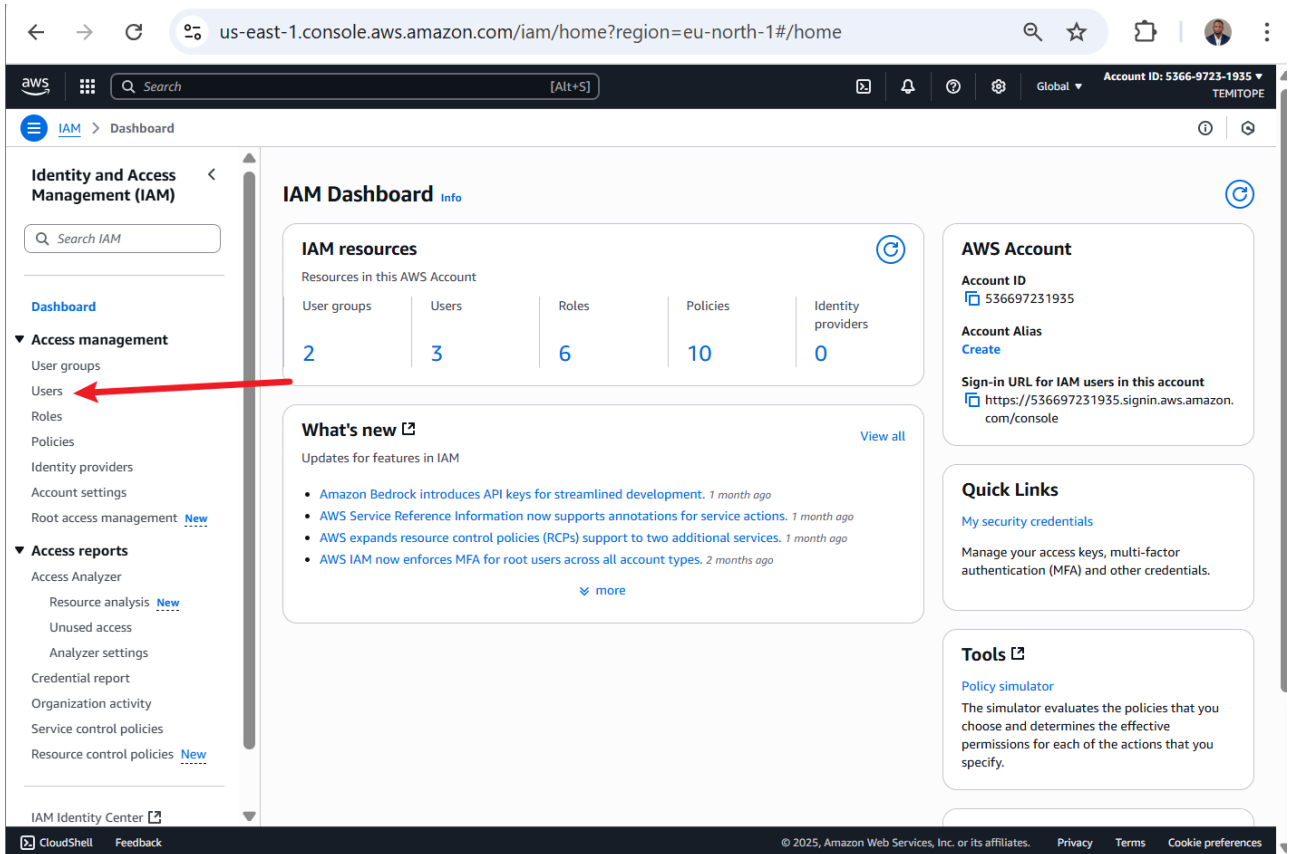
- Search IAM to set MFA for John.



- Login as IAM admin and navigate to IAM dashboard.



- Click on users.



The screenshot shows the AWS IAM Dashboard in the us-east-1 region. The left sidebar under 'Identity and Access Management (IAM)' has a search bar and a list of links. The 'Users' link is highlighted with a red arrow. The main content area shows the 'IAM Dashboard' with a summary of resources (2 user groups, 3 users, 6 roles, 10 policies, 0 identity providers) and a 'What's new' section with updates for features in IAM. The right sidebar contains 'AWS Account' information, 'Quick Links', and 'Tools'.

IAM Dashboard Info

IAM resources
Resources in this AWS Account

User groups	Users	Roles	Policies	Identity providers
2	3	6	10	0

What's new [View all](#)

Updates for features in IAM

- Amazon Bedrock introduces API keys for streamlined development. 1 month ago
- AWS Service Reference Information now supports annotations for service actions. 1 month ago
- AWS expands resource control policies (RCPs) support to two additional services. 1 month ago
- AWS IAM now enforces MFA for root users across all account types. 2 months ago

[more](#)

AWS Account

Account ID
536697231935

Account Alias
[Create](#)

Sign-in URL for IAM users in this account
<https://536697231935.signin.aws.amazon.com/console>

Quick Links

[My security credentials](#)

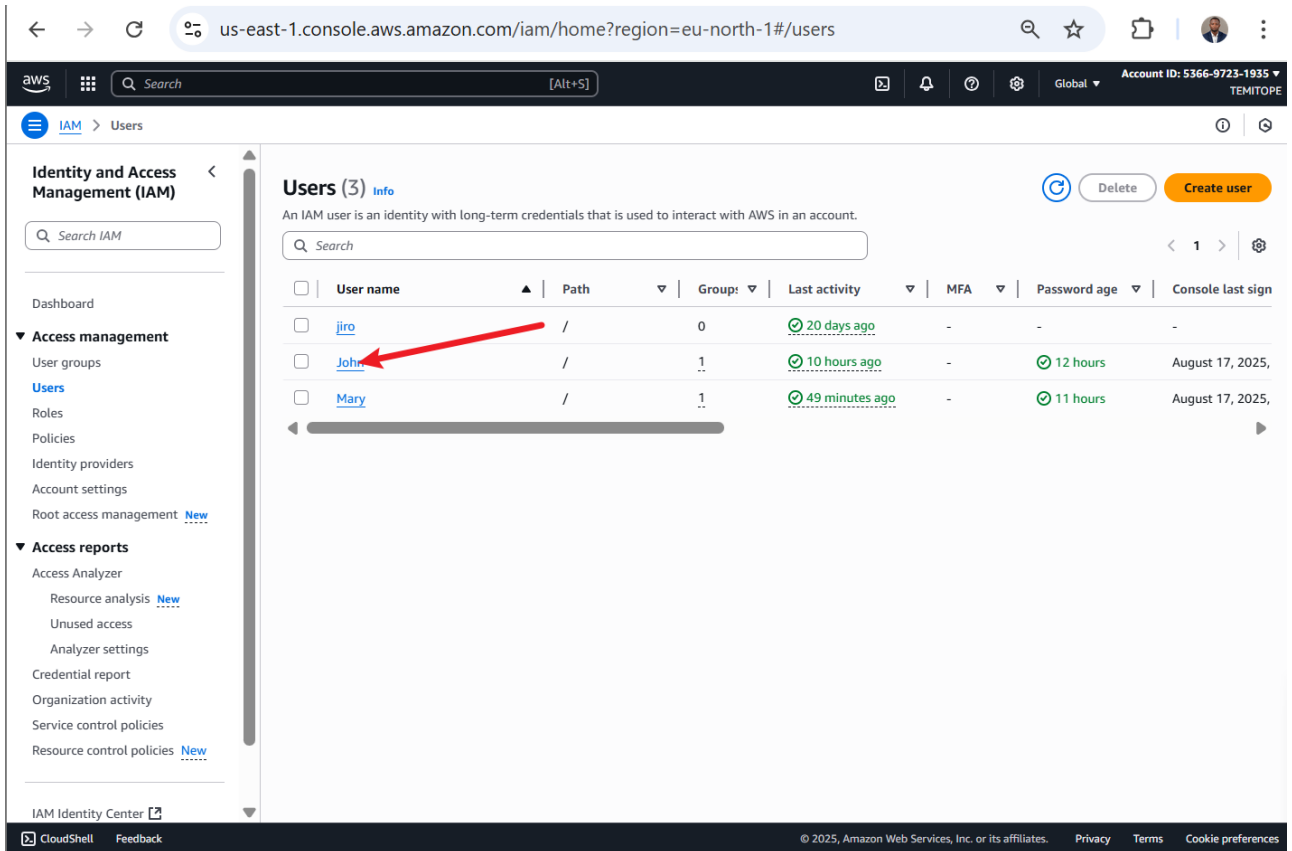
Manage your access keys, multi-factor authentication (MFA) and other credentials.

Tools

[Policy simulator](#)

The simulator evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

- Click on John.



The screenshot shows the AWS IAM Users page in the us-east-1 region. The left sidebar under 'Identity and Access Management (IAM)' has a search bar and a list of links. The 'Users' link is highlighted with a red arrow. The main content area shows the 'Users (3)' page with a description of IAM users and a table of users. The 'John' user is highlighted with a red arrow. The right sidebar contains 'Delete' and 'Create user' buttons.

Users (3) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

[Search](#)

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign
☐	jiro	/	0	20 days ago	-	-	-
☐	John	/	1	10 hours ago	-	12 hours	August 17, 2025,
☐	Mary	/	1	49 minutes ago	-	11 hours	August 17, 2025,

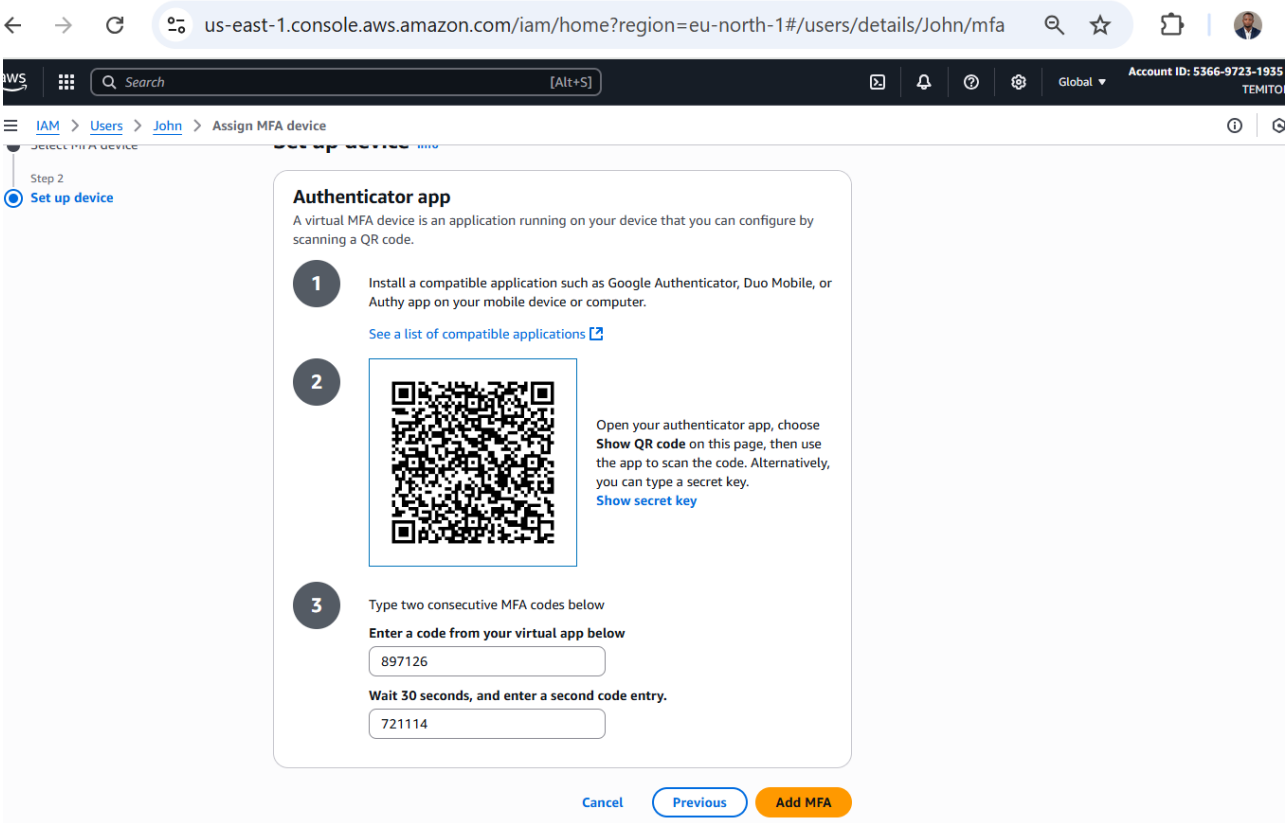
- Click security credential and assign MFA.

The screenshot shows the AWS IAM console interface. On the left is a navigation menu with 'Identity and Access Management (IAM)' selected. The main content area is for user 'John'. The 'Security credentials' tab is active, showing 'Console access' as 'Enabled without MFA' and 'Last console sign-in' as 'Today'. A red arrow points to the 'Security credentials' tab. Below this, the 'Console sign-in' section shows the 'Console sign-in link' and 'Console password' (updated 12 hours ago). Another red arrow points to the 'Assign MFA device' button in the 'Multi-factor authentication (MFA) (0)' section.

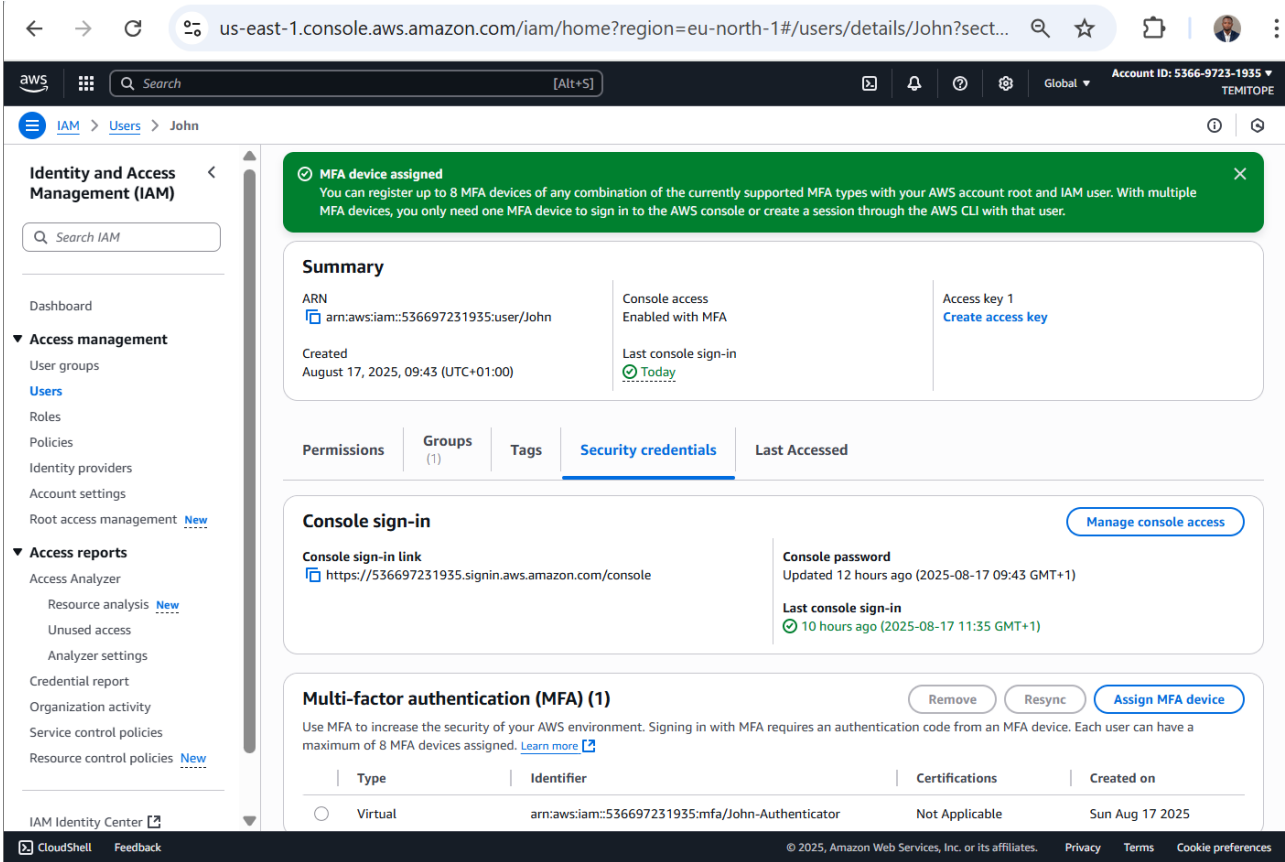
- Set name and choose authenticator app and click next.

The screenshot shows the 'Assign MFA device' setup screen. The 'MFA device name' section has a text input field containing 'John Authenticator', with a red arrow pointing to it. Below this, the 'MFA device' section shows three options: 'Passkey or security key', 'Authenticator app' (which is selected and highlighted with a blue border and a red arrow), and 'Hardware TOTP token'. At the bottom right are 'Cancel' and 'Next' buttons.

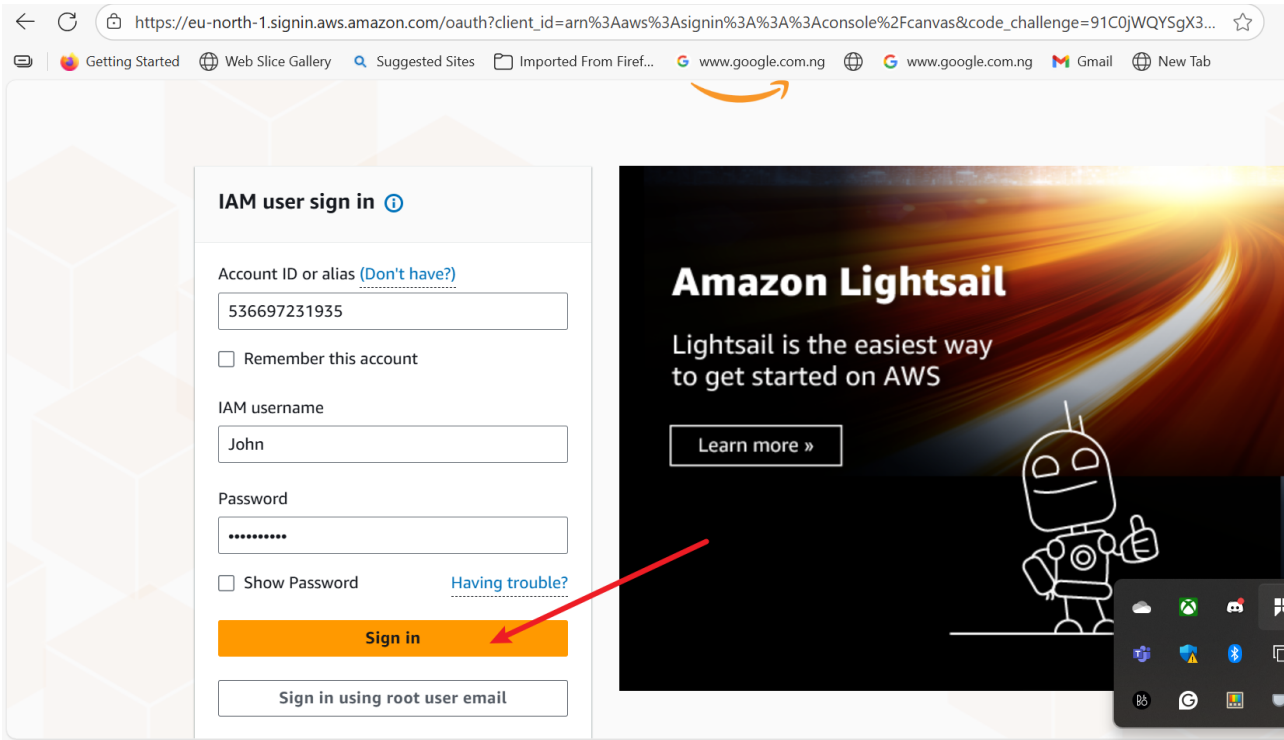
- Show QR, scan it and supply code as instructed.



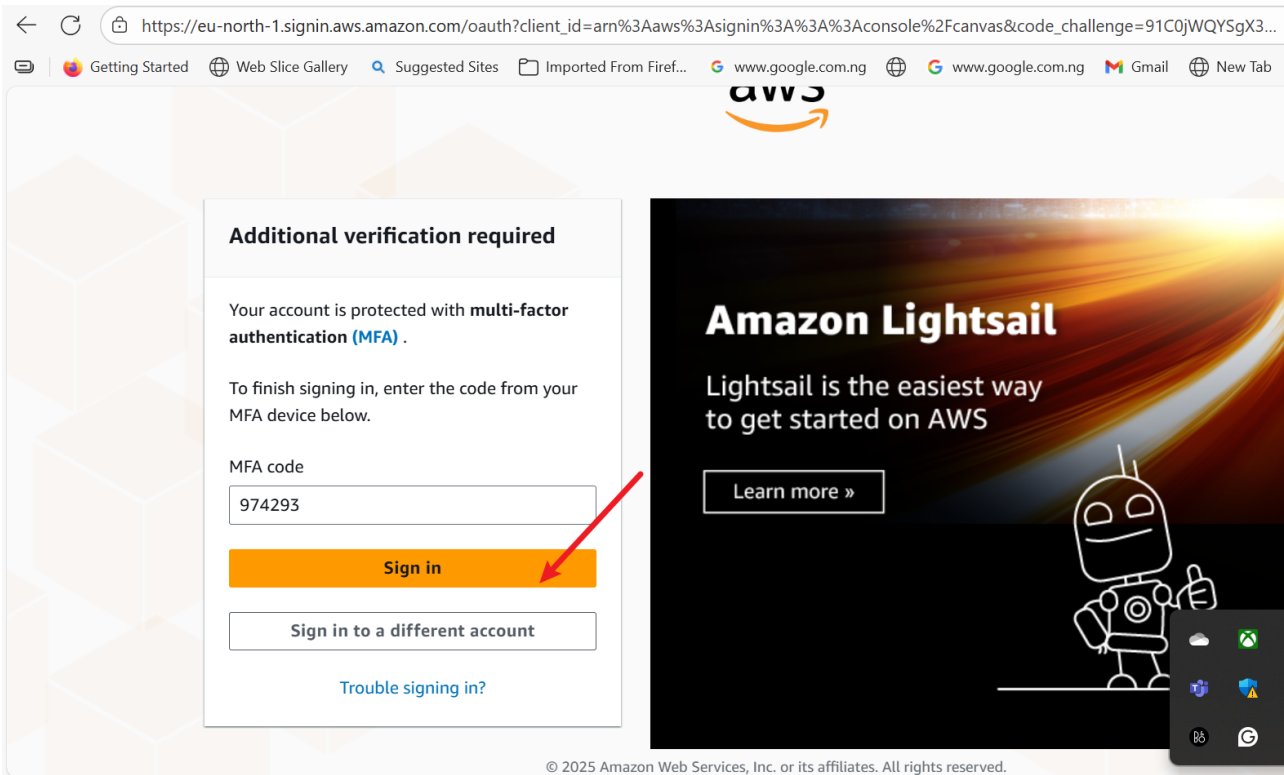
- MFA successfully added.



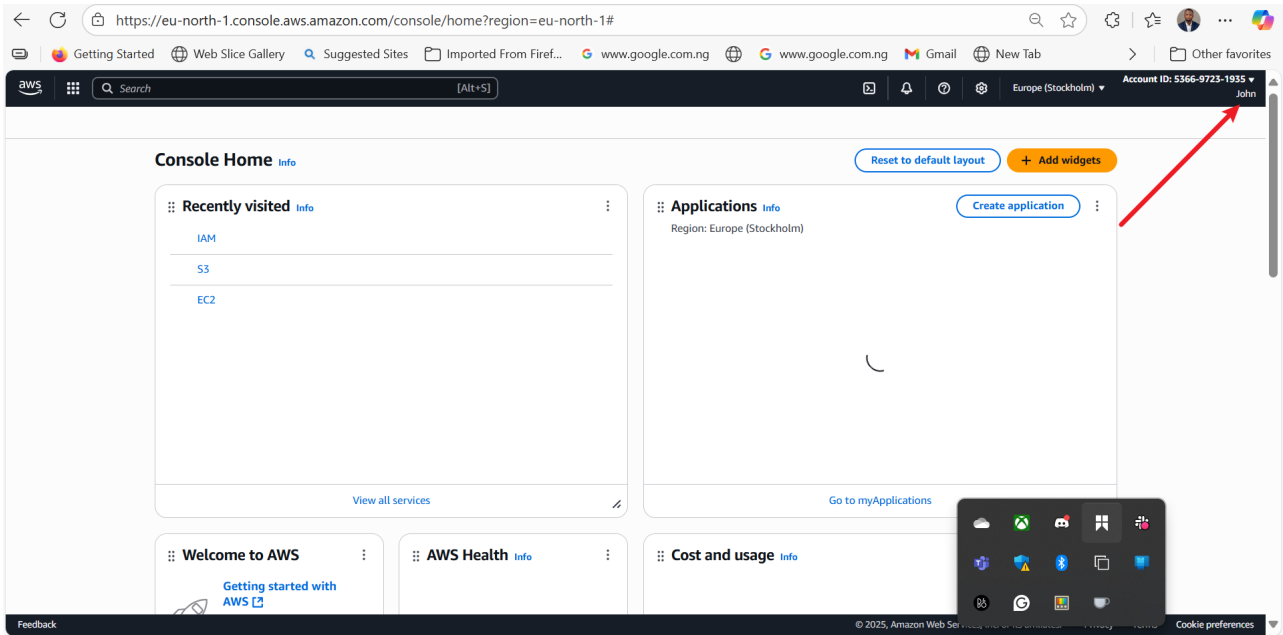
- Sign-in as John.



- Supply required MFA and login.



- Successfully logged in.

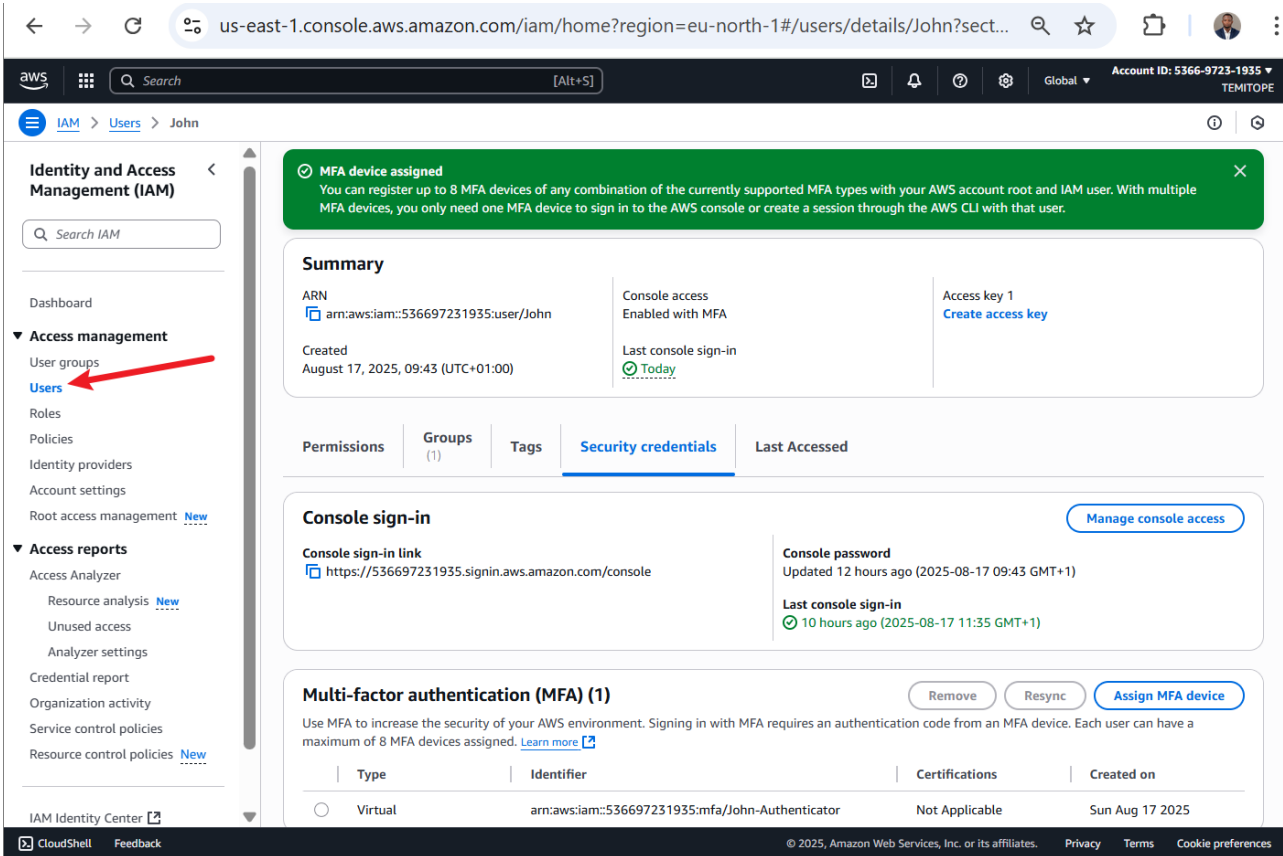


Mary (Data Analyst)

- Enabled MFA.
- Linked with **Microsoft Authenticator**.
- Tested successful login with MFA.

Steps:

- Click users to repeat the same step for Mary.



- Click on Mary.

The screenshot shows the AWS IAM console in the 'us-east-1' region. The 'Users' page displays a list of three users: jiro, John, and Mary. A red arrow points to the user 'Mary' in the list. The table columns are: User name, Path, Group, Last activity, MFA, Password age, and Console last sign.

User name	Path	Group	Last activity	MFA	Password age	Console last sign
jiro	/	0	20 days ago	-	-	-
John	/	1	10 hours ago	-	12 hours	August 17, 2025
Mary	/	1	1 hour ago	-	11 hours	August 17, 2025

- Click security credential and assign MFA.

The screenshot shows the 'Mary' user details page in the AWS IAM console. The 'Security credentials' tab is selected, and a red arrow points to it. The 'Console sign-in' section shows the console sign-in link and password. Another red arrow points to the 'Assign MFA device' button in the 'Multi-factor authentication (MFA)' section.

Summary

- ARN: `arn:aws:iam::536697231935:user/Mary`
- Console access: Enabled without MFA
- Access key 1: [Create access key](#)
- Created: August 17, 2025, 10:53 (UTC+01:00)
- Last console sign-in: Today

Permissions | **Groups (1)** | **Tags** | **Security credentials** | **Last Accessed**

Console sign-in

- Console sign-in link: <https://536697231935.signin.aws.amazon.com/console>
- Console password: Updated 11 hours ago (2025-08-17 10:53 GMT+1)
- Last console sign-in: 1 hour ago (2025-08-17 21:09 GMT+1)

Multi-factor authentication (MFA) (0)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

[Remove](#) [Resync](#) [Assign MFA device](#)

Type	Identifier	Certifications	Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment			

[Assign MFA device](#)

- Set name and choose authenticator app and click next.

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/Mary/mfa

Account ID: 5366-9723-1935

TEMITO

Step 2
Set up device

MFA device name

Device name
This name will be used within the identifying ARN for this device.

Mary-Authenticator

Maximum 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

MFA device

Device options
In addition to username and password, you will use this device to authenticate into your account.

☐ **Passkey or security key**
Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.

☒ **Authenticator app**
Authenticate using a code generated by an app installed on your mobile device or computer.

☐ **Hardware TOTP token**
Authenticate using a code generated by Hardware TOTP token or other hardware devices.

Cancel Next

- Show QR, scan it and supply code as instructed.

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/Mary/mfa

Account ID: 5366-9723-1935

TEMITO

Step 1
Select MFA device

Step 2
Set up device

Set up device

Authenticator app
A virtual MFA device is an application running on your device that you can configure by scanning a QR code.

1 Install a compatible application such as Google Authenticator, Duo Mobile, or Authy app on your mobile device or computer.
[See a list of compatible applications](#)

2 Open your authenticator app, choose **Show QR code** on this page, then use the app to scan the code. Alternatively, you can type a secret key.
[Show secret key](#)

3 Type two consecutive MFA codes below

Enter a code from your virtual app below

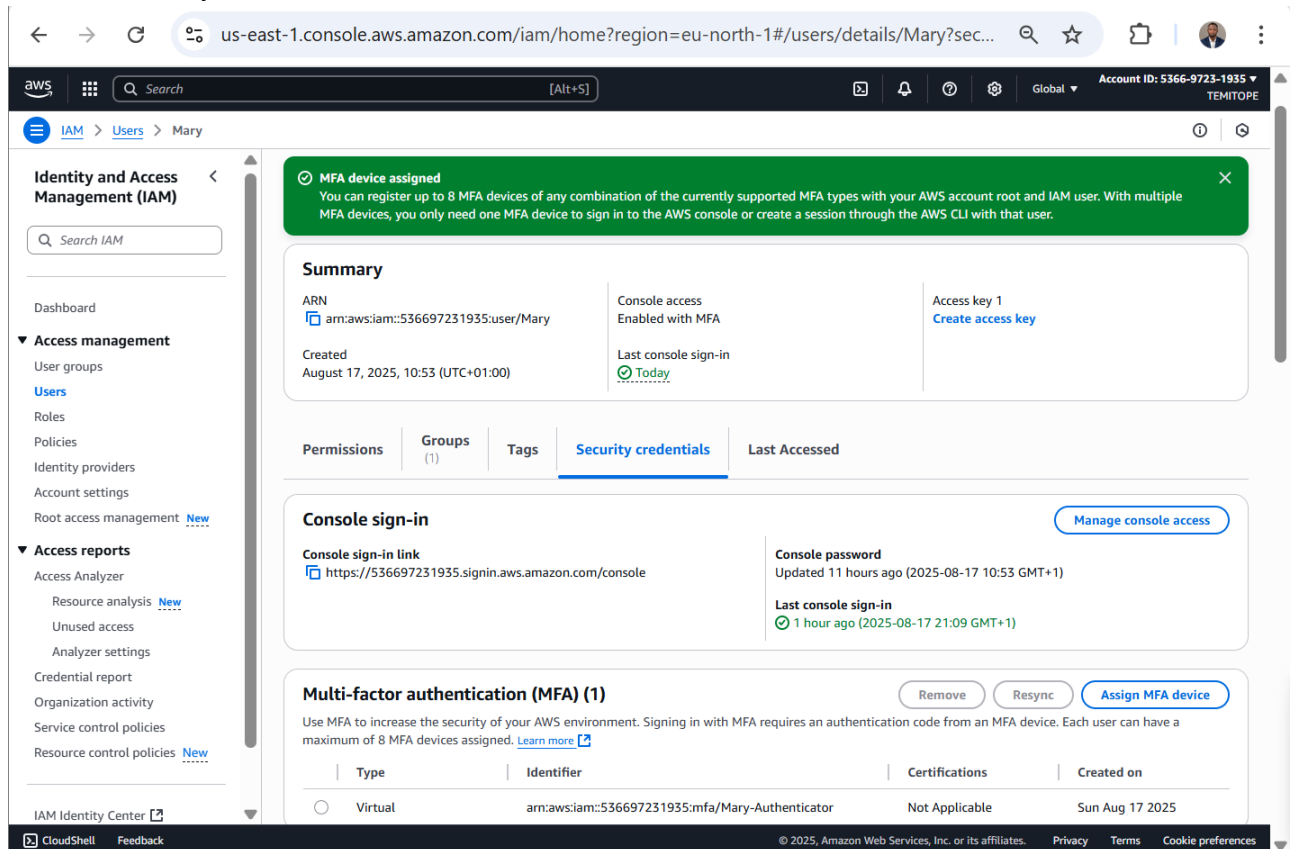
104312

Wait 30 seconds, and enter a second code entry.

458376

Cancel Previous Add MFA

- MFA successfully added.



9. Project Reflection

9.1 Role of IAM in AWS

IAM defines **who** can access AWS and **what actions** they can perform. It enhances security, compliance, and efficient resource management.

9.2 Difference Between IAM Users and Groups

- **IAM Users** → Individual identities (e.g., John, Mary).
- **IAM Groups** → Collections of users with shared permissions (e.g., Development-Team, Analyst-Team). Groups simplify permission management at scale.

9.3 Process of Creating IAM Policies

- Identify required service.
- Select allowed actions.
- Define resources.
- Save as **custom policy**.
- Attach policy to a user or group.

9.4 Significance of Principle of Least Privilege

Granting users only the access they need minimizes risks and prevents accidental or malicious misuse of AWS resources.

9.5 Scenario with John and Mary

- **John** → User in Development-Team → EC2 access only.
 - **Mary** → User in Analyst-Team → S3 access only.
 - Both secured with **MFA**.
 - This setup mirrors real-world fintech security practices.
-

10. Conclusion

This project demonstrated how AWS IAM helps enforce **secure, role-based access control** for Zappy e-Bank. By applying IAM concepts such as **users, groups, policies, and MFA**, the project successfully built a secure environment that aligns with fintech compliance requirements and the **principle of least privilege**.